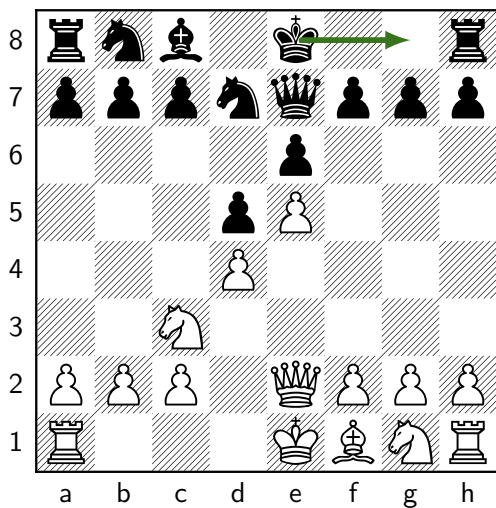


Melanie-BOTS (CCRL engines) — French, Classical

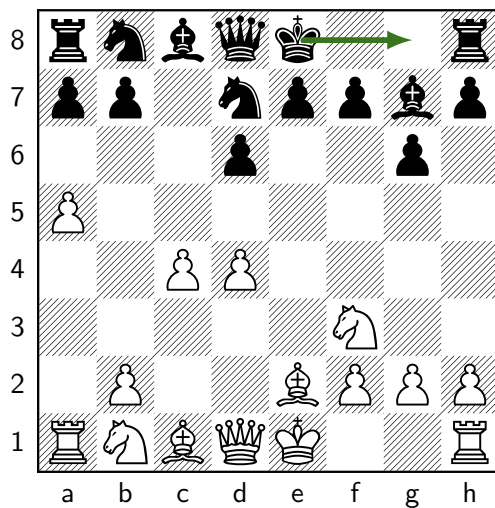
Melanie (Deep_Vision)

Abstract

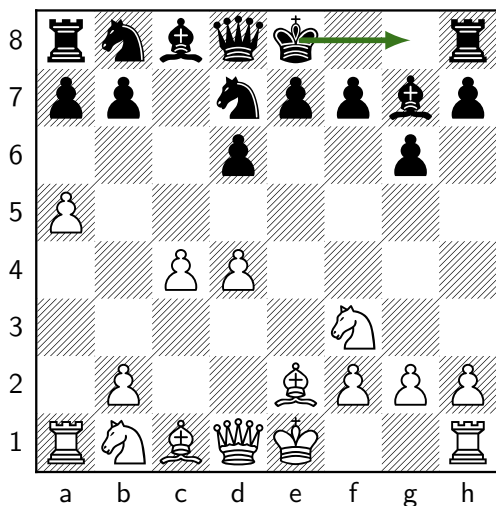
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



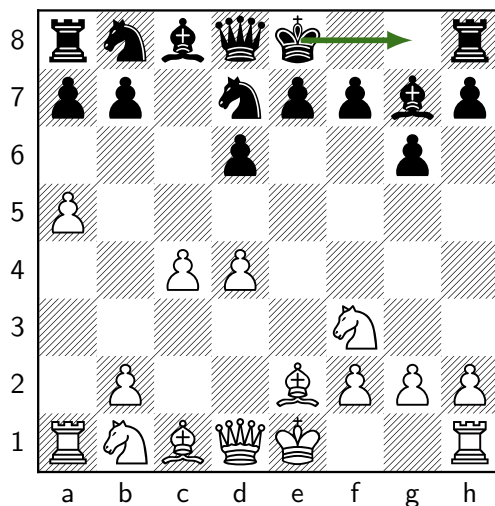
(a)



(b)

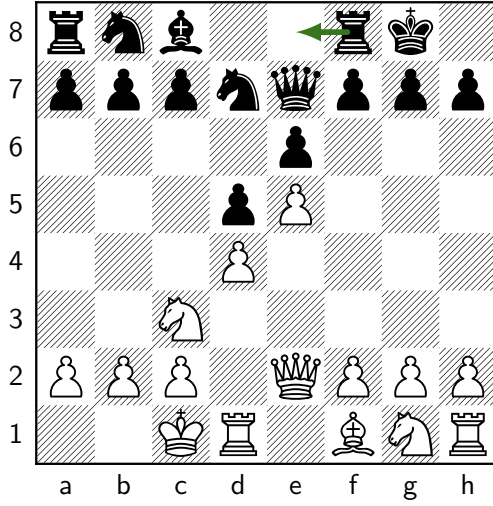


(c)

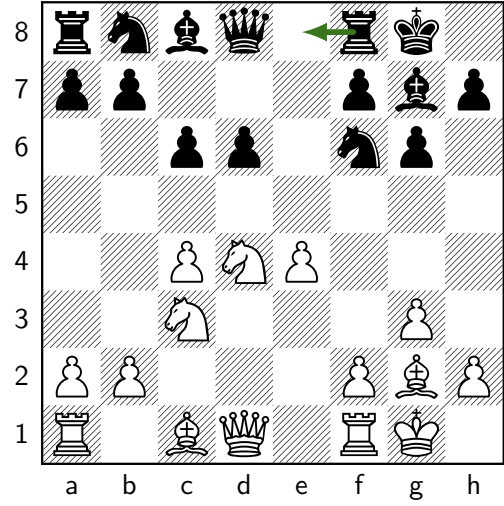


(d)

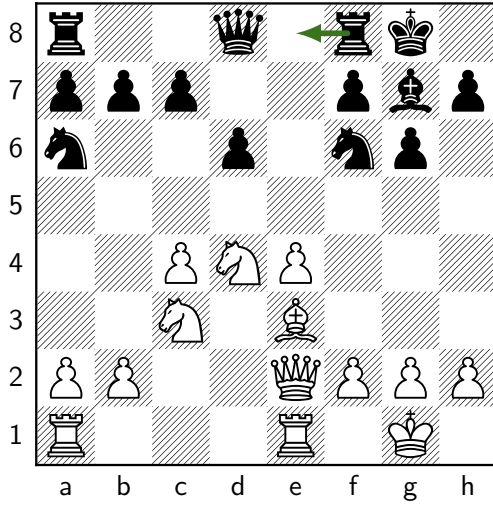
Figure 1: **Move 7, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1707` (ply 17) [structural distance $d=64$, QAP obj=30]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pd4* as their *Ra8*, our *Pf7* as their *Nb8*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1708` (ply 17) [structural distance $d=64$, QAP obj=30]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pd4* as their *Ra8*, our *Pf7* as their *Nb8*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1825` (ply 17) [structural distance $d=64$, QAP obj=30]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pd4* as their *Ra8*, our *Pf7* as their *Nb8*; transferred move e8g8 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to 0-0 (summed QAP objective 181, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



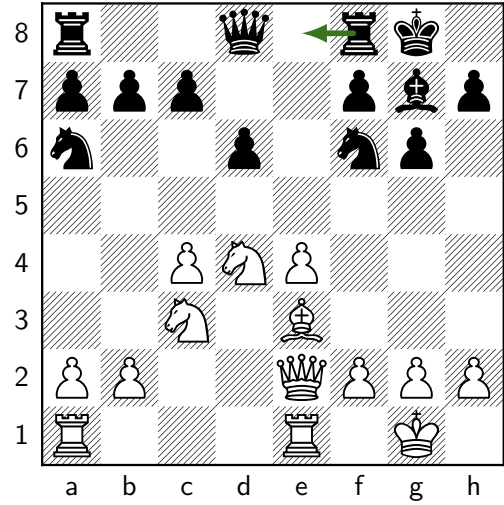
(a)



(b)

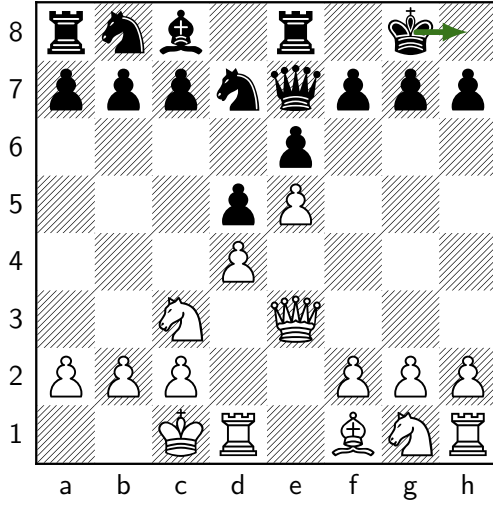


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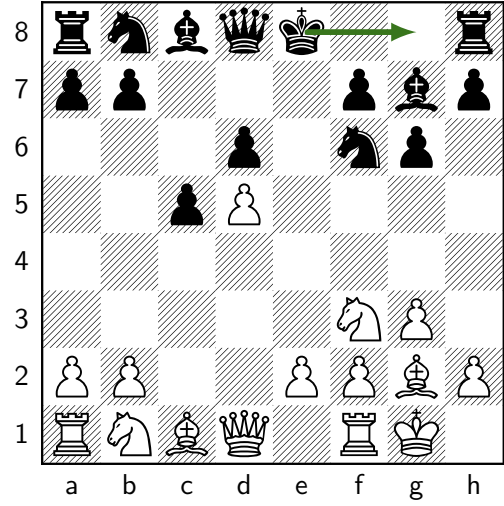


(d)

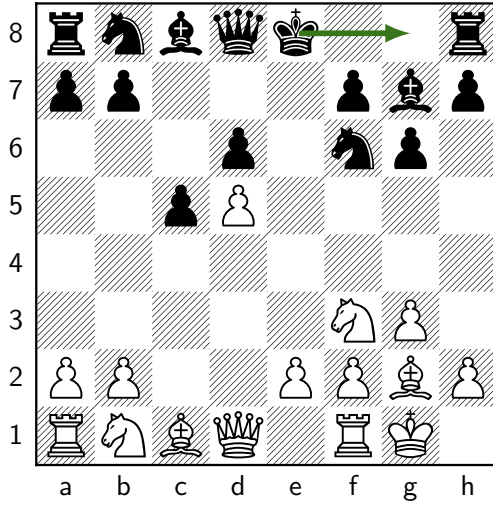
Figure 2: **Move 8, Black: Re8.** (a) Our position; Melanie plays Re8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#352` (ply 17) [structural distance $d=72$, QAP obj=40]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Bc8* as their *Ra8*, our *ph2* as their *Nb8*; transferred move f8e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#660` (ply 21) [structural distance $d=72$, QAP obj=23]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Bc8* as their *Pa7*, our *pe5* as their *Bg7*; transferred move f8e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1453` (ply 21) [structural distance $d=72$, QAP obj=23]: the master played Re8; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Bc8* as their *Pa7*, our *pe5* as their *Bg7*; transferred move f8e8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Re8 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



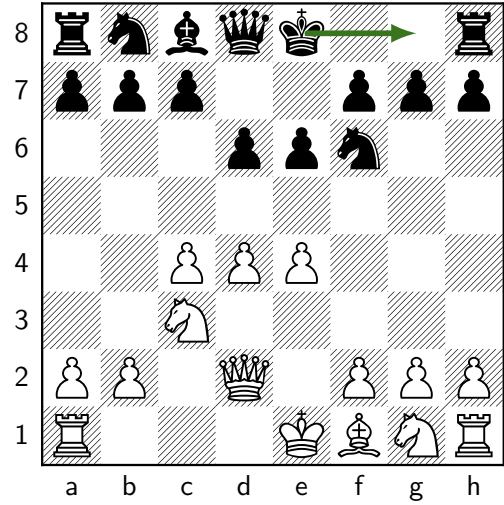
(a)



(b)

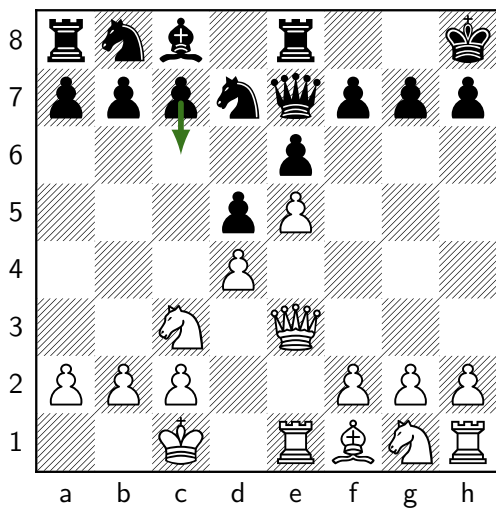


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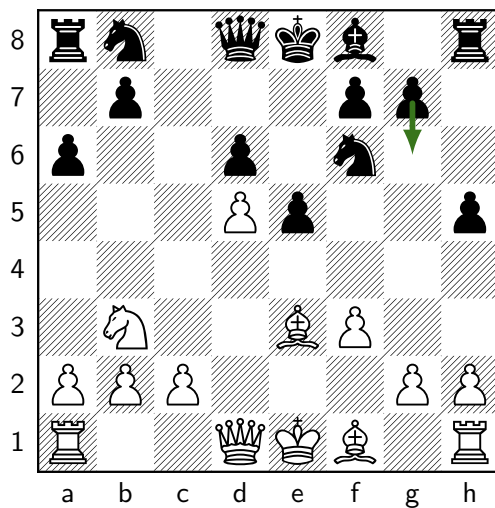


(d)

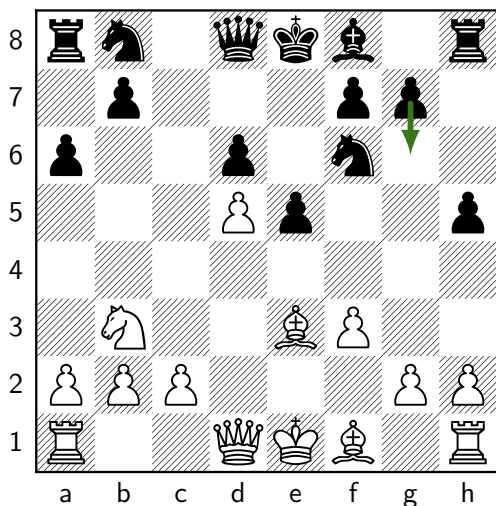
Figure 3: **Move 9, Black: Kh8.** (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2672` (ply 15) [structural distance $d=69$, QAP obj=22]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Pc7* as their *Nb8*; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2673` (ply 15) [structural distance $d=69$, QAP obj=22]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Pc7* as their *Nb8*; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3463` (ply 11) [structural distance $d=71$, QAP obj=22]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Nb8* as their *Ra8*, our *Bc8* as their *Nb8*; transferred move g8h8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kh8 (summed QAP objective 102, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



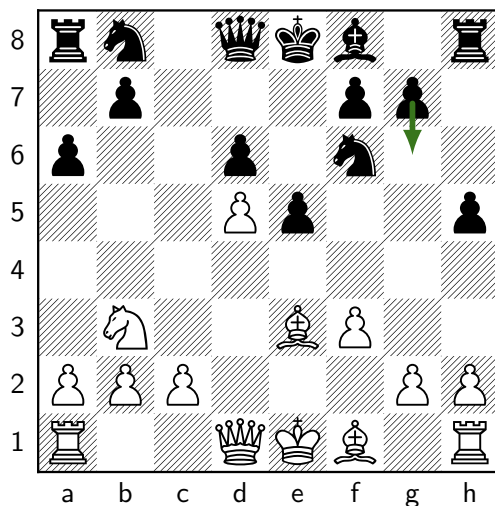
(a)



(b)

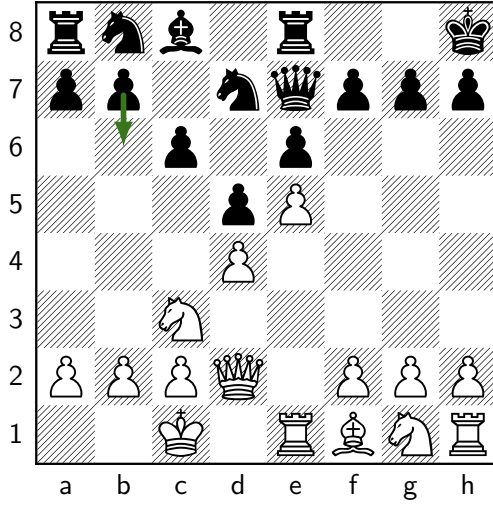


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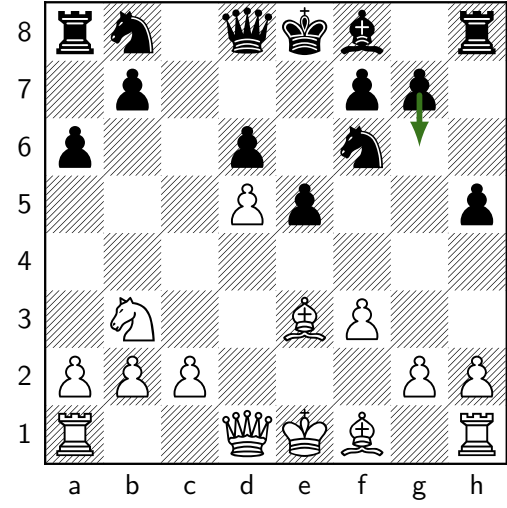


(d)

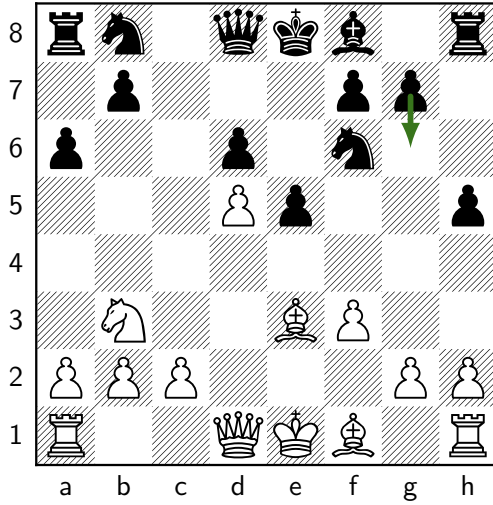
Figure 4: **Move 10, Black: c6.** (a) Our position; Melanie plays c6 (c7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2358` (ply 19) [structural distance $d=75$, QAP obj=30]: the master played g6; under μ our $Pc2$ plays the role of that game’s $Pg2$; cross-type roles: our $Pe6$ as their $Ra8$, our $ph2$ as their $Nb8$; transferred move c7c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2359` (ply 19) [structural distance $d=75$, QAP obj=30]: the master played g6; under μ our $Pc2$ plays the role of that game’s $Pg2$; cross-type roles: our $Pe6$ as their $Ra8$, our $ph2$ as their $Nb8$; transferred move c7c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3616` (ply 19) [structural distance $d=75$, QAP obj=30]: the master played g6; under μ our $Pc2$ plays the role of that game’s $Pg2$; cross-type roles: our $Pe6$ as their $Ra8$, our $ph2$ as their $Nb8$; transferred move c7c6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to c6 (summed QAP objective 149, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



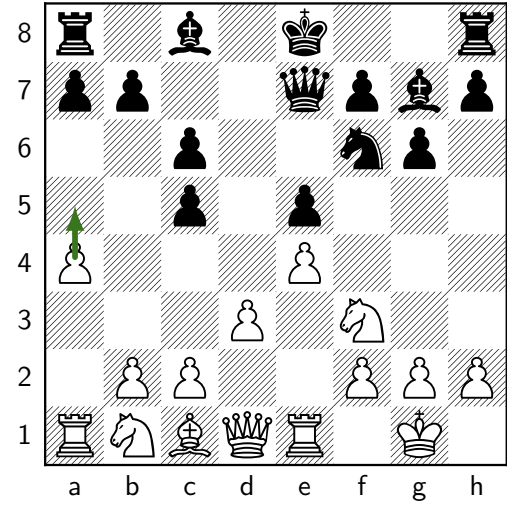
(a)



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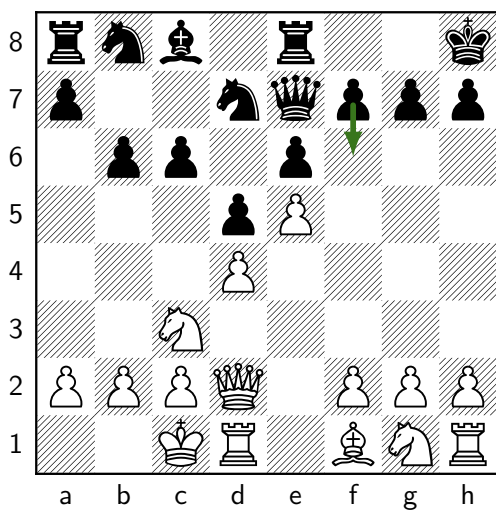


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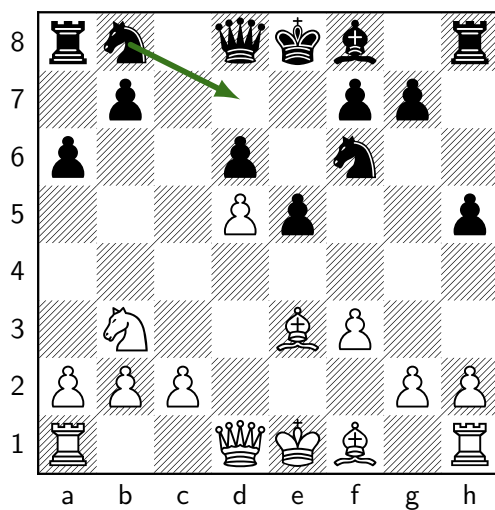


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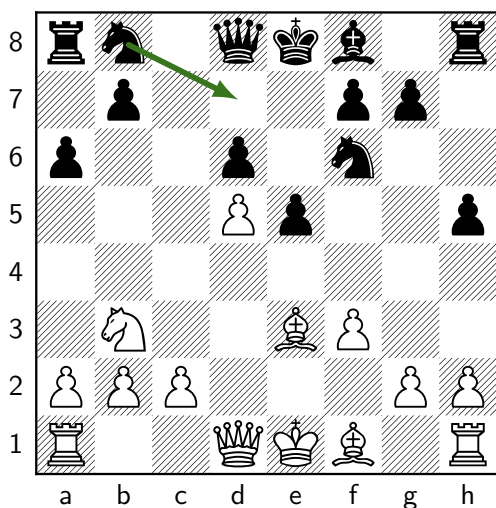
Figure 5: **Move 11, Black: b6.** (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2358` (ply 19) [structural distance $d=78$, QAP obj=30]: the master played g6; under μ our $Pb2$ plays the role of that game’s $Pg2$; cross-type roles: our $pg2$ as their $Nb8$, our $re1$ as their $Bf8$; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2359` (ply 19) [structural distance $d=78$, QAP obj=30]: the master played g6; under μ our $Pb2$ plays the role of that game’s $Pg2$; cross-type roles: our $pg2$ as their $Nb8$, our $re1$ as their $Bf8$; transferred move b7b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1355` (ply 16) [structural distance $d=75$, QAP obj=39]: the master played a5; under μ our $Pc3$ plays the role of that game’s $Pa4$; cross-type roles: our $Ph7$ as their $Ra1$, our $Ra8$ as their $Nb1$; transferred move c6c5. Synthesis: 2 retrieved analogs transferred to b6 (summed QAP objective 60, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



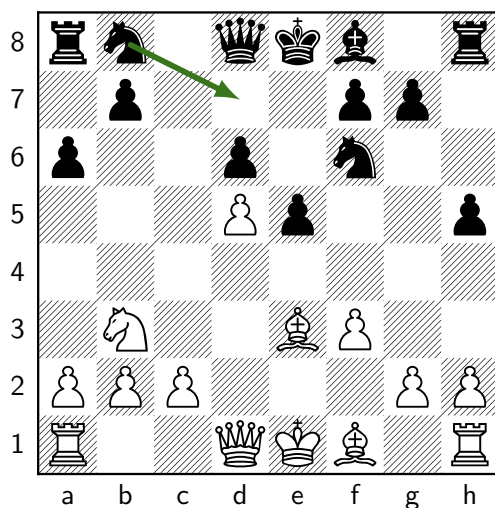
(a)



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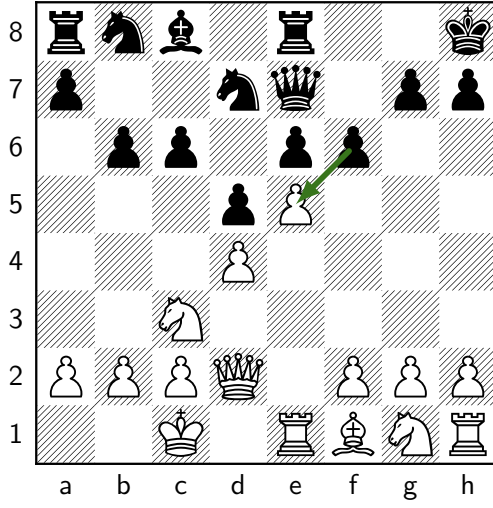


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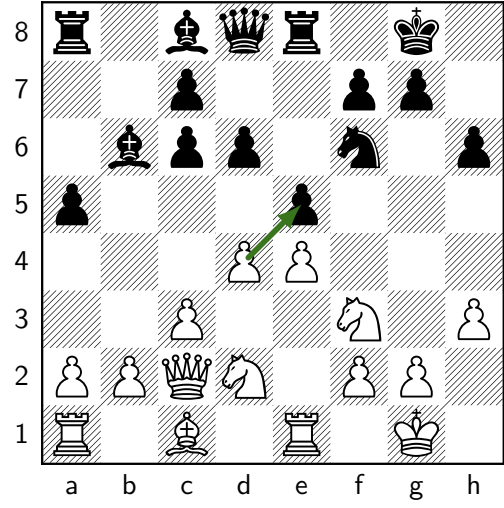


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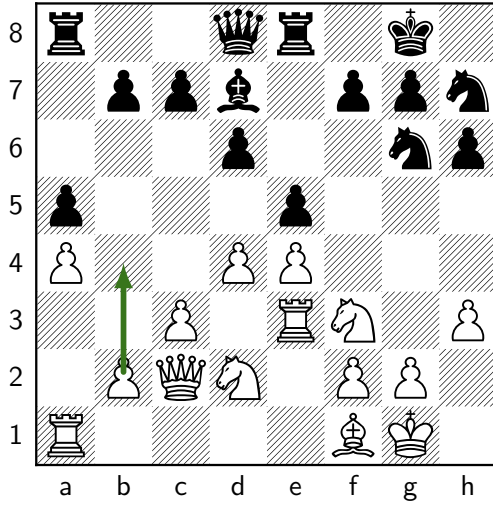
Figure 6: **Move 12, Black: f6.** (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#407` (ply 19) [structural distance $d=80$, QAP obj=30]: the master played Nbd7; under μ our Pf2 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra8, our Pg7 as their Bf8; transferred move f7f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#755` (ply 19) [structural distance $d=80$, QAP obj=30]: the master played Nbd7; under μ our Pf2 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra8, our Pg7 as their Bf8; transferred move f7f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1281` (ply 19) [structural distance $d=80$, QAP obj=30]: the master played Nbd7; under μ our Pf2 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra8, our Pg7 as their Bf8; transferred move f7f6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to f6 (summed QAP objective 151, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



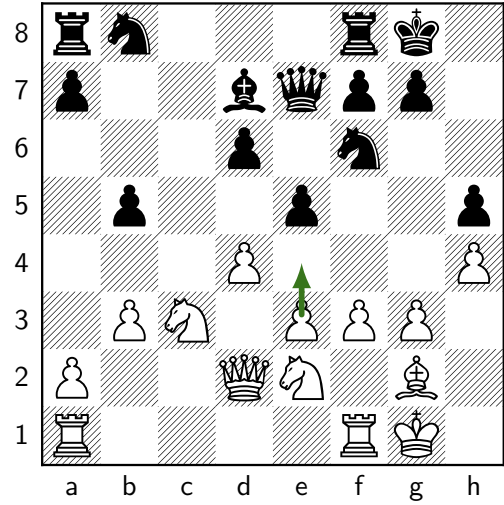
(a)



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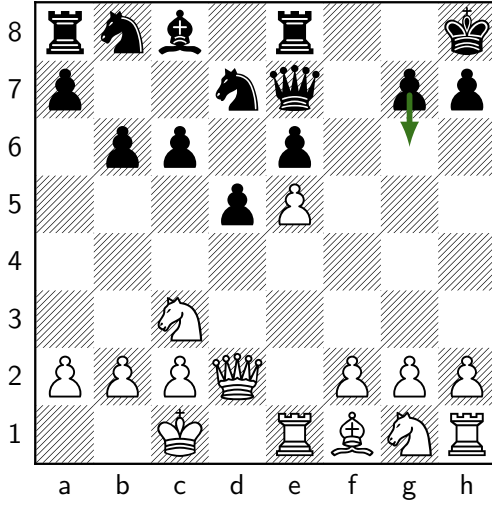


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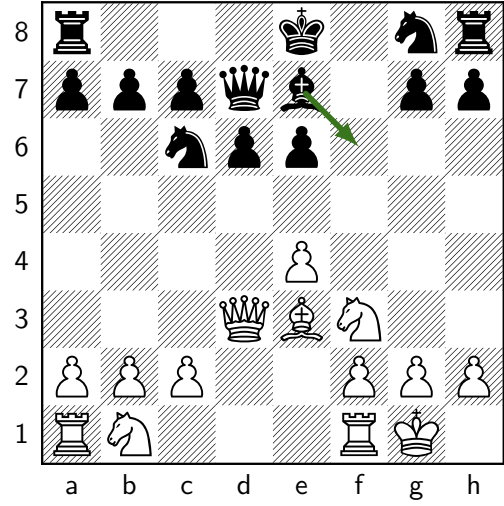


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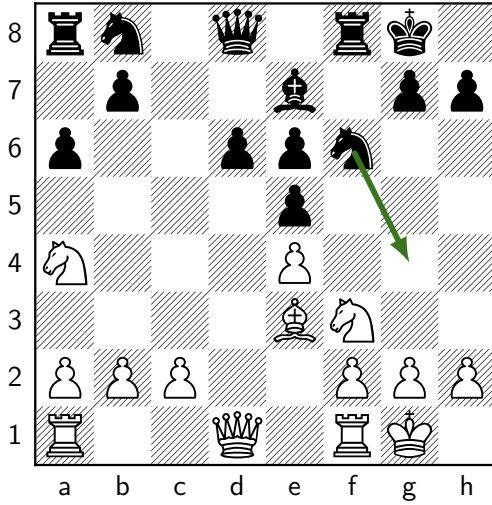
Figure 7: **Move 13, Black: fxe5.** (a) Our position; Melanie plays fxe5 (f6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3942` (ply 24) [structural distance $d=77$, QAP obj=79]: the master played dxe5; under μ our Pf3 plays the role of that game’s Pd4; cross-type roles: our Ph7 as their Bc1, our Pc6 as their Re1; transferred move f6e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3867` (ply 32) [structural distance $d=85$, QAP obj=73]: the master played b4; under μ our Nb1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Pe6 as their Ra1, our Pb6 as their Bf1; transferred move b8a6. (d) Retrieved from `ccr1_top_p0002#3050` (ply 28) [structural distance $d=85$, QAP obj=54]: the master played e4; under μ our Pg2 plays the role of that game’s Pe3; cross-type roles: our Nb8 as their Ra1, our pd4 as their Bg2; transferred move g7g5. Synthesis: 1 retrieved analog transferred to fxe5 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



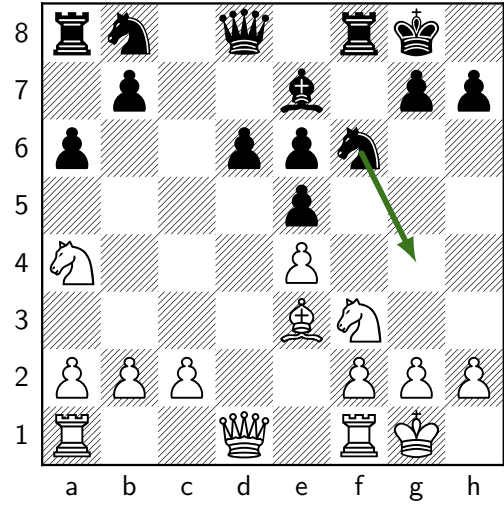
(a)



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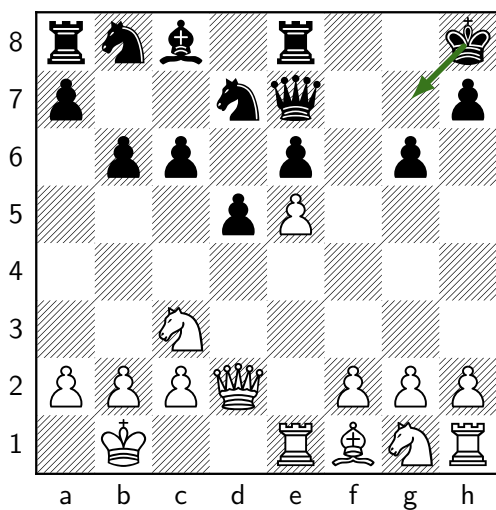


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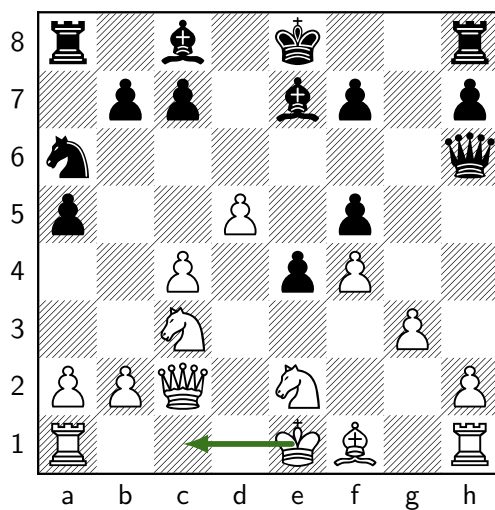


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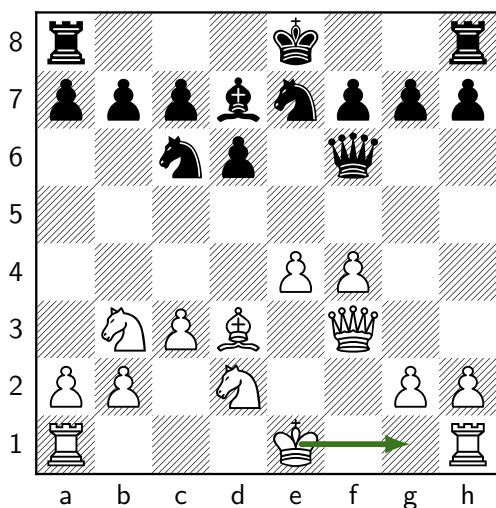
Figure 8: **Move 14, Black: g6.** (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2879` (ply 21) [structural distance $d=83$, QAP obj=47]: the master played Bf6; under μ our Pg2 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our rh1 as their Ng8, our Pc6 as their Qd7; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3513` (ply 21) [structural distance $d=73$, QAP obj=58]: the master played Ng4; under μ our Nd2 plays the role of that game’s Nf3; cross-type roles: our Qe7 as their Ra8, our Ra8 as their Nb8; transferred move d7e5. (d) Retrieved from `ccr1_top_p0003#1668` (ply 21) [structural distance $d=73$, QAP obj=58]: the master played Ng4; under μ our Nd2 plays the role of that game’s Nf3; cross-type roles: our Qe7 as their Ra8, our Ra8 as their Nb8; transferred move d7e5. Synthesis: 1 retrieved analog transferred to g6 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



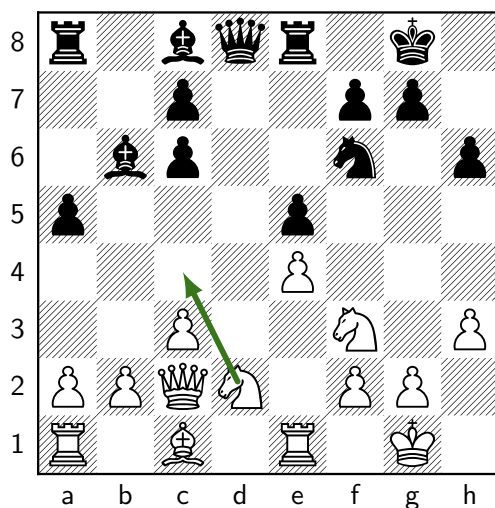
(a)



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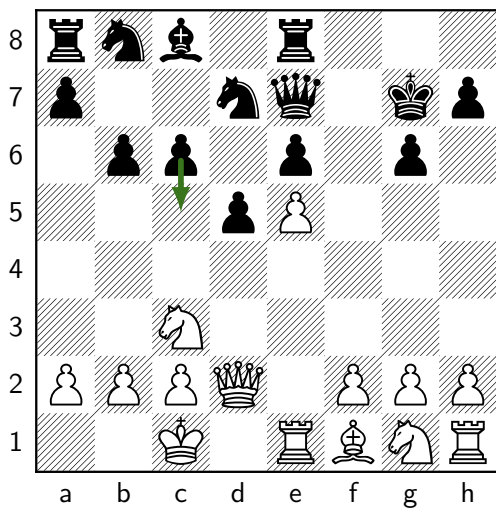


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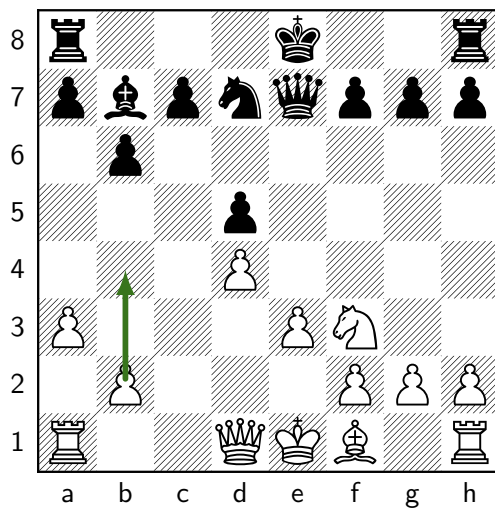


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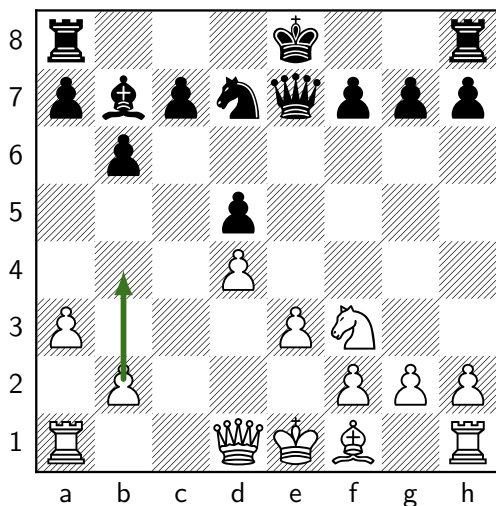
Figure 9: **Move 15, Black: Kg7**. (a) Our position; Melanie plays Kg7 (h8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#802 (ply 26) [structural distance $d=88$, QAP obj=71]: the master played 0-0-0; under μ our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Pb6* as their *Bf1*, our *Pc6* as their *Rh1*; transferred move h8g7 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#275 (ply 24) [structural distance $d=74$, QAP obj=42]: the master played 0-0; under μ our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Qe7* as their *Pa2*, our *Nb8* as their *Pb2*; transferred move h8g7 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3942 (ply 26) [structural distance $d=82$, QAP obj=64]: the master played Nc4; under μ our *Pd4* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *Nb8* as their *Ra1*, our *Qe7* as their *Bc1*; transferred move d5d4. Synthesis: 2 retrieved analogs transferred to Kg7 (summed QAP objective 113, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



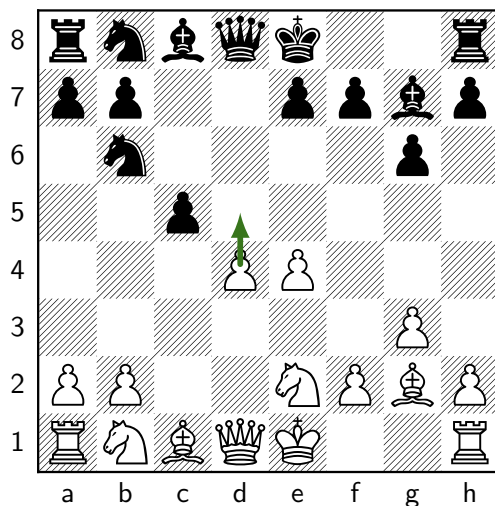
(a)



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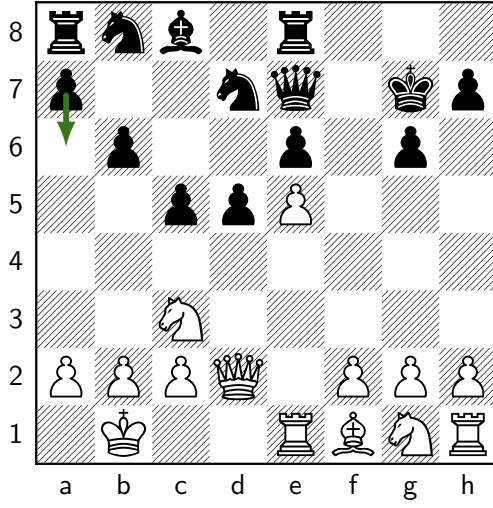


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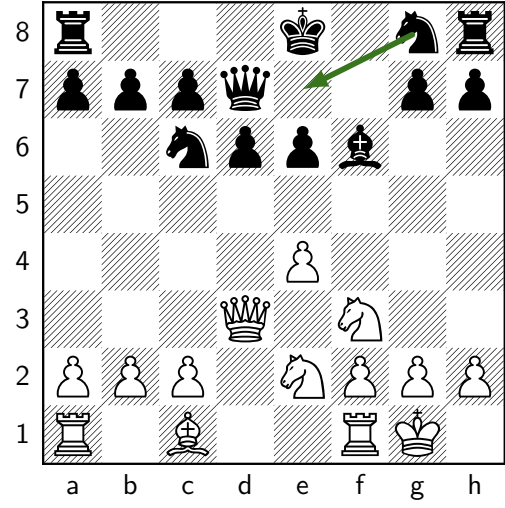


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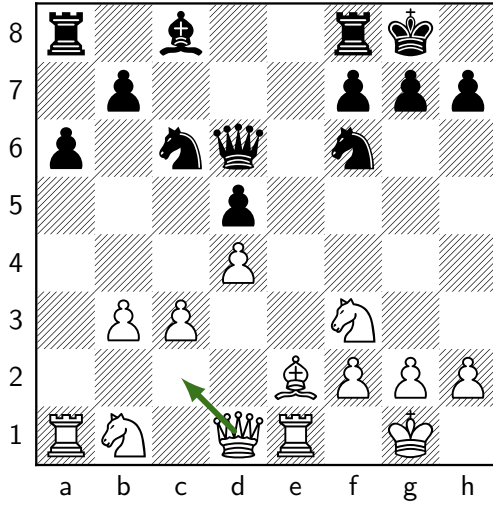
Figure 10: **Move 16, Black: c5**. (a) Our position; Melanie plays c5 (c6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#475` (ply 20) [structural distance $d=83$, QAP obj=44]: the master played b4; under μ our $Pc3$ plays the role of that game’s $Pb2$; cross-type roles: our $pb2$ as their $Ra1$, our $ng1$ as their $Qd1$; transferred move c6c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#12` (ply 20) [structural distance $d=83$, QAP obj=44]: the master played b4; under μ our $Pc3$ plays the role of that game’s $Pb2$; cross-type roles: our $pb2$ as their $Ra1$, our $ng1$ as their $Qd1$; transferred move c6c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2916` (ply 14) [structural distance $d=84$, QAP obj=55]: the master played d5; under μ our $Pd4$ plays the role of that game’s $Pd4$; cross-type roles: our $pe5$ as their $Nb1$, our $Pc6$ as their $Bc1$; transferred move d5d4. Synthesis: 2 retrieved analogs transferred to c5 (summed QAP objective 88, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



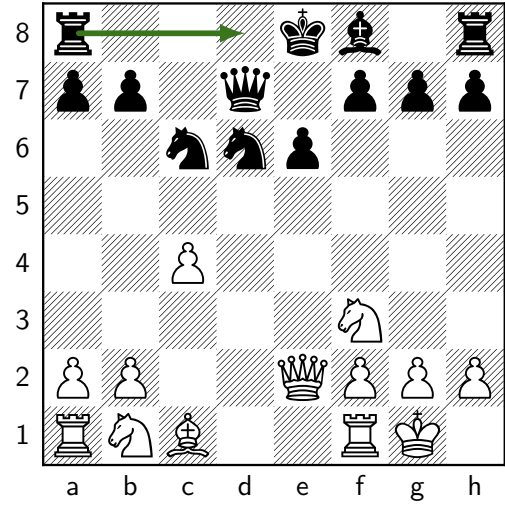
(a)



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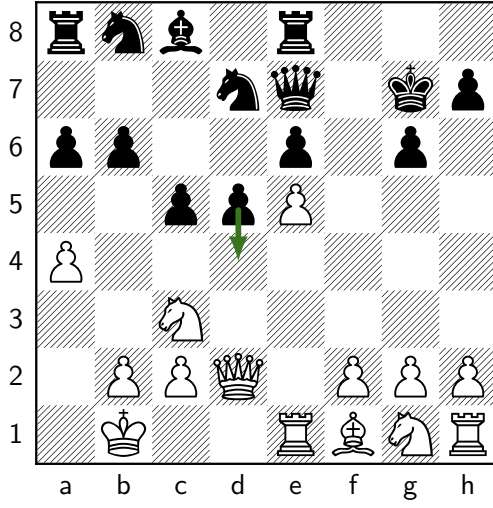


(c)

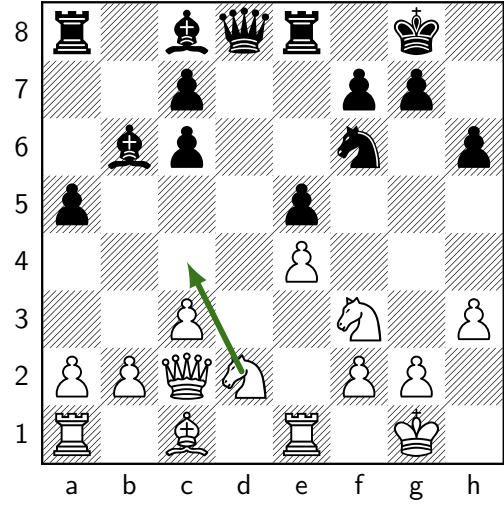


(d)

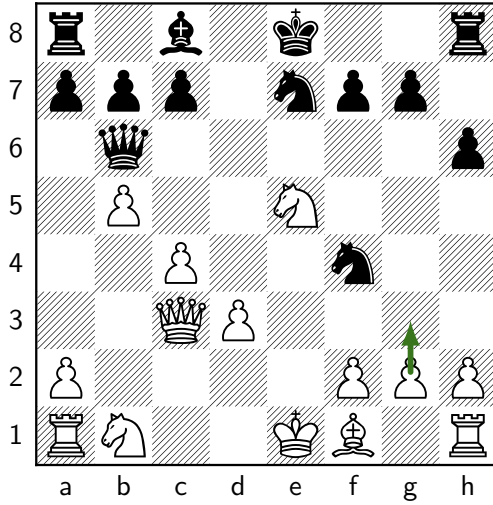
Figure 11: **Move 17, Black: a6.** (a) Our position; Melanie plays a6 (a7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2878` (ply 23) [structural distance $d=78$, QAP obj=59]: the master played Nge7; under μ our Pa2 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Qe7 as their Ra8, our Nb8 as their Rh8; transferred move a7a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1338` (ply 24) [structural distance $d=73$, QAP obj=52]: the master played Qc2; under μ our Re1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our qd2 as their Ra1, our Pa7 as their Nb1; transferred move e8d8. (d) Retrieved from `ccr1_top_p0000#3662` (ply 19) [structural distance $d=81$, QAP obj=51]: the master played Rd8; under μ our Qe2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Re8 as their Bf8, our Ph7 as their Rh8; transferred move e7f8. Synthesis: 1 retrieved analog transferred to a6 (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



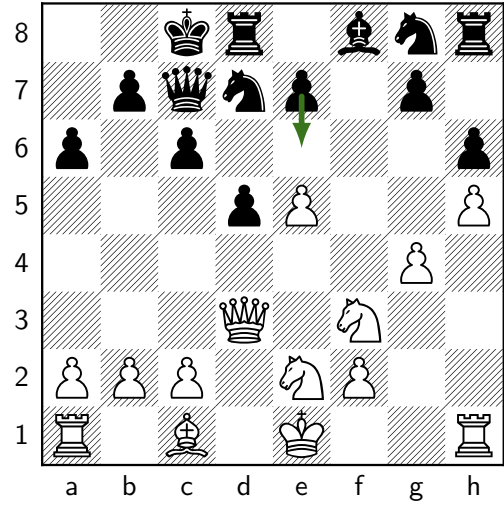
(a)



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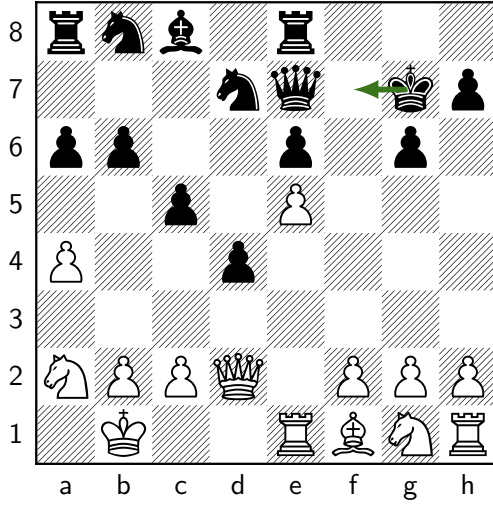


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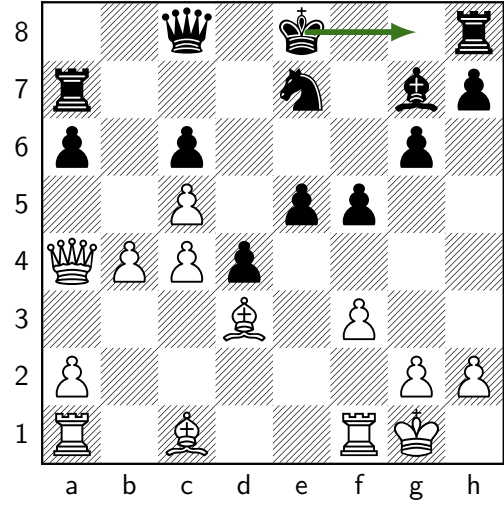


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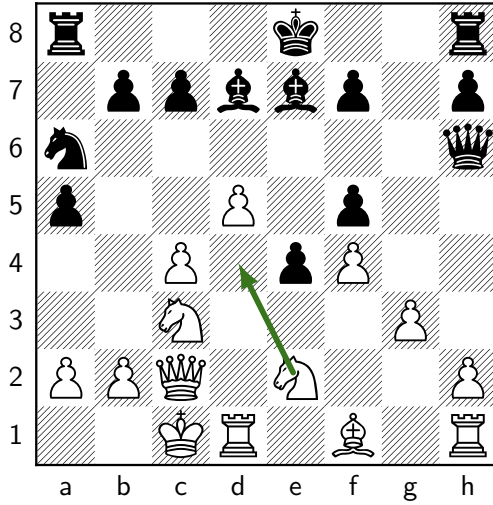
Figure 12: **Move 18, Black: d4**. (a) Our position; Melanie plays d4 (d5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3942` (ply 26) [structural distance $d=57$, QAP obj=67]: the master played Nc4; under μ our $Pd4$ plays the role of that game’s $Nd2$ (a **cross-type** correspondence); cross-type roles: our Nb8 as their Ra1, our Qe7 as their Re1; transferred move d5d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3155` (ply 26) [structural distance $d=84$, QAP obj=41]: the master played g3; under μ our $Pd4$ plays the role of that game’s $Pg2$; cross-type roles: our Pc5 as their Ra1, our Re8 as their Nb1; transferred move d5d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#929` (ply 25) [structural distance $d=64$, QAP obj=86]: the master played e6; under μ our $Bc1$ plays the role of that game’s $Pe2$ (a **cross-type** correspondence); cross-type roles: our Qe7 as their Rd8, our Re8 as their Bf8; transferred move c8b7. Synthesis: 2 retrieved analogs transferred to d4 (summed QAP objective 107, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



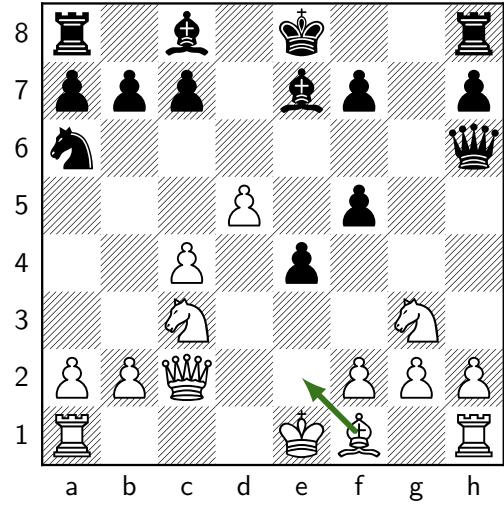
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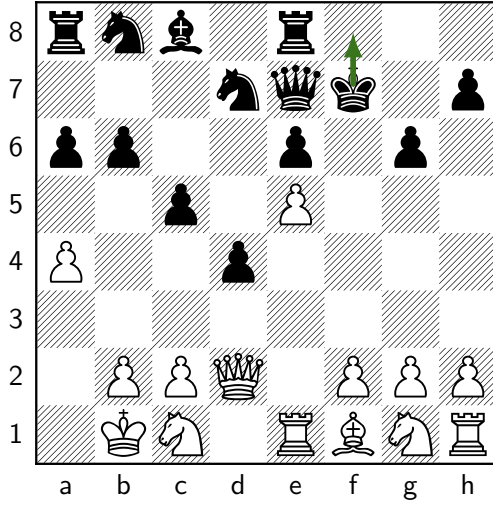


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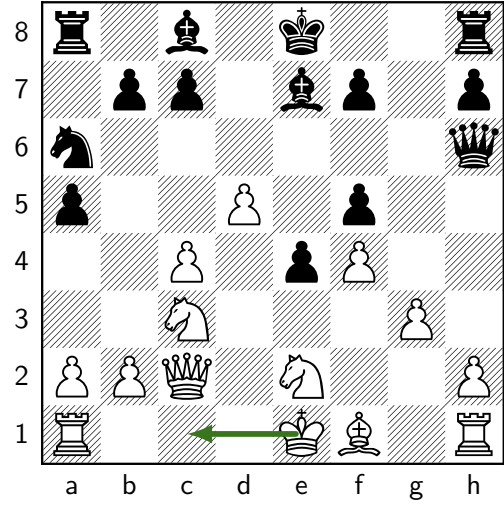


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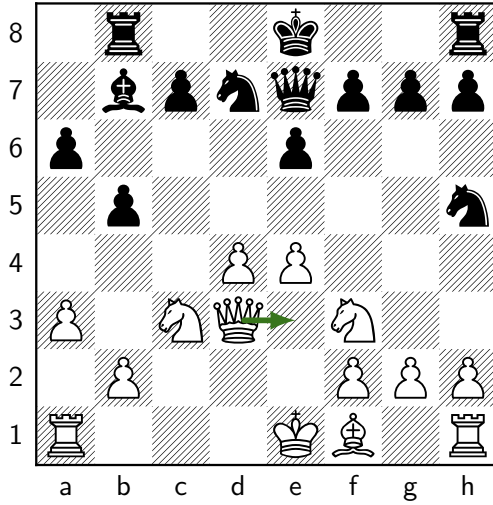
Figure 13: **Move 19, Black: Kf7**. (a) Our position; Melanie plays Kf7 (g7f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2244` (ply 35) [structural distance $d=90$, QAP obj=101]: the master played 0-0; under μ our *Kg2* plays the role of that game’s *Ke1*; cross-type roles: our *Ph7* as their *Rh8*, our *Pc5* as their *Ra7*; transferred move g7f7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#802` (ply 28) [structural distance $d=100$, QAP obj=90]: the master played Nd4; under μ our *Pg3* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *Qe7* as their *Rd1*, our *Re8* as their *Bf1*; transferred move g6g5. (d) Retrieved from `ccrl_top_p0000#3059` (ply 24) [structural distance $d=102$, QAP obj=83]: the master played Be2; under μ our *Pc4* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Qe7* as their *Ra1*, our *Bc8* as their *Pb2*; transferred move c5c4. Synthesis: 1 retrieved analog transferred to Kf7 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



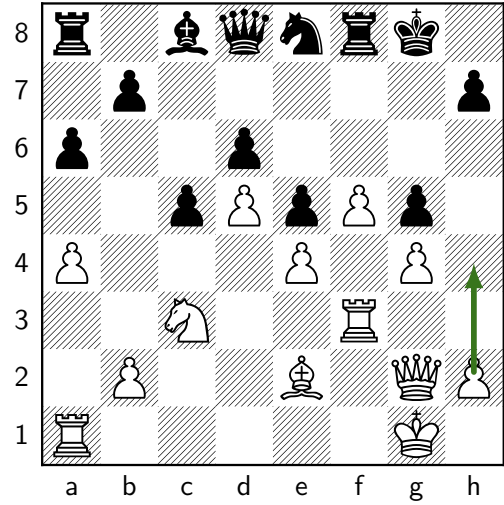
(a)



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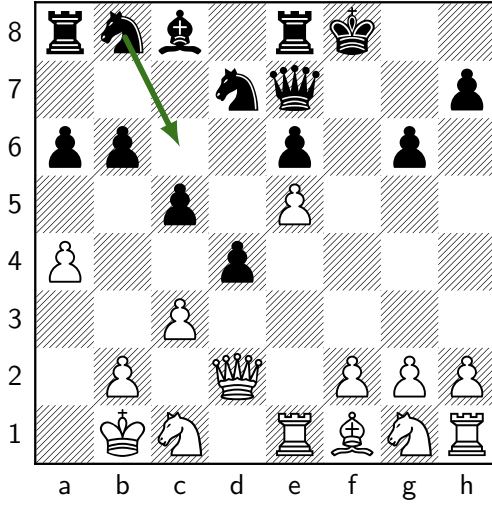


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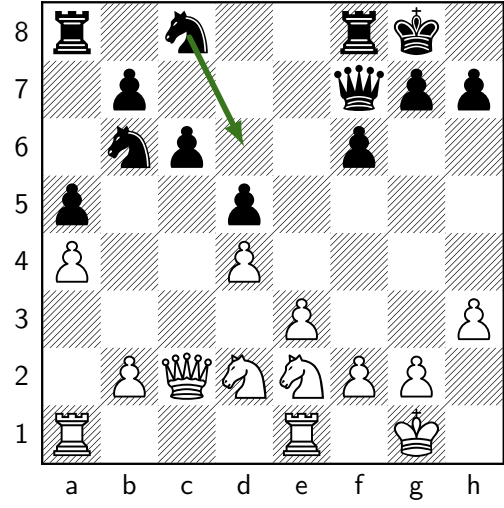


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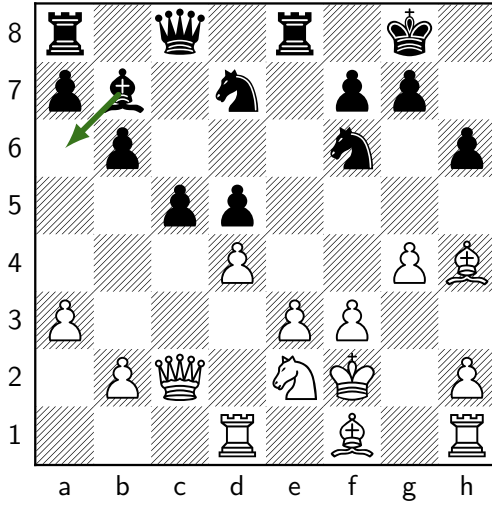
Figure 14: **Move 20, Black: Kf8.** (a) Our position; Melanie plays Kf8 (f7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#802` (ply 26) [structural distance $d=101$, QAP obj=70]: the master played 0-0-0; under μ our Kf2 plays the role of that game’s Ke1; cross-type roles: our Qe7 as their Ra1, our Ph7 as their Bf1; transferred move f7f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1343` (ply 24) [structural distance $d=111$, QAP obj=62]: the master played Qe3; under μ our Nd2 plays the role of that game’s Qd3 (a **cross-type** correspondence); cross-type roles: our Qe7 as their Ra1, our Nb8 as their Bf1; transferred move d7e5. (d) Retrieved from `ccr1_top_p0003#1309` (ply 44) [structural distance $d=113$, QAP obj=49]: the master played h4; under μ our Re1 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our Qe7 as their Ra1, our Nb8 as their Pb2; transferred move e8h8. Synthesis: 1 retrieved analog transferred to Kf8 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



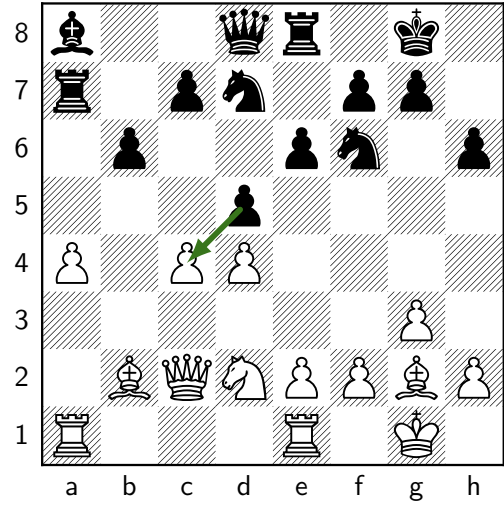
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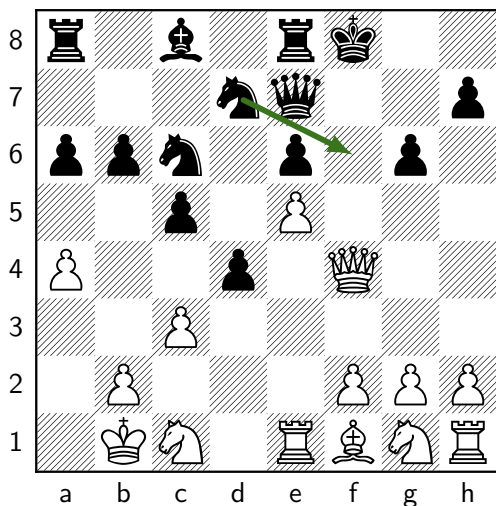


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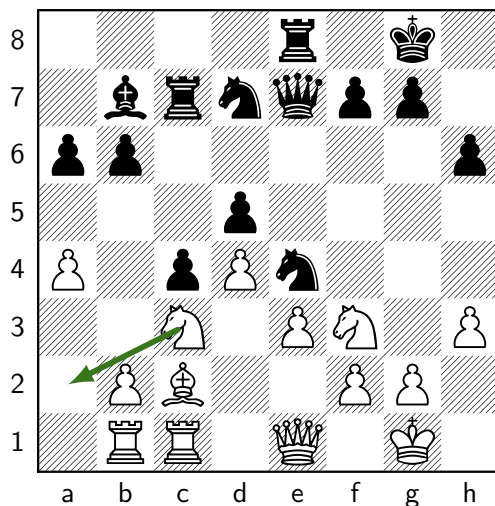


(d)

Figure 15: **Move 21, Black: Nc6.** (a) Our position; Melanie plays Nc6 (b8c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3061` (ply 39) [structural distance $d=104$, QAP obj=31]: the master played Nd6; under μ our Nb1 plays the role of that game’s Nc1; cross-type roles: our Pa6 as their Ra8, our pg2 as their Qf7; transferred move b8c6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1841` (ply 31) [structural distance $d=103$, QAP obj=88]: the master played Ba6; under μ our Pb3 plays the role of that game’s Bb2 (a **cross-type** correspondence); cross-type roles: our pe5 as their Qc8, our Qe7 as their Re8; transferred move b6b5. (d) Retrieved from `ccrl_top_p0002#684` (ply 35) [structural distance $d=91$, QAP obj=79]: the master played dxc4; under μ our Pd5 plays the role of that game’s Pd4; cross-type roles: our Pa6 as their Ba8, our Nd7 as their Re8; transferred move d4d3. Synthesis: 1 retrieved analog transferred to Nc6 (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

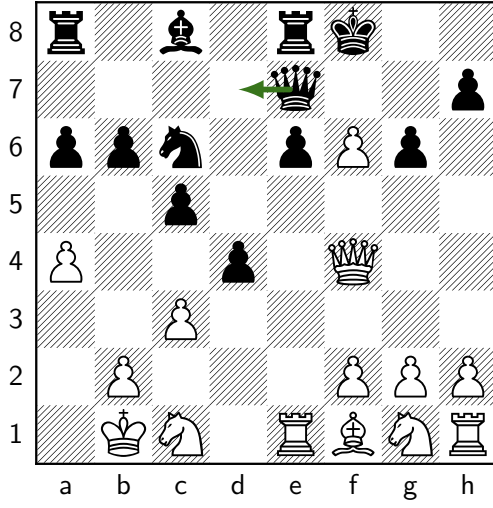


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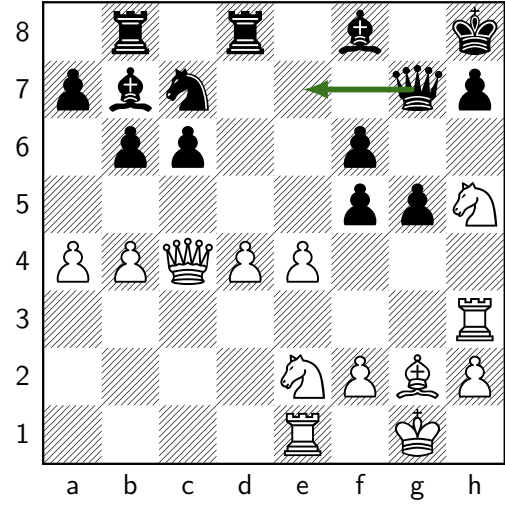


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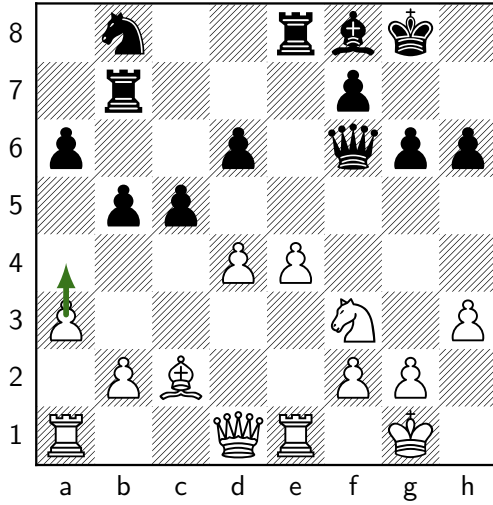
Figure 16: **Move 22, Black: Nf6.** (a) Our position; Melanie plays Nf6 (d7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#781` (ply 38) [structural distance $d=108$, QAP obj=9]: the master played Na2; under μ our Nd2 plays the role of that game’s Nc3; cross-type roles: our Pe6 as their Rc1, our rh1 as their Qe1; transferred move d7f6 (**endorses this move**). Synthesis: 1 retrieved analog transferred to Nf6 (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



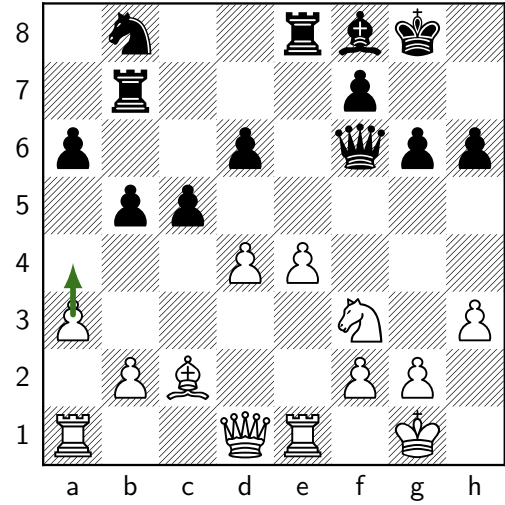
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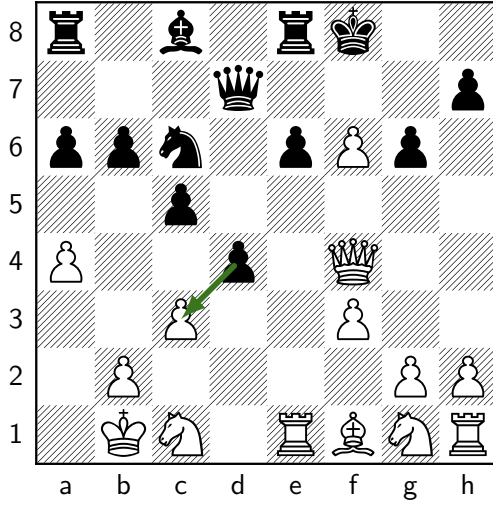


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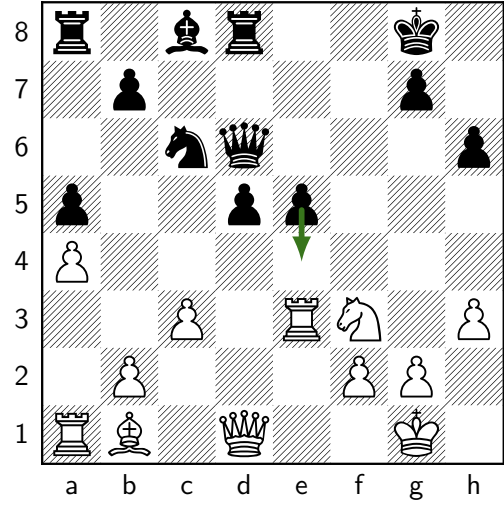


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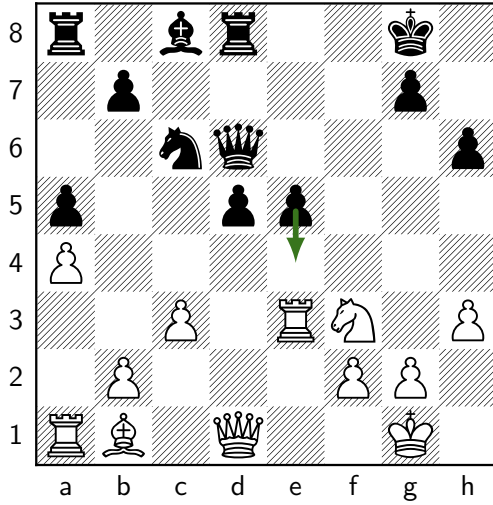
Figure 17: **Move 23, Black: Qd7.** (a) Our position; Melanie plays Qd7 (e7d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2058` (ply 55) [structural distance $d=101$, QAP obj=25]: the master played Qe7; under μ our Qe2 plays the role of that game's Qg2; cross-type roles: our ph2 as their Rb8, our pc3 as their Rd8; transferred move e7d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2091` (ply 38) [structural distance $d=102$, QAP obj=23]: the master played a4; under μ our Pb3 plays the role of that game's Pa3; cross-type roles: our Pa6 as their Qd1, our Qe7 as their Re1; transferred move b6b5. (d) Retrieved from `ccr1_top_p0003#2093` (ply 38) [structural distance $d=102$, QAP obj=23]: the master played a4; under μ our Pb3 plays the role of that game's Pa3; cross-type roles: our Pa6 as their Qd1, our Qe7 as their Re1; transferred move b6b5. Synthesis: 1 retrieved analog transferred to Qd7 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



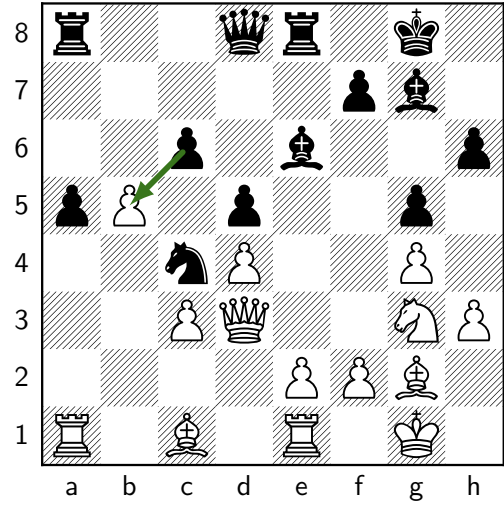
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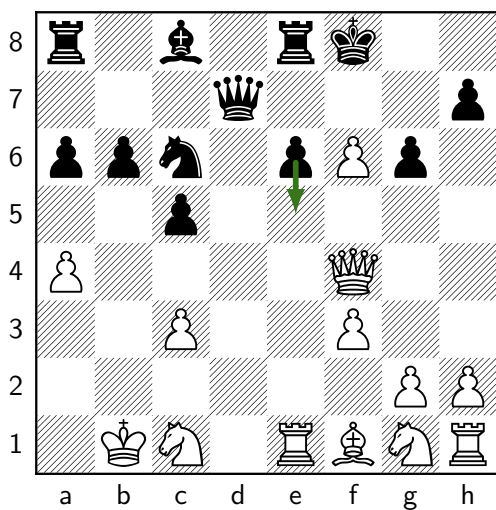


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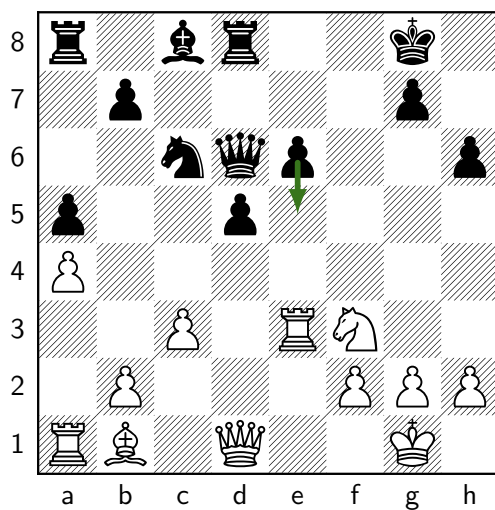


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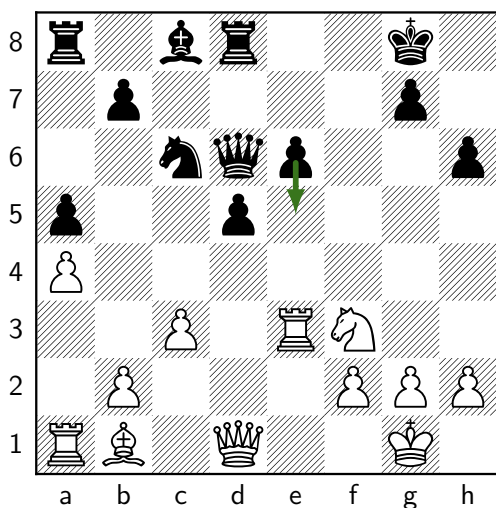
Figure 18: **Move 24, Black: dxc3.** (a) Our position; Melanie plays dxc3 (d4c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2516` (ply 41) [structural distance $d=84$, QAP obj=49]: the master played e4; under μ our $Pd5$ plays the role of that game’s $Pe4$; cross-type roles: our $Pb6$ as their $Ra8$, our $Pg6$ as their $Bc8$; transferred move d4c3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1430` (ply 41) [structural distance $d=84$, QAP obj=49]: the master played e4; under μ our $Pd5$ plays the role of that game’s $Pe4$; cross-type roles: our $Pb6$ as their $Ra8$, our $Pg6$ as their $Bc8$; transferred move d4c3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2670` (ply 47) [structural distance $d=91$, QAP obj=33]: the master played cxb5; under μ our $Pd5$ plays the role of that game’s $Pc3$; cross-type roles: our $pf6$ as their $Ra8$, our $Nc6$ as their $Qd8$; transferred move d4c3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to dxc3 (summed QAP objective 193, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



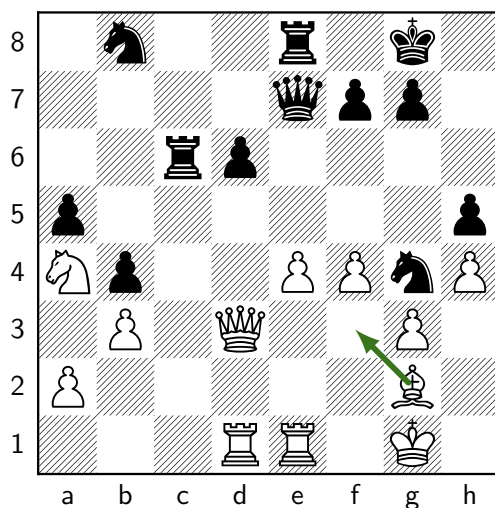
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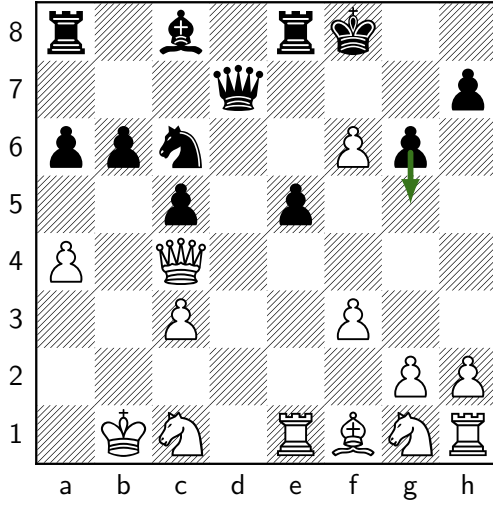


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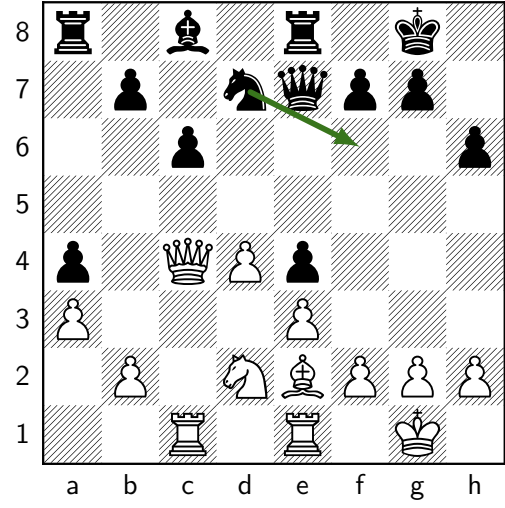


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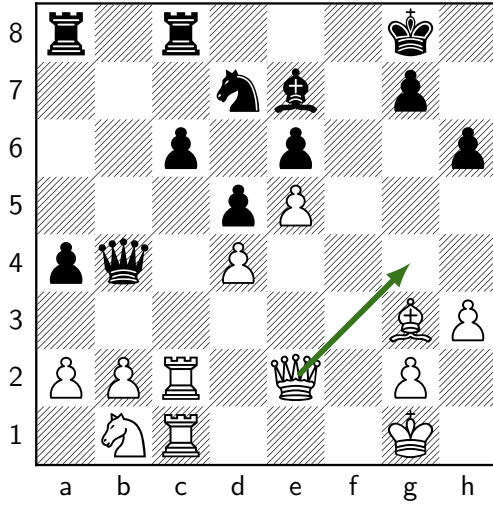
Figure 19: **Move 25, Black: e5**. (a) Our position; Melanie plays **e5** (**e6e5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2516` (ply 39) [structural distance $d=75$, QAP obj=43]: the master played **e5**; under μ our *Pe3* plays the role of that game’s *Pe3*; cross-type roles: our *pf3* as their *Ra8*, our *Ra8* as their *Pg7*; transferred move **e6e5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1430` (ply 39) [structural distance $d=75$, QAP obj=43]: the master played **e5**; under μ our *Pe3* plays the role of that game’s *Pe3*; cross-type roles: our *pf3* as their *Ra8*, our *Ra8* as their *Pg7*; transferred move **e6e5** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3050` (ply 48) [structural distance $d=96$, QAP obj=46]: the master played **Bf3**; under μ our *Bc1* plays the role of that game’s *Bg2*; cross-type roles: our *pg2* as their *Rd1*, our *bf1* as their *Qd3*; transferred move **c8b7**. Synthesis: 2 retrieved analogs transferred to **e5** (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



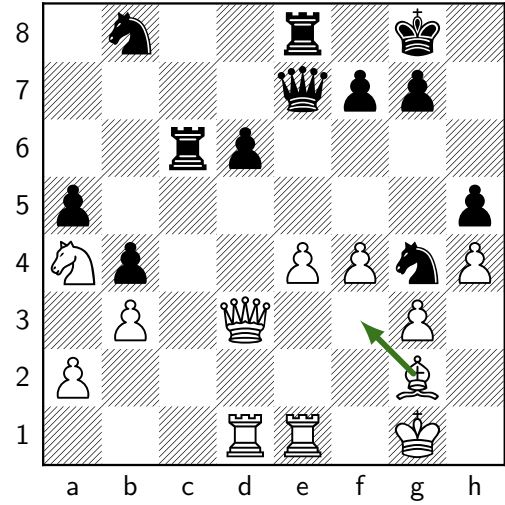
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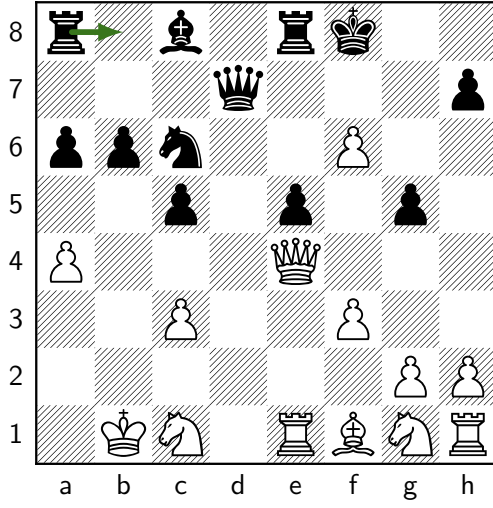


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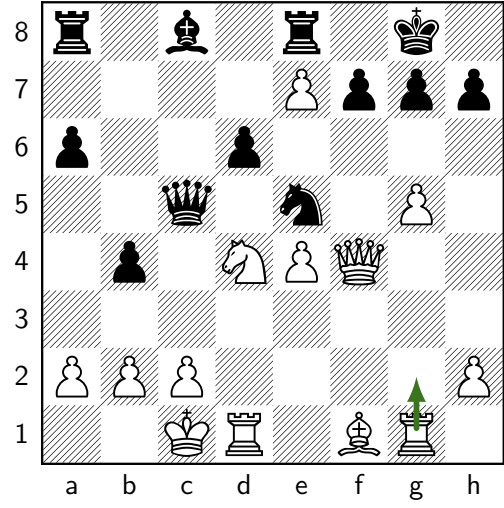


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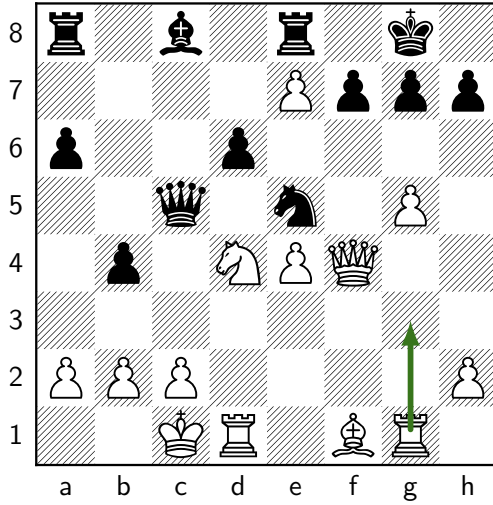
Figure 20: **Move 26, Black: g5.** (a) Our position; Melanie plays g5 (g6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3307` (ply 35) [structural distance $d=103$, QAP obj=78]: the master played Nf6; under μ our Pg3 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra8, our Qd7 as their Bc8; transferred move g6g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#706` (ply 60) [structural distance $d=98$, QAP obj=65]: the master played Qg4; under μ our Ra1 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Nb1, our nc1 as their Rc1; transferred move a8b8. (d) Retrieved from `ccr1_top_p0002#3050` (ply 48) [structural distance $d=89$, QAP obj=46]: the master played Bf3; under μ our Bc1 plays the role of that game’s Bg2; cross-type roles: our Pb6 as their Rd1, our pc3 as their Re1; transferred move c8b7. Synthesis: 1 retrieved analog transferred to g5 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



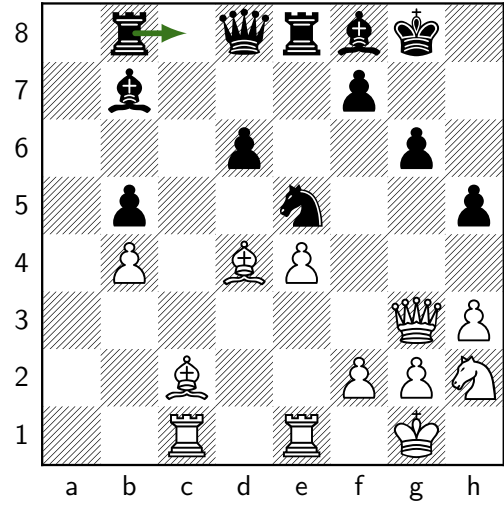
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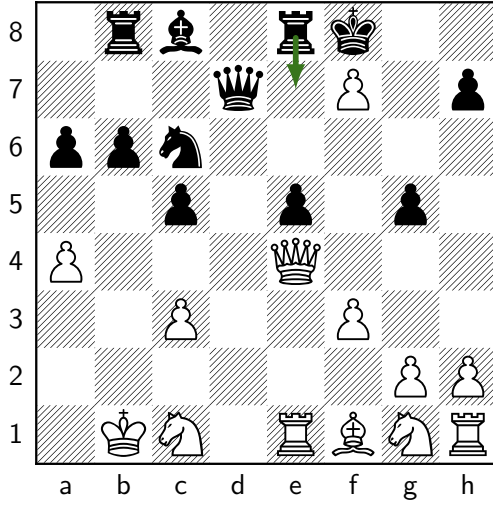


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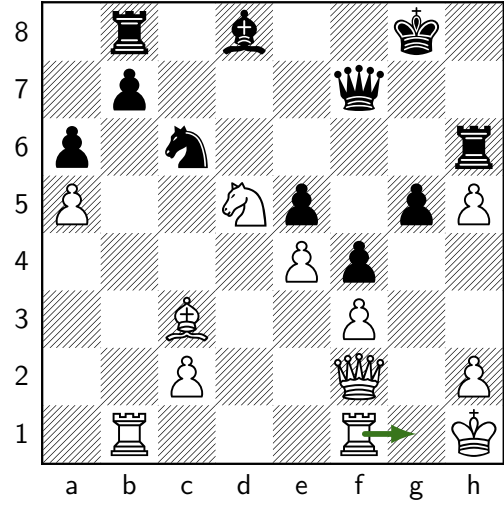


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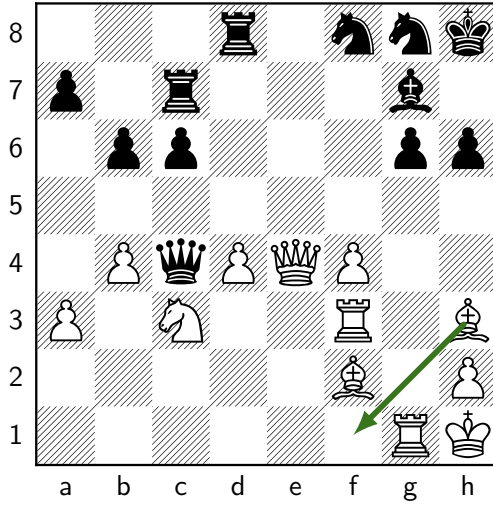
Figure 21: **Move 27, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2763` (ply 38) [structural distance $d=99$, QAP obj=49]: the master played Rg2; under μ our *Ra1* plays the role of that game’s *Rg1*; cross-type roles: our *Qd7* as their *Bf1*, our *Bc8* as their *Pa2*; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2010` (ply 38) [structural distance $d=99$, QAP obj=49]: the master played Rg3; under μ our *Ra1* plays the role of that game’s *Rg1*; cross-type roles: our *Qd7* as their *Bf1*, our *Bc8* as their *Pa2*; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#224` (ply 57) [structural distance $d=97$, QAP obj=36]: the master played Rc8; under μ our *Ra1* plays the role of that game’s *Rb1*; cross-type roles: our *Pa6* as their *Qd8*, our *Qd7* as their *Re8*; transferred move a8b8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rb8 (summed QAP objective 134, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



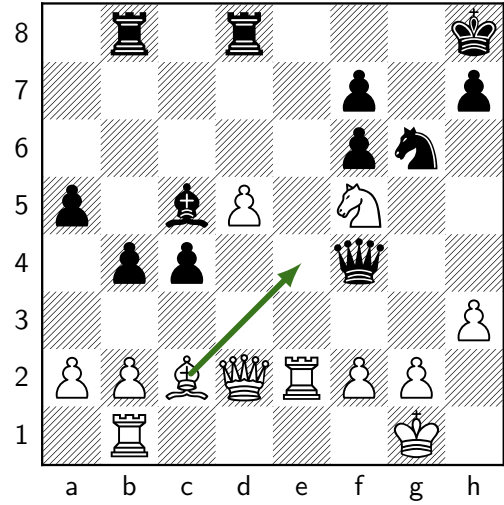
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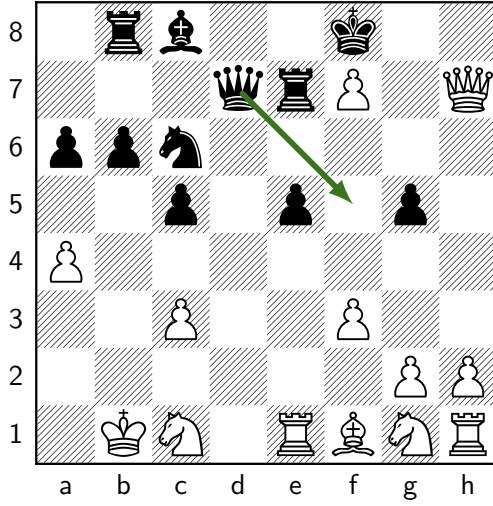


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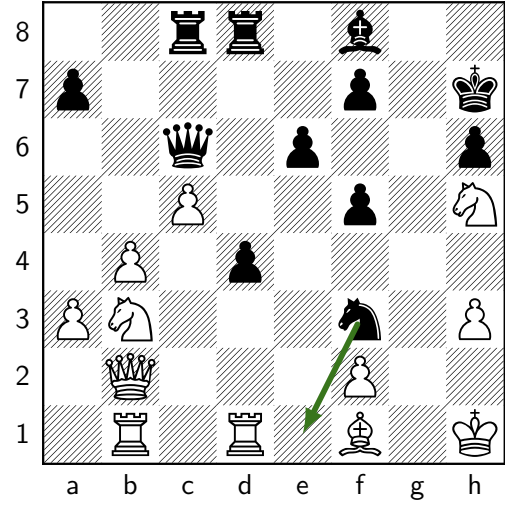


(d)

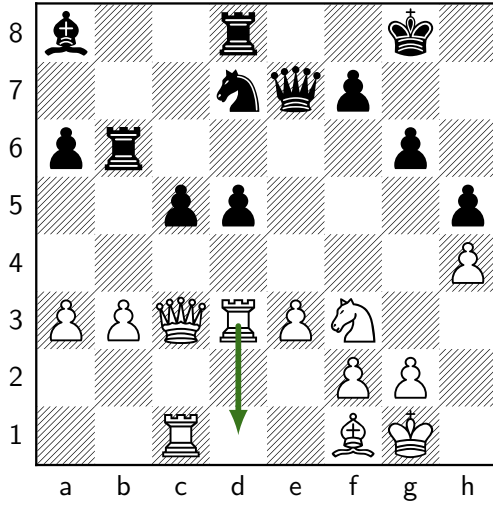
Figure 22: **Move 28, Black: Re7.** (a) Our position; Melanie plays Re7 (e8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#300` (ply 58) [structural distance $d=95$, QAP obj=12]: the master played Rg1; under μ our Re1 plays the role of that game’s Rf1; cross-type roles: our pf3 as their Rb1, our Rb8 as their Pc2; transferred move e8e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3279` (ply 62) [structural distance $d=87$, QAP obj=53]: the master played Bf1; under μ our Pa3 plays the role of that game’s Bh3 (a **cross-type** correspondence); cross-type roles: our re1 as their pc6, our ng1 as their pg6; transferred move a6a5. (d) Retrieved from `ccr1_top_p0000#3445` (ply 56) [structural distance $d=85$, QAP obj=36]: the master played Be4; under μ our Nc3 plays the role of that game’s Bc2 (a **cross-type** correspondence); cross-type roles: our Qd7 as their Pa2, our Re8 as their Qd2; transferred move c6b4. Synthesis: 1 retrieved analog transferred to Re7 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



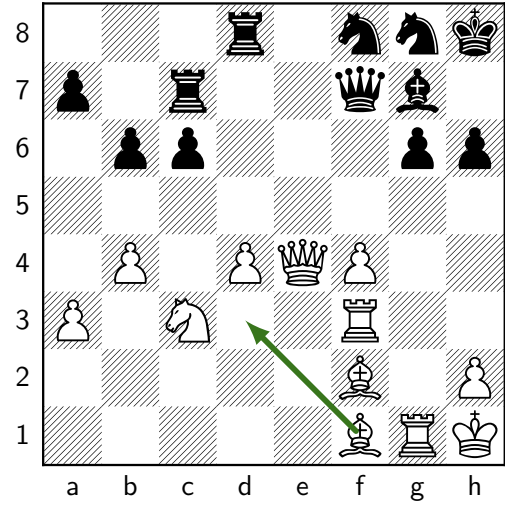
(a)



(b)

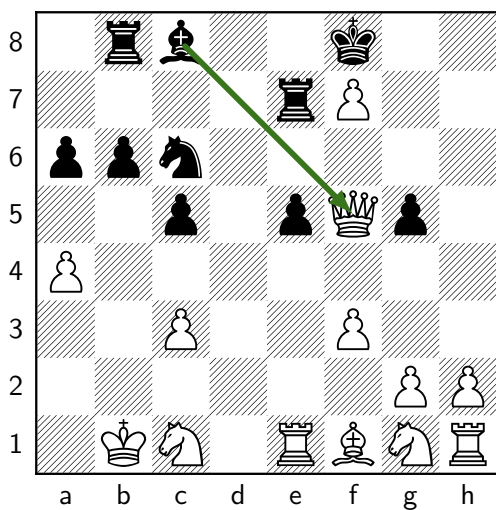


(c)

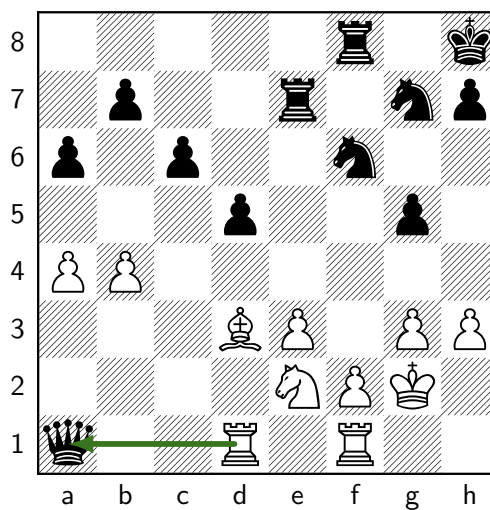


(d)

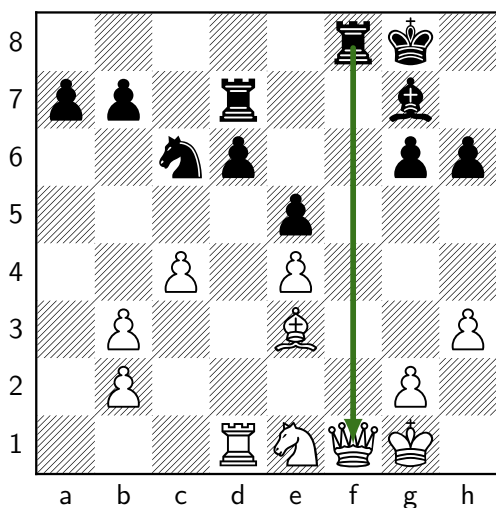
Figure 23: **Move 29, Black: Qf5+.** (a) Our position; Melanie plays Qf5+ (d7f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3196` (ply 69) [structural distance $d=91$, QAP obj=46]: the master played Ne1+; under μ our Nc3 plays the role of that game’s Nf6; cross-type roles: our Qd7 as their Rc8, our qh7 as their Rd8; transferred move c6b4. (c) Retrieved from `ccr1_top_p0001#2930` (ply 50) [structural distance $d=76$, QAP obj=33]: the master played Rdd1; under μ our Re2 plays the role of that game’s Rd3; cross-type roles: our Nc6 as their Rc1, our pg2 as their Bf1; transferred move e7e6. (d) Retrieved from `ccr1_top_p0000#3279` (ply 64) [structural distance $d=82$, QAP obj=29]: the master played Bd3; under μ our Pc4 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Pg5 as their Rg1, our pf3 as their Bf2; transferred move c5c4. Synthesis: no retrieved analog transferred to Qf5+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



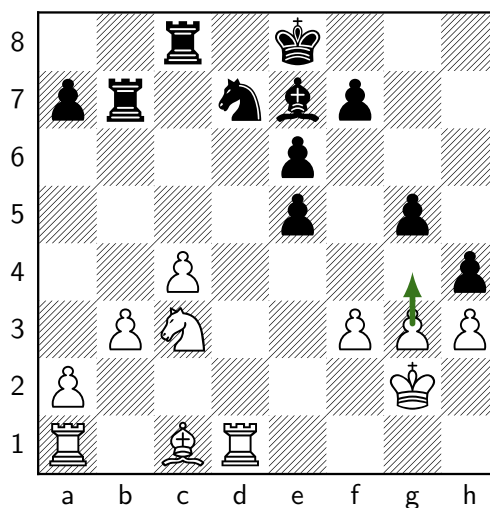
(a)



(b)

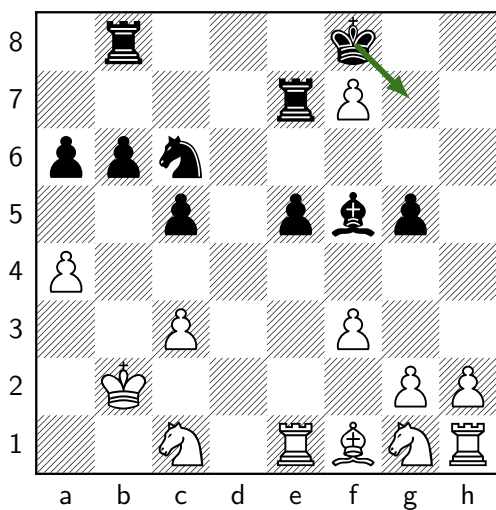


(c)

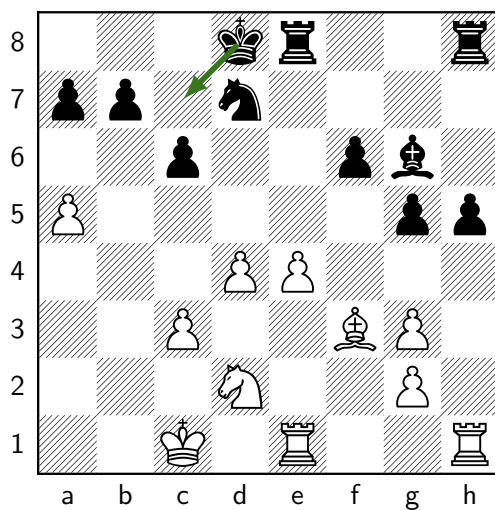


(d)

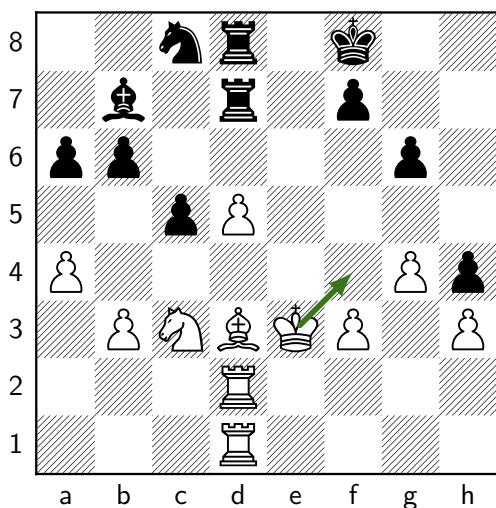
Figure 24: **Move 30, Black: Bxf5+.** (a) Our position; Melanie plays Bxf5+ (c8f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#3038 (ply 58) [structural distance $d=104$, QAP obj=50]: the master played Rxa1; under μ our Bc1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our rh1 as their Ne2, our Nc6 as their Pf2; transferred move c8f5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#2289 (ply 43) [structural distance $d=104$, QAP obj=37]: the master played Rxf1+; under μ our Bc1 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pb6 as their Bg7, our Rb8 as their Ph6; transferred move c8f5 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2062 (ply 42) [structural distance $d=105$, QAP obj=36]: the master played g4; under μ our Bc1 plays the role of that game’s Pg3 (a **cross-type** correspondence); cross-type roles: our Pa6 as their Bc1, our Nc6 as their Rd1; transferred move c8f5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bxf5+ (summed QAP objective 177, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



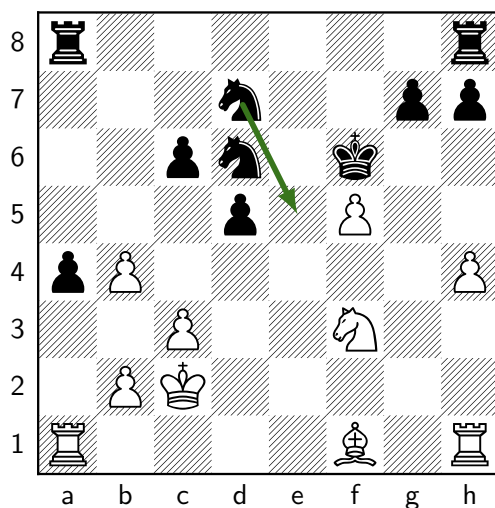
(a)



(b)

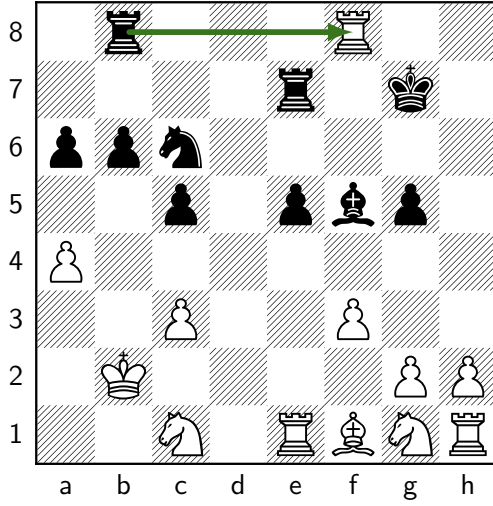


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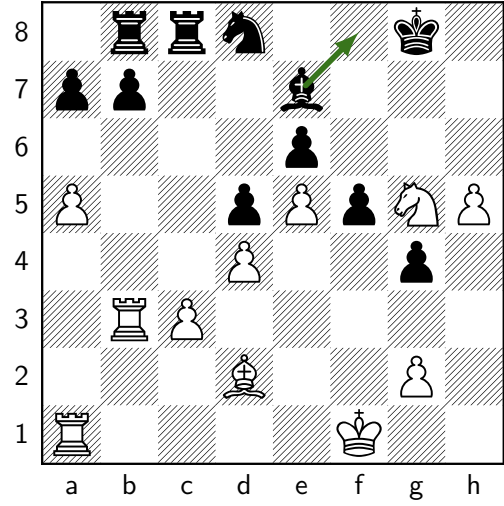


(d)

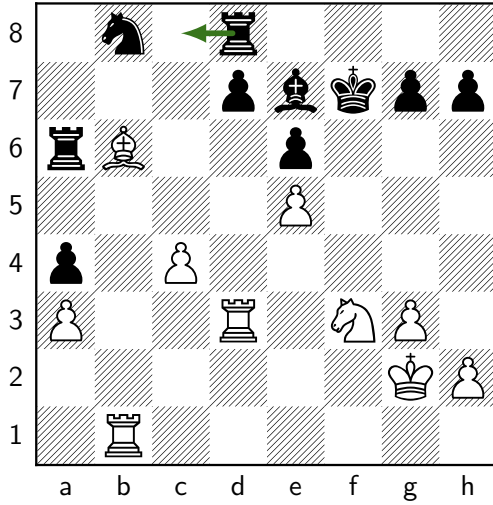
Figure 25: **Move 31, Black: Kg7 (f8g7)**. (a) Our position; Melanie plays Kg7 (f8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2657` (ply 47) [structural distance $d=72$, QAP obj=50]: the master played Kc7; under μ our Kf1 plays the role of that game’s Kd1; cross-type roles: our bf1 as their Rh8, our Bf5 as their Pa7; transferred move f8g7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#382` (ply 60) [structural distance $d=69$, QAP obj=26]: the master played Kf4; under μ our Kf1 plays the role of that game’s Ke3; cross-type roles: our Nc6 as their Rd1, our ng1 as their Pb3; transferred move f8g7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3294` (ply 53) [structural distance $d=74$, QAP obj=57]: the master played Ne5; under μ our Rb1 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Nc6 as their Ra8, our pa4 as their Rh8; transferred move b8e8. Synthesis: 2 retrieved analogs transferred to Kg7 (summed QAP objective 76, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



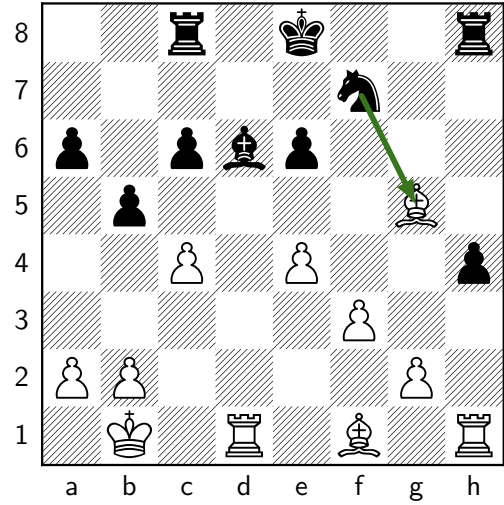
(a)



(b)

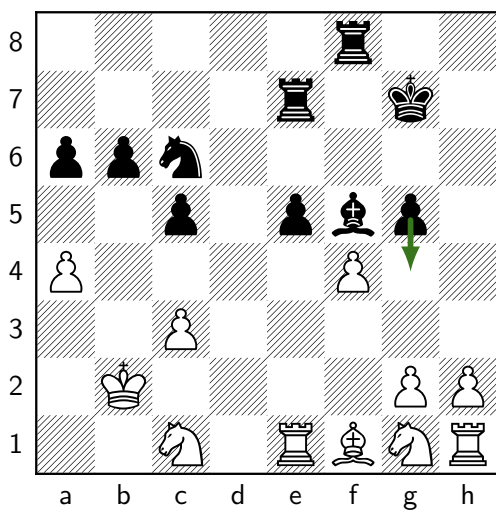


(c)

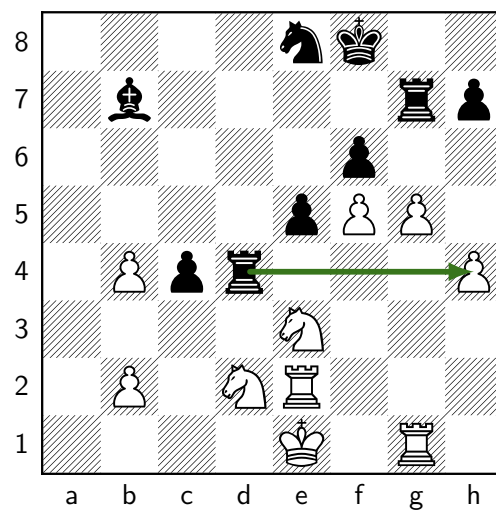


(d)

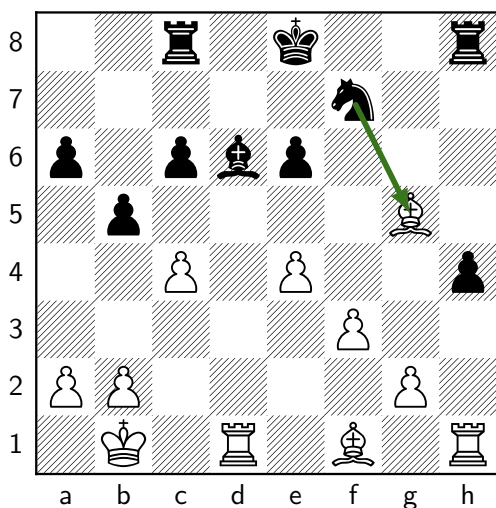
Figure 26: **Move 32, Black: Rxf8.** (a) Our position; Melanie plays Rxf8 (b8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1975` (ply 63) [structural distance $d=89$, QAP obj=32]: the master played Bf8; under μ our Rb1 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our bf1 as their Rb8, our Bf5 as their Pa7; transferred move b8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1871` (ply 55) [structural distance $d=82$, QAP obj=41]: the master played Rc8; under μ our Rb1 plays the role of that game’s Rd1; cross-type roles: our Re7 as their Nb8, our ng1 as their Pg7; transferred move b8a8. (d) Retrieved from `ccrl_top_p0003#1485` (ply 49) [structural distance $d=79$, QAP obj=30]: the master played Nxg5; under μ our Rb1 plays the role of that game’s Nf2 (a **cross-type** correspondence); cross-type roles: our Nc6 as their Rh8, our rf8 as their bg5; transferred move b8b7. Synthesis: 1 retrieved analog transferred to Rxf8 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



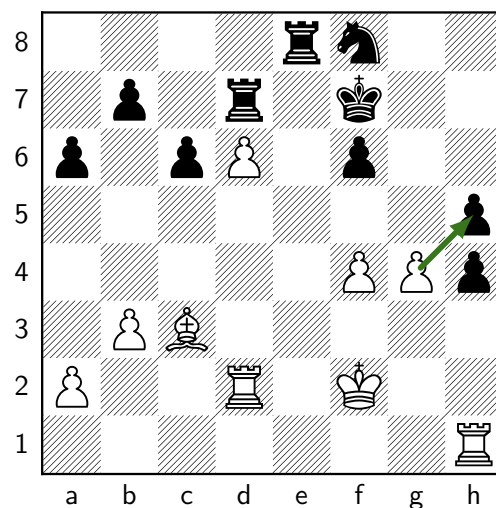
(a)



(b)

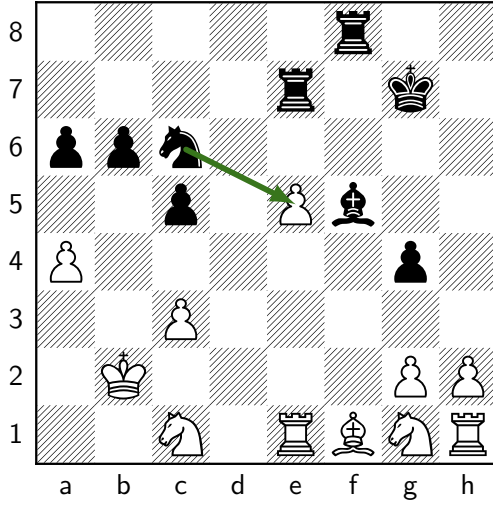


(c)

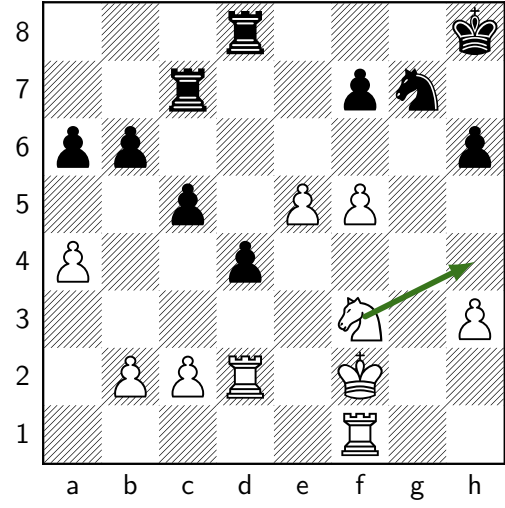


(d)

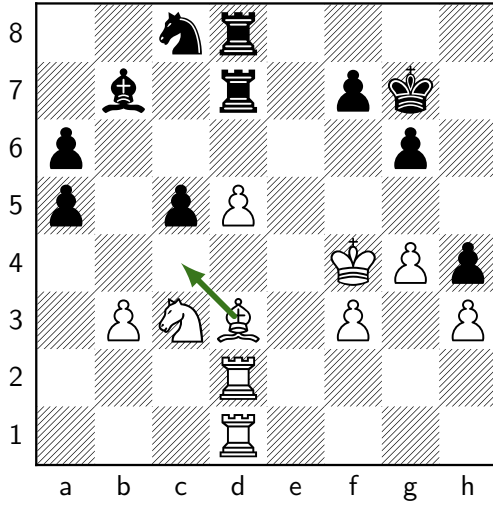
Figure 27: **Move 33, Black: g4 (g5g4).** (a) Our position; Melanie plays g4 (g5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3362` (ply 75) [structural distance $d=81$, QAP obj=79]: the master played $Rxh4$; under μ our $Pg4$ plays the role of that game’s $Rd5$ (a **cross-type** correspondence); cross-type roles: our $Pc5$ as their $Ne8$, our $nc1$ as their $Bb7$; transferred move $g5g4$ (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1485` (ply 49) [structural distance $d=75$, QAP obj=74]: the master played $Nxg5$; under μ our $Pg4$ plays the role of that game’s $Nf2$ (a **cross-type** correspondence); cross-type roles: our $Pb6$ as their $Rc8$, our $Nc6$ as their $Pa6$; transferred move $g5g4$ (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1625` (ply 68) [structural distance $d=78$, QAP obj=55]: the master played $gxh5$; under μ our $Pg4$ plays the role of that game’s $Pg4$; cross-type roles: our $Bf5$ as their $Pb3$, our $Nc6$ as their $Bc3$; transferred move $g5g4$ (**endorses this move**). Synthesis: 4 retrieved analogs transferred to g4 (summed QAP objective 242, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



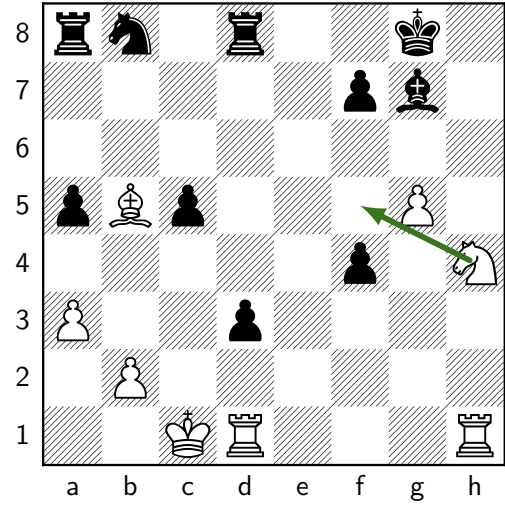
(a)



(b)

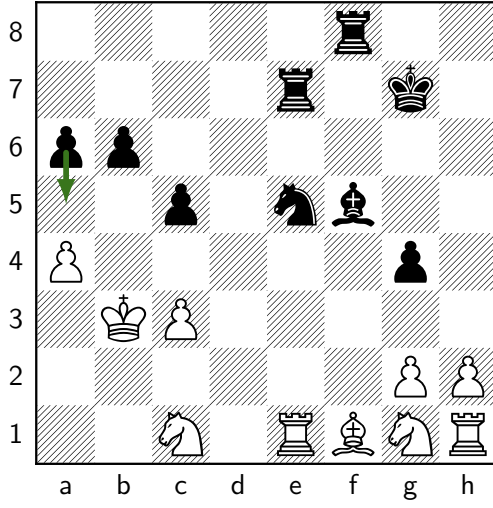


(c)

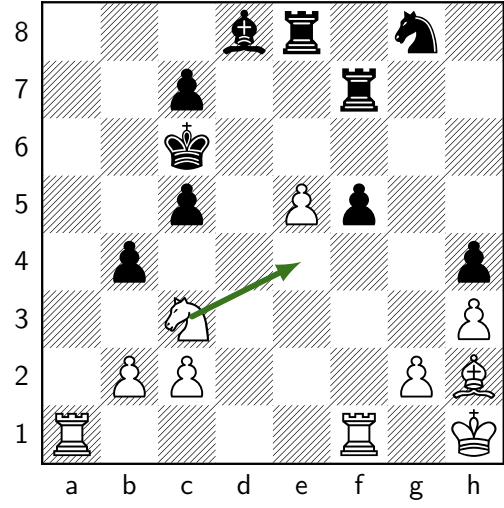


(d)

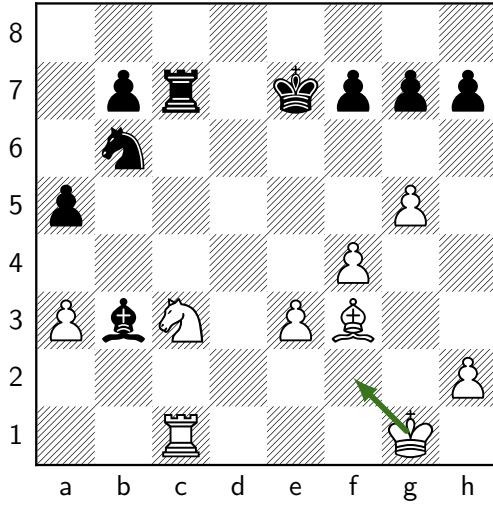
Figure 28: **Move 34, Black: Nxe5.** (a) Our position; Melanie plays Nxe5 (c6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#538` (ply 70) [structural distance $d=58$, QAP obj=96]: the master played Nh4; under μ our Nc3 plays the role of that game’s Nf3; cross-type roles: our Pc5 as their Rf1, our Rf8 as their Pb2; transferred move c6a7. (c) Retrieved from `ccr1_top_p0000#382` (ply 64) [structural distance $d=67$, QAP obj=87]: the master played Bc4; under μ our Re2 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our Bf5 as their Rd1, our pc3 as their Rd2; transferred move e7e6. (d) Retrieved from `ccr1_top_p0001#486` (ply 62) [structural distance $d=52$, QAP obj=79]: the master played Nf5; under μ our Bf4 plays the role of that game’s Nh4 (a **cross-type** correspondence); cross-type roles: our nc1 as their pf4, our Nc6 as their Bb5; transferred move f5e6. Synthesis: no retrieved analog transferred to Nxe5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



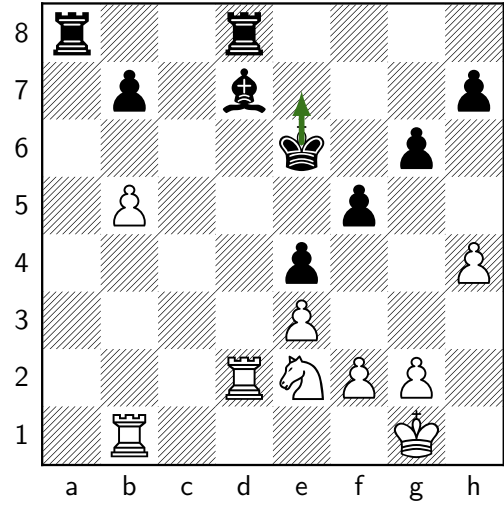
(a)



(b)

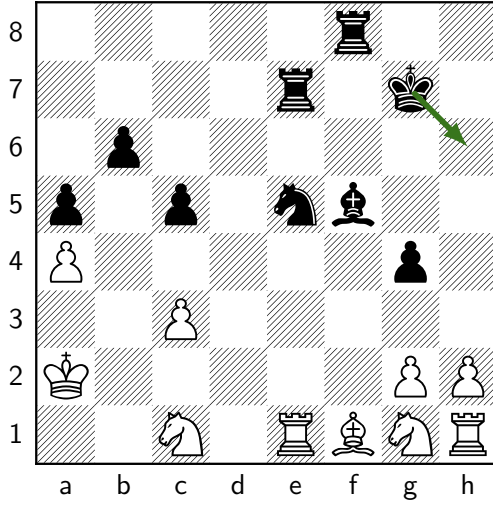


(c)

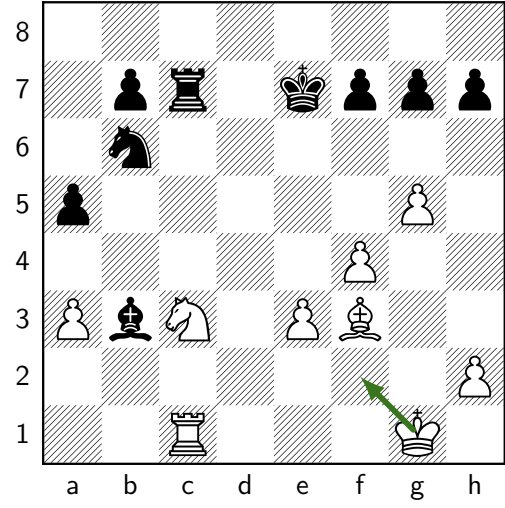


(d)

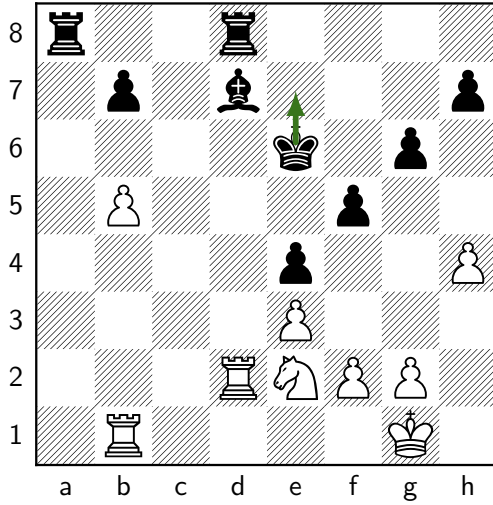
Figure 29: **Move 35, Black: a5.** (a) Our position; Melanie plays a5 (a6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#87` (ply 80) [structural distance $d=55$, QAP obj=14]: the master played Ne4; under μ our Pa3 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our ph2 as their Ra1, our rh1 as their Pg2; transferred move a6a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3838` (ply 60) [structural distance $d=53$, QAP obj=52]: the master played Kf2; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Rf8 as their Ph2, our pg2 as their bb3; transferred move g7h6. (d) Retrieved from `ccr1_top_p0003#2210` (ply 71) [structural distance $d=56$, QAP obj=40]: the master played Ke7; under μ our Kg2 plays the role of that game’s Ke3; cross-type roles: our Bf5 as their Ra8, our Ne5 as their Bd7; transferred move g7h6. Synthesis: 1 retrieved analog transferred to a5 (summed QAP objective 14, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



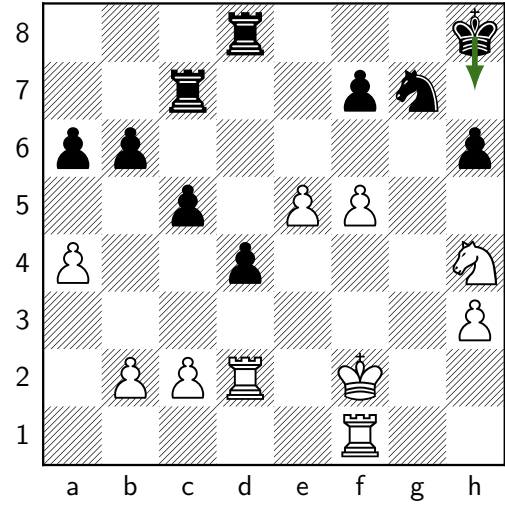
(a)



(b)

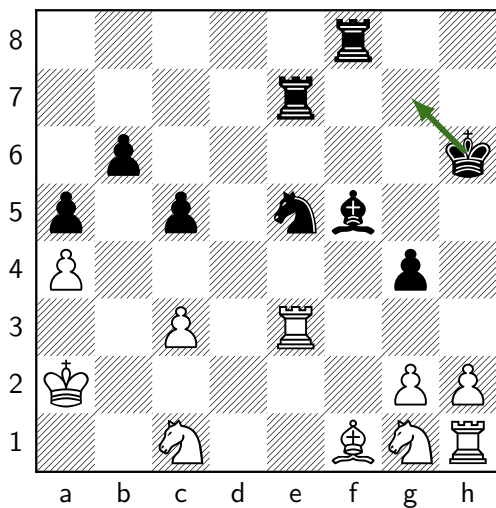


(c)

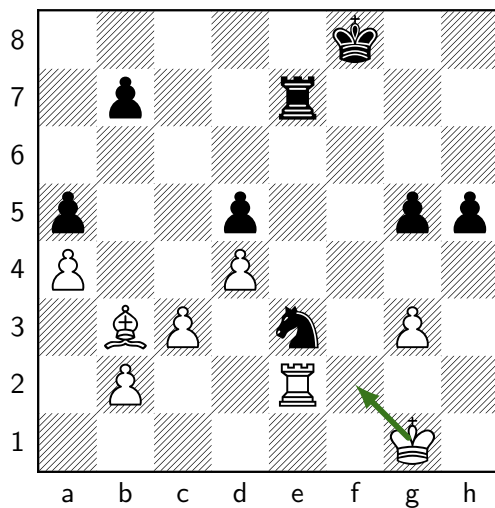


(d)

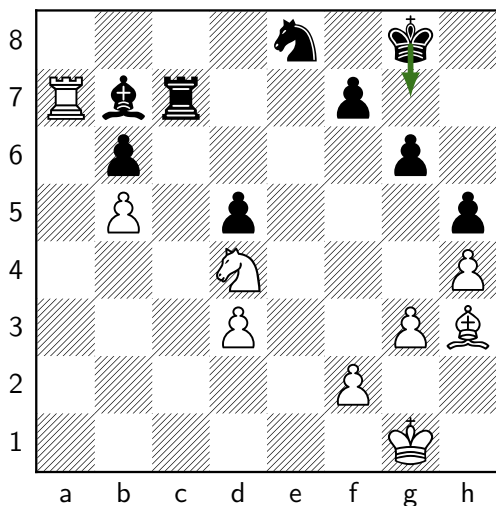
Figure 30: **Move 36, Black: Kh6**. (a) Our position; Melanie plays Kh6 (g7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3838` (ply 60) [structural distance $d=52$, QAP obj=51]: the master played Kf2; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Rf8 as their Ph2, our ng1 as their bb3; transferred move g7h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2210` (ply 71) [structural distance $d=60$, QAP obj=40]: the master played Ke7; under μ our Kg2 plays the role of that game’s Ke3; cross-type roles: our Bf5 as their Ra8, our Ne5 as their Bd7; transferred move g7h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#538` (ply 71) [structural distance $d=56$, QAP obj=34]: the master played Kh7; under μ our Kg2 plays the role of that game’s Kh1; cross-type roles: our Pb6 as their Rc7, our re1 as their Ng7; transferred move g7h6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kh6 (summed QAP objective 150, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



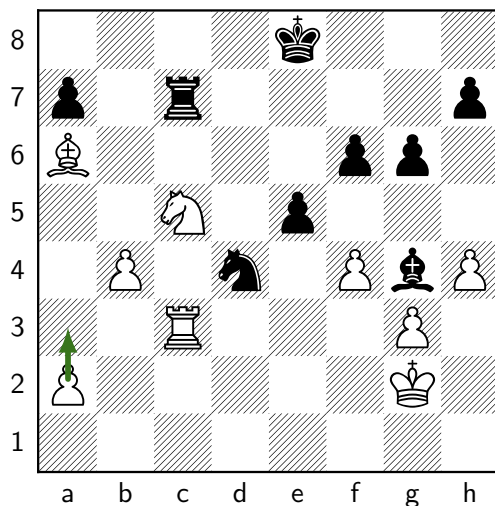
(a)



(b)

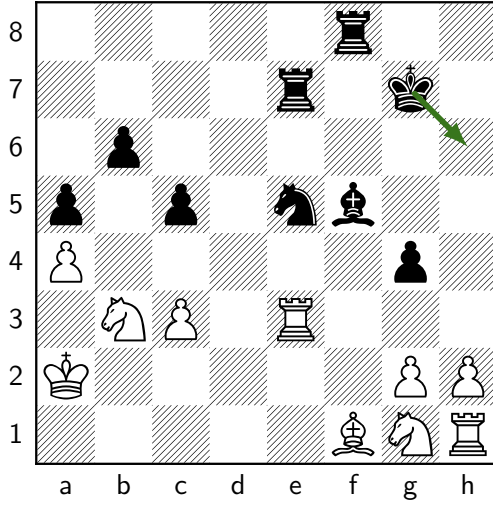


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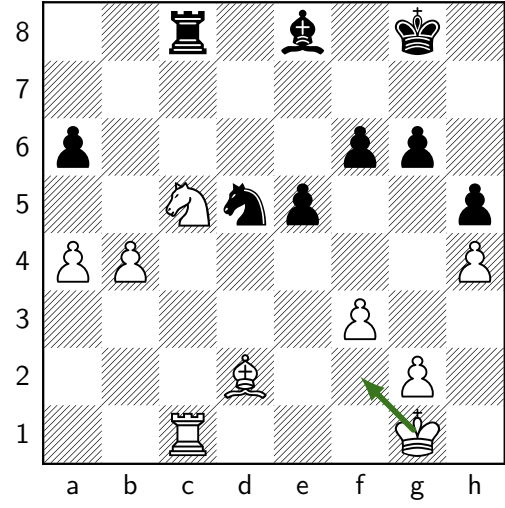


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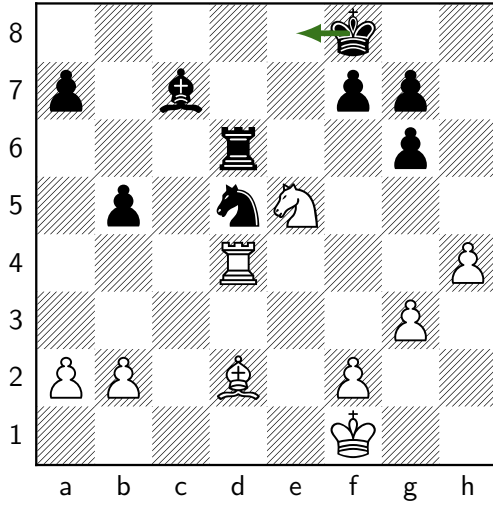
Figure 31: **Move 37, Black: Kg7 (h6g7)**. (a) Our position; Melanie plays Kg7 (h6g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0000#3628` (ply 60) [structural distance $d=61$, QAP obj=30]: the master played Kf2; under μ our Kh3 plays the role of that game’s Kg1; cross-type roles: our nc1 as their Bb3, our ng1 as their Pc3; transferred move h6g7 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#939` (ply 65) [structural distance $d=61$, QAP obj=26]: the master played Kg7; under μ our Kh3 plays the role of that game’s Kg1; cross-type roles: our Pc5 as their Ne8, our Ne5 as their Bb7; transferred move h6g7 (**endorses this move**). (d) Retrieved from `ccl1_top_p0003#33` (ply 56) [structural distance $d=48$, QAP obj=47]: the master played a3; under μ our Rf1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Bf5 as their Ph4, our Pg4 as their Ba6; transferred move f8a8. Synthesis: 2 retrieved analogs transferred to Kg7 (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



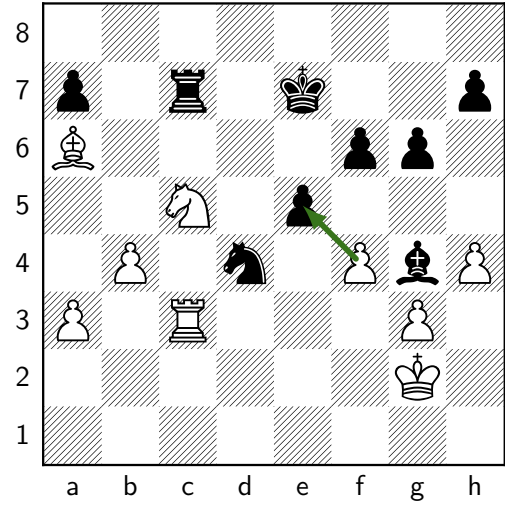
(a)



(b)

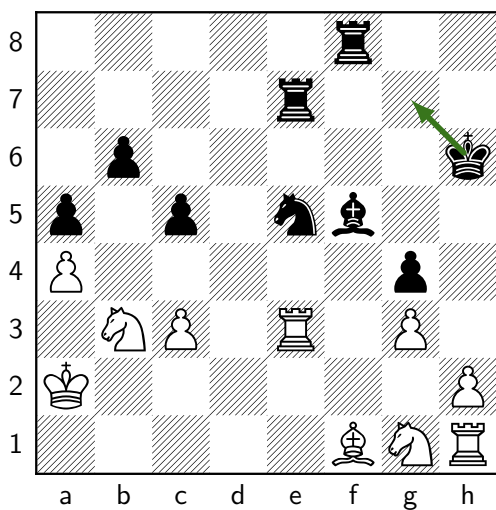


(c)

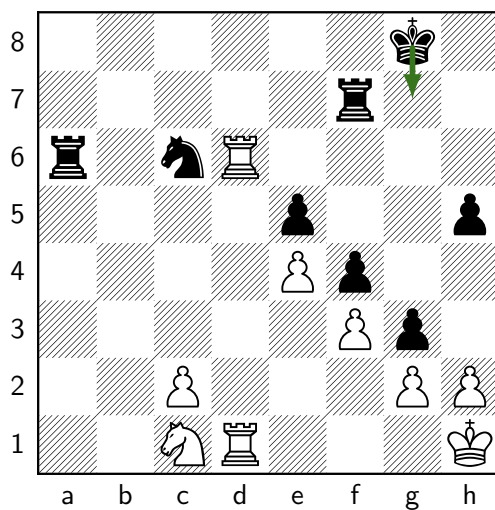


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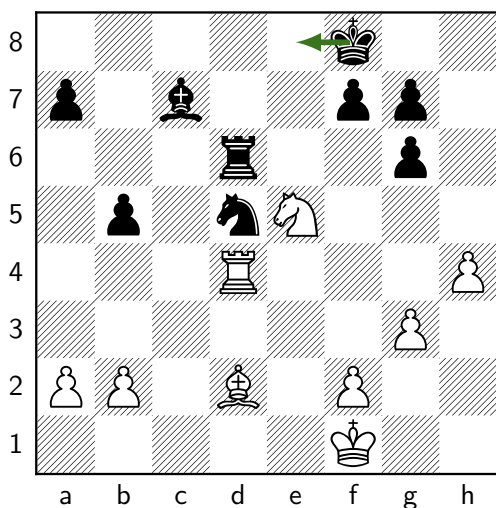
Figure 32: **Move 38, Black: Kh6**. (a) Our position; Melanie plays Kh6 (g7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#540` (ply 58) [structural distance $d=62$, QAP obj=39]: the master played Kf2; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Rf8 as their Bd2, our ng1 as their Ph4; transferred move g7h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3831` (ply 67) [structural distance $d=64$, QAP obj=26]: the master played Ke8; under μ our Kg2 plays the role of that game’s Kf1; cross-type roles: our Re7 as their Pa7, our Pg4 as their Bc7; transferred move g7h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#33` (ply 58) [structural distance $d=62$, QAP obj=52]: the master played fxe5; under μ our Pc4 plays the role of that game’s Pf4; cross-type roles: our bf1 as their Pb4, our pc3 as their bg4; transferred move c5c4. Synthesis: 2 retrieved analogs transferred to Kh6 (summed QAP objective 65, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



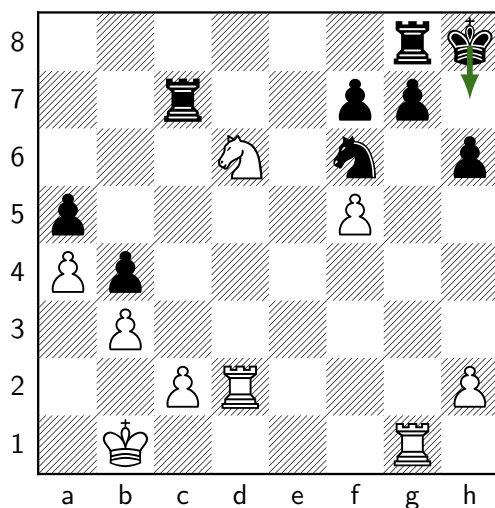
(a)



(b)

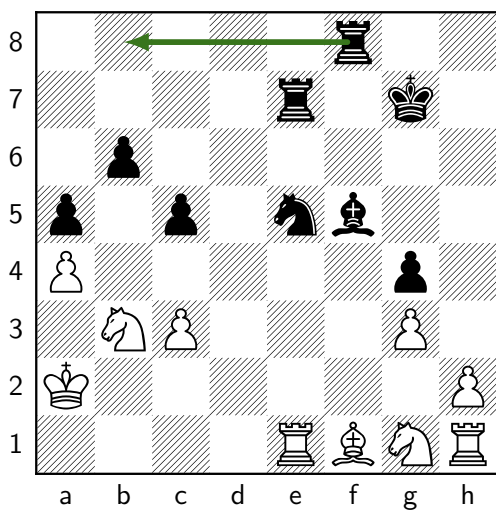


(c)

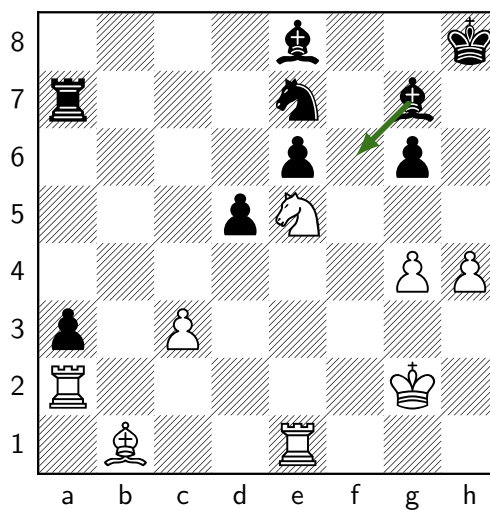


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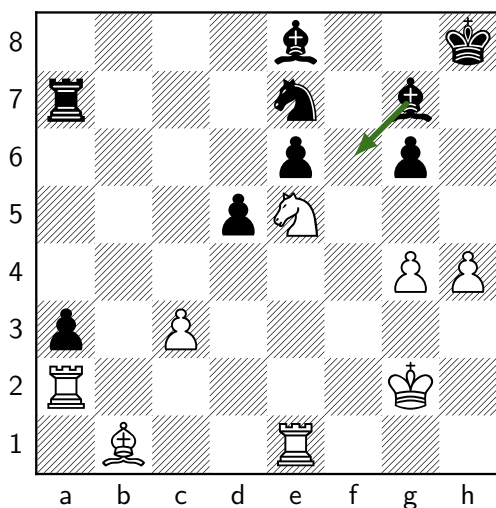
Figure 33: **Move 39, Black: Kg7.** (a) Our position; Melanie plays Kg7 (h6g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0003#2642` (ply 75) [structural distance $d=63$, QAP obj=31]: the master played Kg7; under μ our Kh3 plays the role of that game’s Kg1; cross-type roles: our Bf5 as their Rf7, our nb3 as their Ph5; transferred move h6g7 (**endorses this move**). (c) Retrieved from `cctl_top_p0002#3831` (ply 67) [structural distance $d=64$, QAP obj=27]: the master played Ke8; under μ our Kh3 plays the role of that game’s Kf1; cross-type roles: our Rf8 as their Pa7, our Pg4 as their Bc7; transferred move h6g7 (**endorses this move**). (d) Retrieved from `cctl_top_p0003#1589` (ply 75) [structural distance $d=67$, QAP obj=11]: the master played Kh7; under μ our Kh3 plays the role of that game’s Kh1; cross-type roles: our Bf5 as their Rg8, our pa4 as their Rc7; transferred move h6g7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg7 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



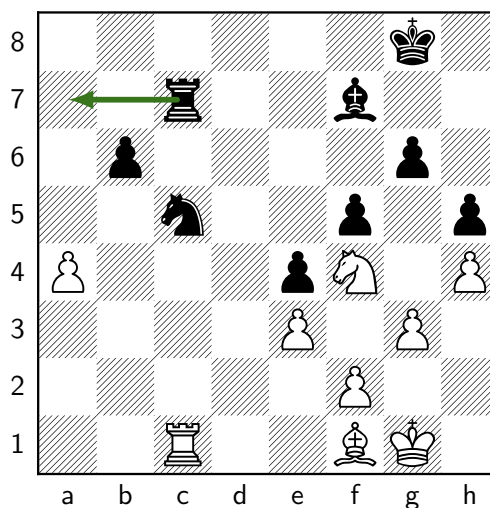
(a)



(b)

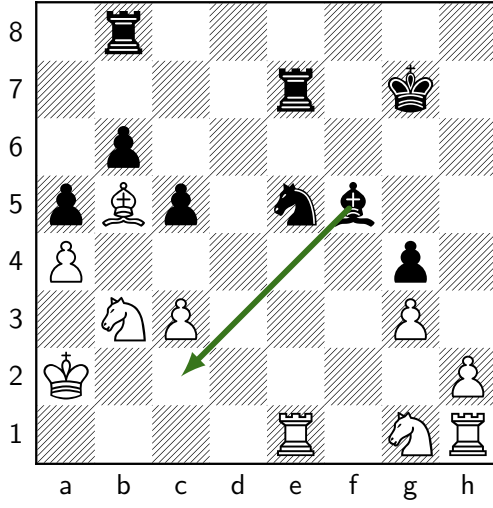


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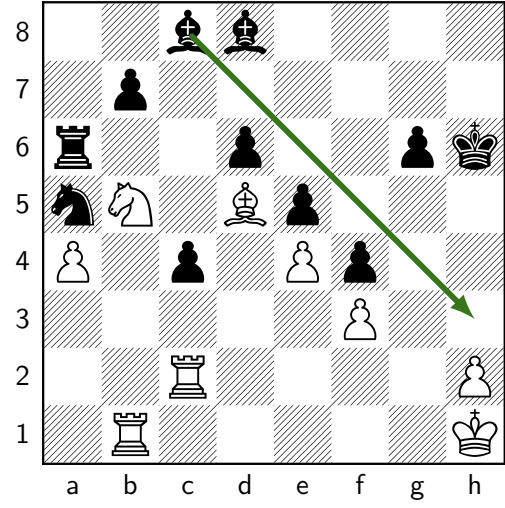


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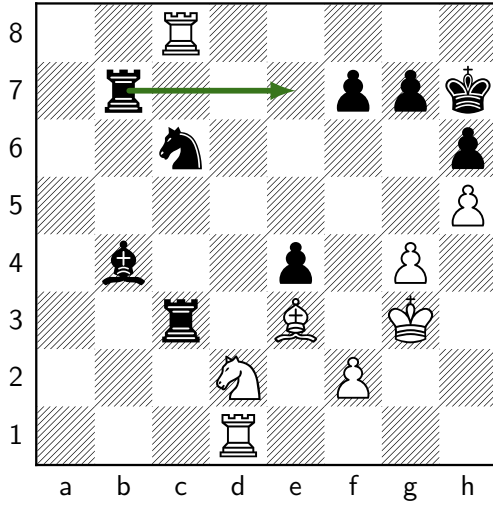
Figure 34: **Move 40, Black: Rb8.** (a) Our position; Melanie plays Rb8 (f8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#274` (ply 65) [structural distance $d=60$, QAP obj=35]: the master played Bf6; under μ our Rf1 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Pb6 as their Ne7, our bf1 as their ph4; transferred move f8b8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#275` (ply 65) [structural distance $d=60$, QAP obj=35]: the master played Bf6; under μ our Rf1 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Pb6 as their Ne7, our bf1 as their ph4; transferred move f8b8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3674` (ply 83) [structural distance $d=61$, QAP obj=80]: the master played Ra7; under μ our Re2 plays the role of that game’s Rc2; cross-type roles: our Pb6 as their Bf7, our Bf5 as their Pb6; transferred move e7a7. Synthesis: 2 retrieved analogs transferred to Rb8 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



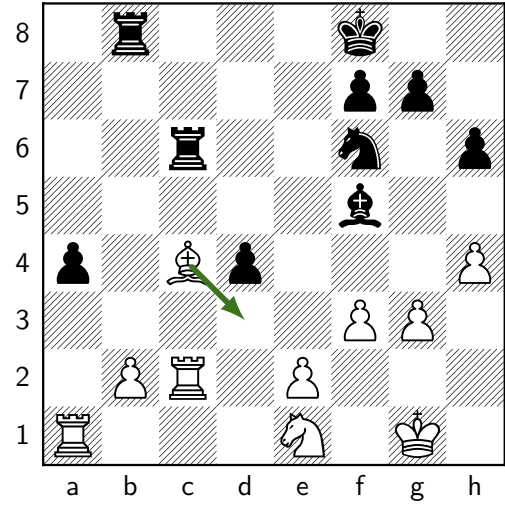
(a)



(b)

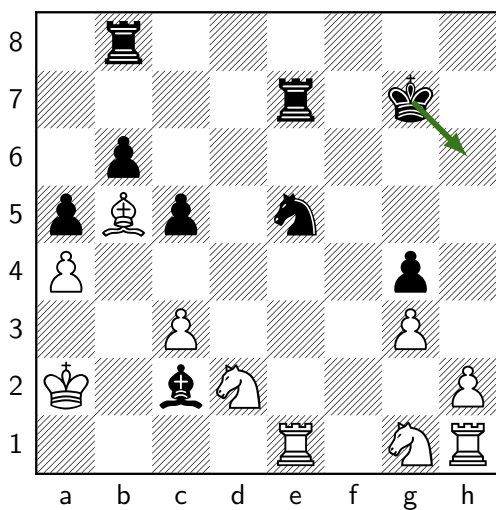


(c)

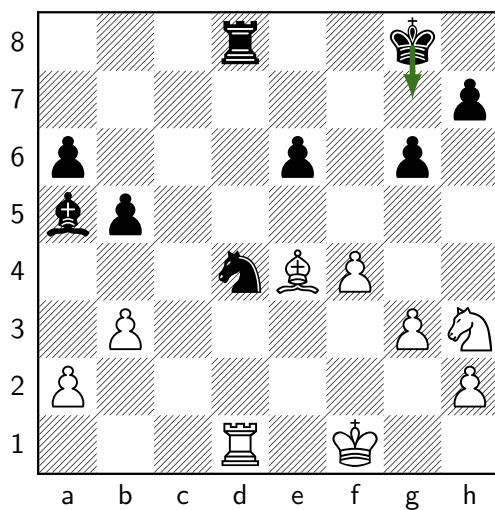


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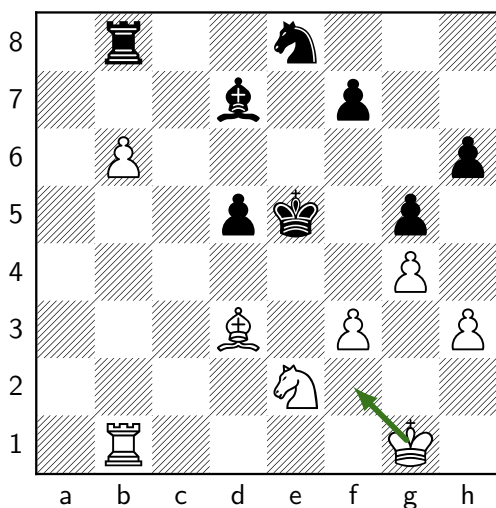
Figure 35: **Move 41, Black: Bc2.** (a) Our position; Melanie plays Bc2 (f5c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#397` (ply 71) [structural distance $d=61$, QAP obj=51]: the master played Bh3; under μ our Bf4 plays the role of that game’s Bc1; cross-type roles: our pc3 as their Bd8, our Ne5 as their Pb7; transferred move f5c2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1228` (ply 77) [structural distance $d=67$, QAP obj=33]: the master played Re7; under μ our Re2 plays the role of that game’s Rb2; cross-type roles: our bb5 as their Pg7, our pg3 as their Bb4; transferred move e7d7. (d) Retrieved from `ccr1_top_p0001#1990` (ply 72) [structural distance $d=61$, QAP obj=28]: the master played Bd3; under μ our Ne4 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our Pc5 as their Ne1, our Re7 as their Pb2; transferred move e5d3. Synthesis: 1 retrieved analog transferred to Bc2 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



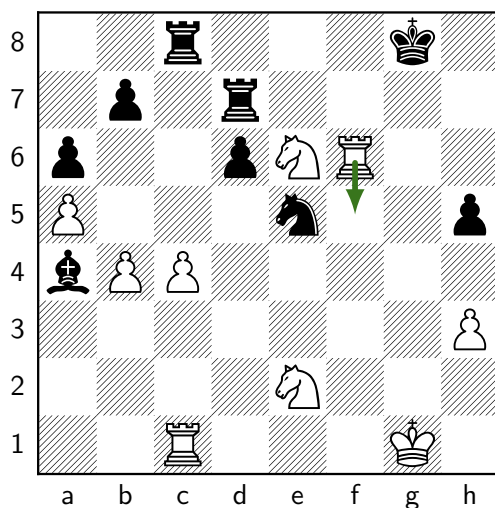
(a)



(b)

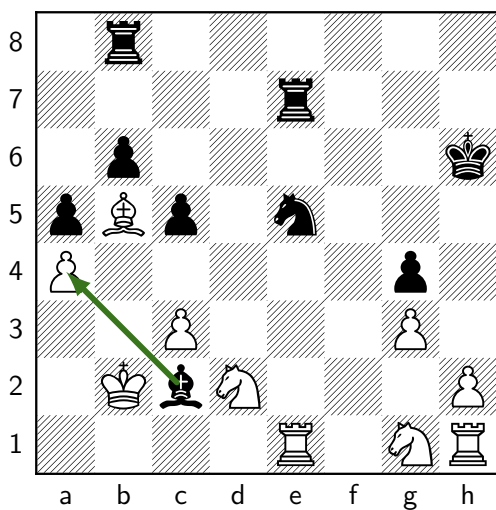


(c)

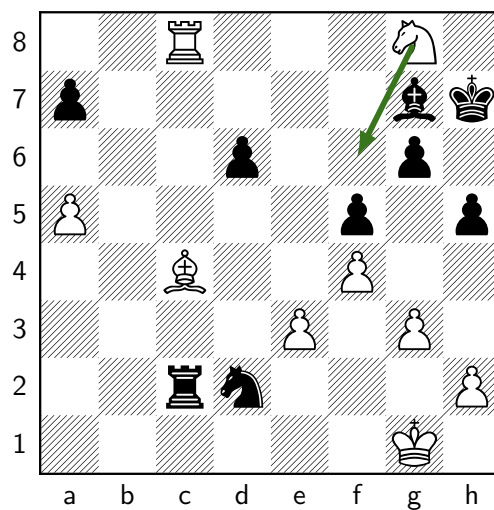


(d)

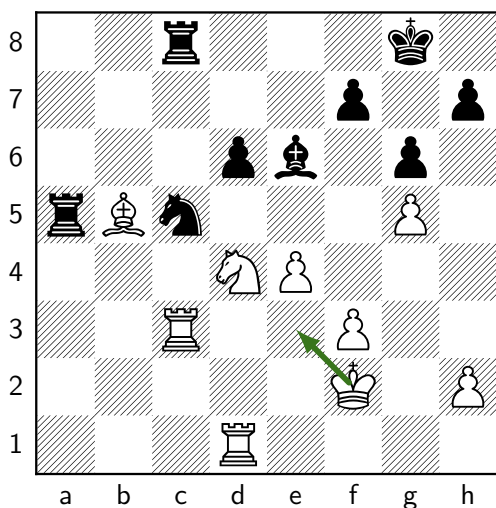
Figure 36: **Move 42, Black: Kh6**. (a) Our position; Melanie plays Kh6 (g7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1036` (ply 67) [structural distance $d=61$, QAP obj=49]: the master played Kg7; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our nd2 as their Ph7, our rh1 as their Pe6; transferred move g7h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2032` (ply 68) [structural distance $d=67$, QAP obj=43]: the master played Kf2; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our pg3 as their Ne2, our Pc5 as their Bd3; transferred move g7h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1266` (ply 92) [structural distance $d=60$, QAP obj=32]: the master played Rf5; under μ our Rb1 plays the role of that game’s Rf6; cross-type roles: our rh1 as their Ne2, our Ne5 as their Pc4; transferred move b8a8. Synthesis: 2 retrieved analogs transferred to Kh6 (summed QAP objective 92, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



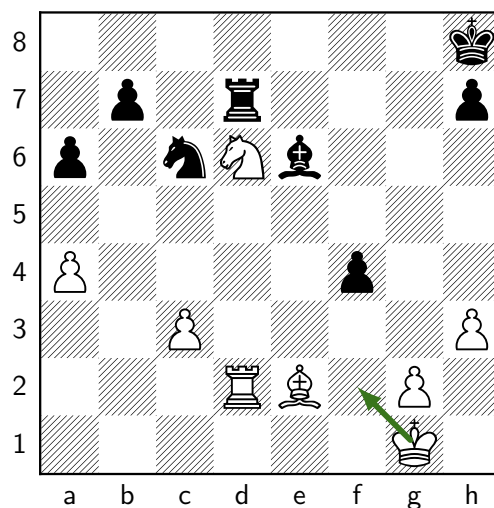
(a)



(b)

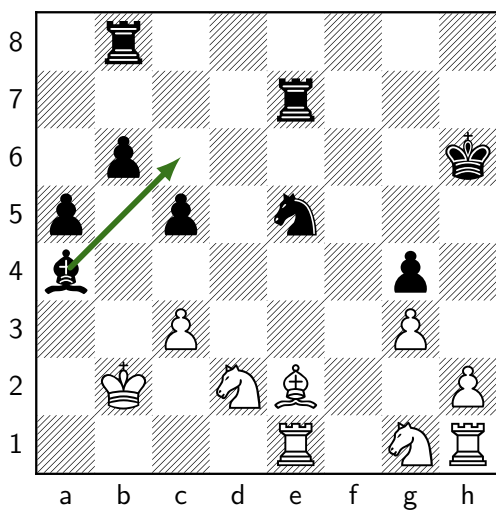


(c)

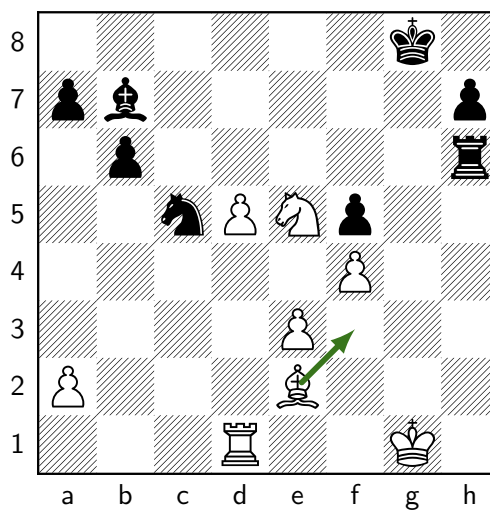


(d)

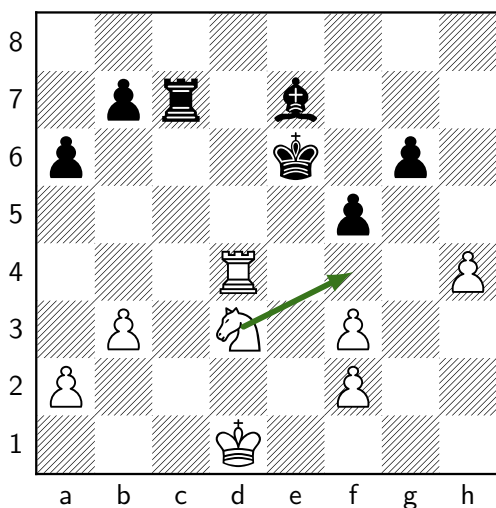
Figure 37: **Move 43, Black: Bxa4.** (a) Our position; Melanie plays Bxa4 (c2a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2125` (ply 98) [structural distance $d=55$, QAP obj=44]: the master played Nf6+; under μ our Bc7 plays the role of that game’s Ng8 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Ph2, our Ne5 as their Bc4; transferred move c2e4. (c) Retrieved from `ccr1_top_p0002#3531` (ply 68) [structural distance $d=53$, QAP obj=35]: the master played Ke3; under μ our Kh3 plays the role of that game’s Kf2; cross-type roles: our kb2 as their Rc3, our kb2 as their ra5; transferred move h6g7. (d) Retrieved from `ccr1_top_p0001#1221` (ply 78) [structural distance $d=64$, QAP obj=35]: the master played Kf2; under μ our Kh3 plays the role of that game’s Kg1; cross-type roles: our ng1 as their Be2, our Re7 as their Pg2; transferred move h6h5. Synthesis: no retrieved analog transferred to Bxa4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



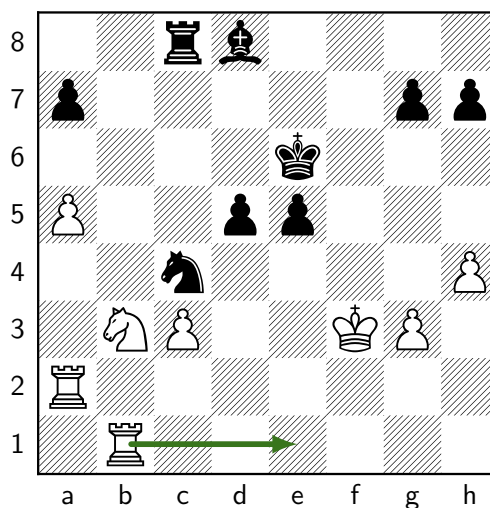
(a)



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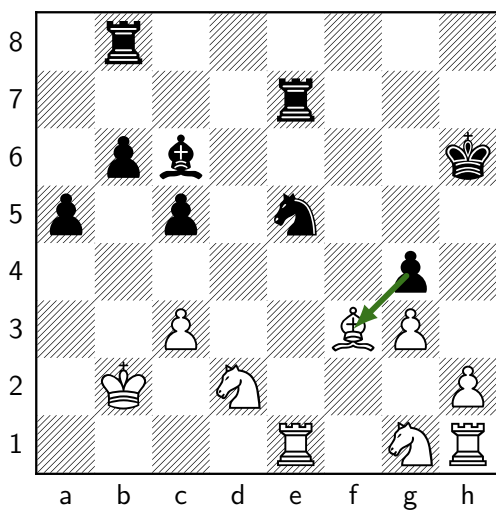


(c)

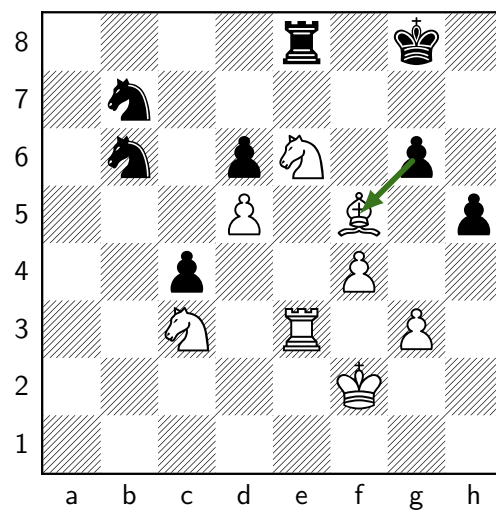


(d)

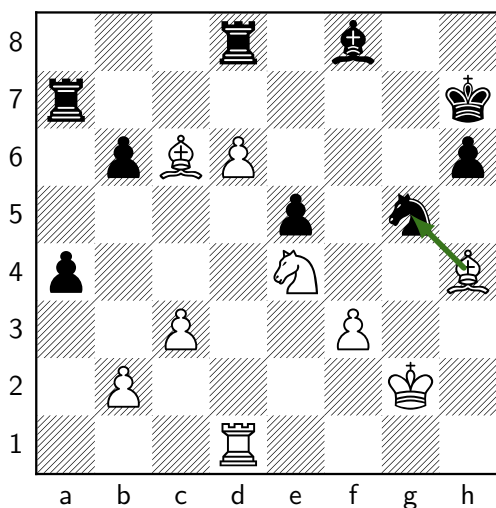
Figure 38: **Move 44, Black: Bc6.** (a) Our position; Melanie plays Bc6 (a4c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2153` (ply 80) [structural distance $d=61$, QAP obj=22]: the master played Bf3; under μ our Ba5 plays the role of that game’s Be2; cross-type roles: our Ne5 as their Rd1, our Pc5 as their Ne5; transferred move a4c6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3929` (ply 102) [structural distance $d=66$, QAP obj=19]: the master played Nf4+; under μ our Ba5 plays the role of that game’s Nd3 (a **cross-type** correspondence); cross-type roles: our rh1 as their Pa2, our Ne5 as their Rd4; transferred move a4c6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1414` (ply 74) [structural distance $d=46$, QAP obj=43]: the master played Re1; under μ our Re2 plays the role of that game’s Rb1; cross-type roles: our Ba4 as their Ra2, our be2 as their nc4; transferred move e7f7. Synthesis: 2 retrieved analogs transferred to Bc6 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



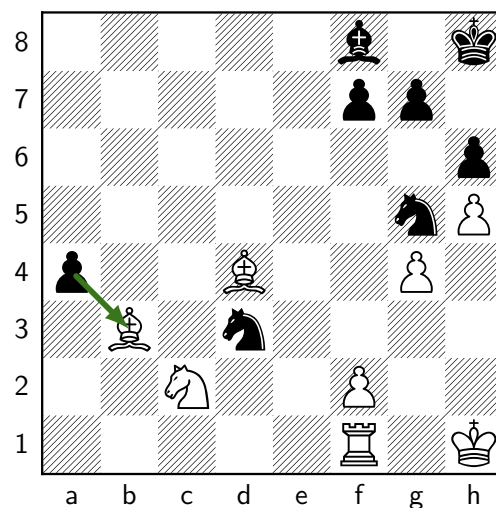
(a)



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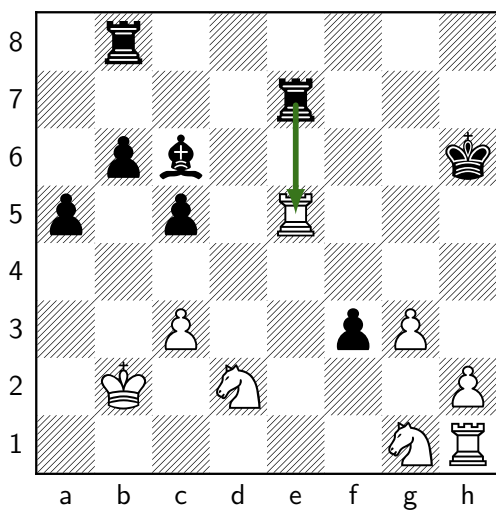


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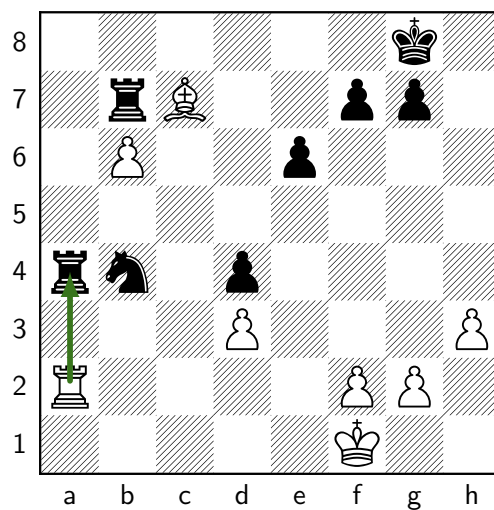


(d)

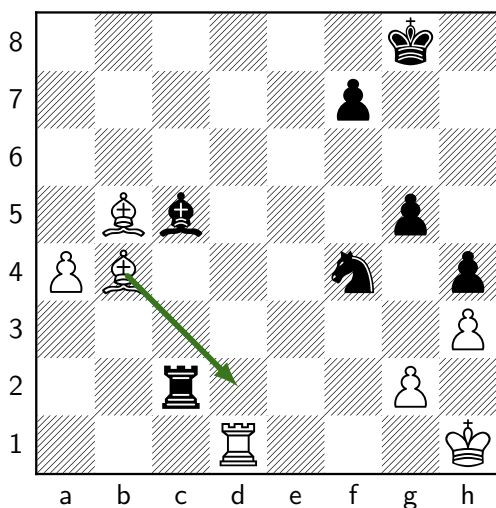
Figure 39: **Move 45, Black: gxf3.** (a) Our position; Melanie plays **gxf3** (**g4f3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#67` (ply 75) [structural distance $d=72$, QAP obj=47]: the master played **gxf5**; under μ our $Pg5$ plays the role of that game’s $Pg3$; cross-type roles: our $Bc6$ as their $Re8$, our $Rb8$ as their $Nb7$; transferred move **g4f3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3715` (ply 88) [structural distance $d=78$, QAP obj=46]: the master played **Bxg5**; under μ our $Pg5$ plays the role of that game’s $Bh4$ (a **cross-type** correspondence); cross-type roles: our $nd2$ as their $Rd1$, our $Re7$ as their $Pb2$; transferred move **g4f3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#167` (ply 89) [structural distance $d=77$, QAP obj=39]: the master played **axb3**; under μ our $Pg5$ plays the role of that game’s $Pa5$; cross-type roles: our $Re7$ as their $Bf8$, our $nd2$ as their $Pf7$; transferred move **g4f3** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **gxf3** (summed QAP objective 198, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



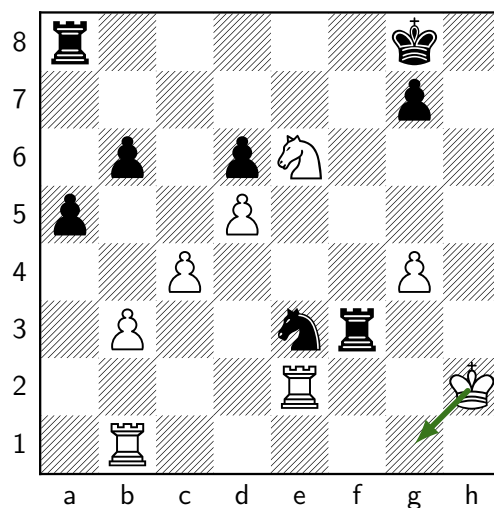
(a)



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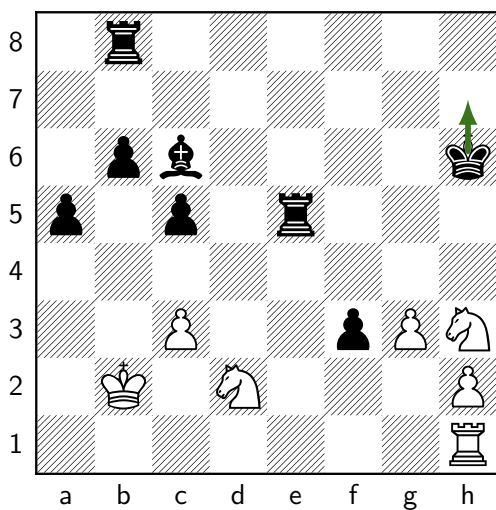


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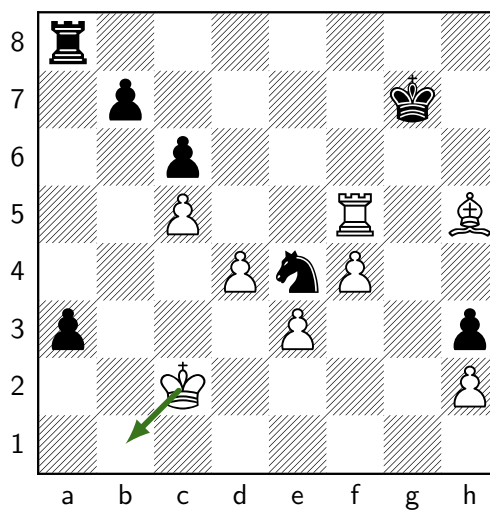


(d)

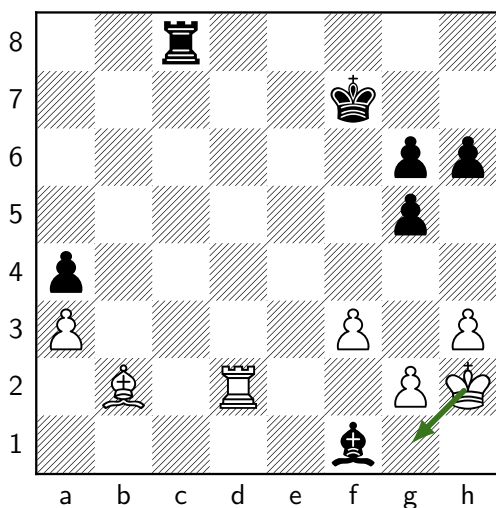
Figure 40: **Move 46, Black: Rxe5**. (a) Our position; Melanie plays **Rxe5** (**e7e5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3883` (ply 82) [structural distance $d=30$, QAP obj=53]: the master played **Rxa4**; under μ our *Re2* plays the role of that game’s *Ra2*; cross-type roles: our *ng1* as their *Pf2*, our *re5* as their *nb4*; transferred move **e7a7**. (c) Retrieved from `ccr1_top_p0001#856` (ply 88) [structural distance $d=38$, QAP obj=33]: the master played **Bd2**; under μ our *Re2* plays the role of that game’s *Bb4* (a **cross-type** correspondence); cross-type roles: our *Bc6* as their *Rd1*, our *nd2* as their *rc2*; transferred move **e7c7**. (d) Retrieved from `ccr1_top_p0002#3159` (ply 78) [structural distance $d=38$, QAP obj=19]: the master played **Kg1**; under μ our *Kh3* plays the role of that game’s *Kh2*; cross-type roles: our *Pb6* as their *Rb1*, our *re5* as their *ne3*; transferred move **h6h7**. Synthesis: no retrieved analog transferred to **Rxe5** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



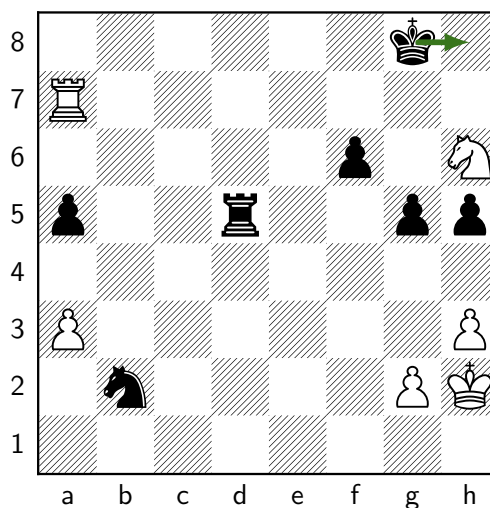
(a)



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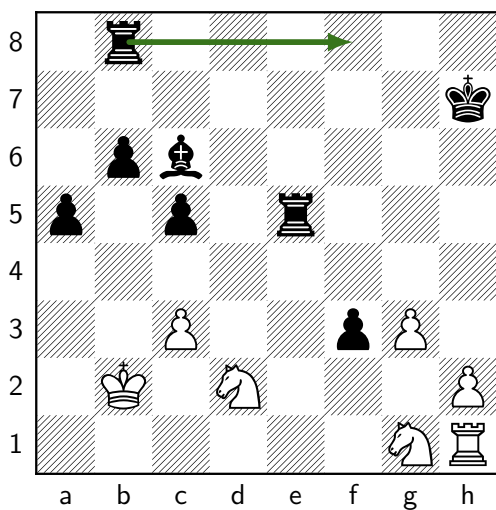


(c)

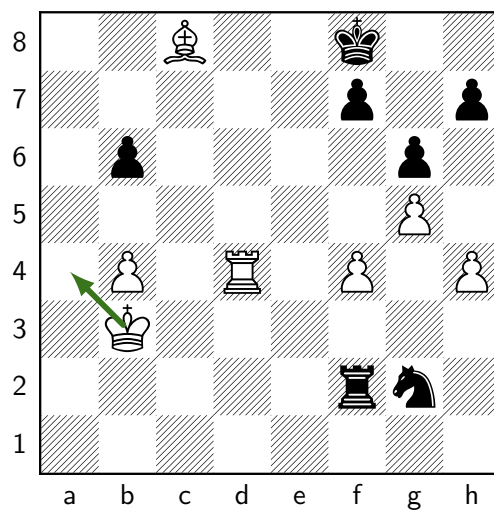


(d)

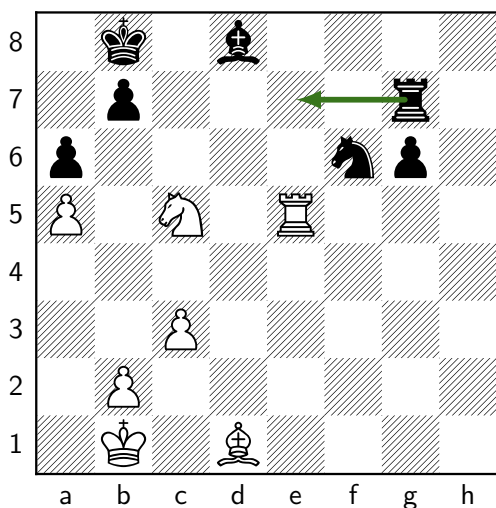
Figure 41: **Move 47, Black: Kh7 (h6h7)**. (a) Our position; Melanie plays Kh7 (h6h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2378` (ply 90) [structural distance $d=29$, QAP obj=22]: the master played Kb1; under μ our Kh3 plays the role of that game’s Kc2; cross-type roles: our nh3 as their Ph2, our rh1 as their pa3; transferred move h6h7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#721` (ply 78) [structural distance $d=31$, QAP obj=13]: the master played Kg1; under μ our Kh3 plays the role of that game’s Kh2; cross-type roles: our nd2 as their bf1, our ph2 as their Bb2; transferred move h6h7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#9` (ply 109) [structural distance $d=27$, QAP obj=5]: the master played Kh8; under μ our Kh3 plays the role of that game’s Kg1; cross-type roles: our nh3 as their Pa5, our pg3 as their Nb2; transferred move h6h7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kh7 (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



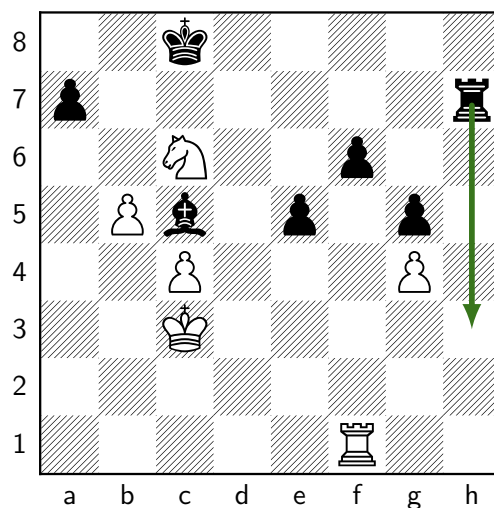
(a)



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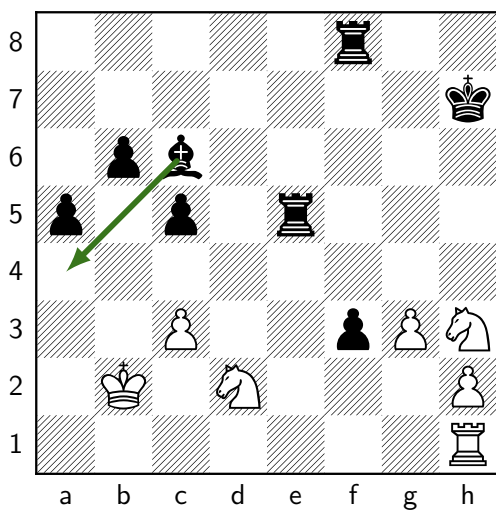


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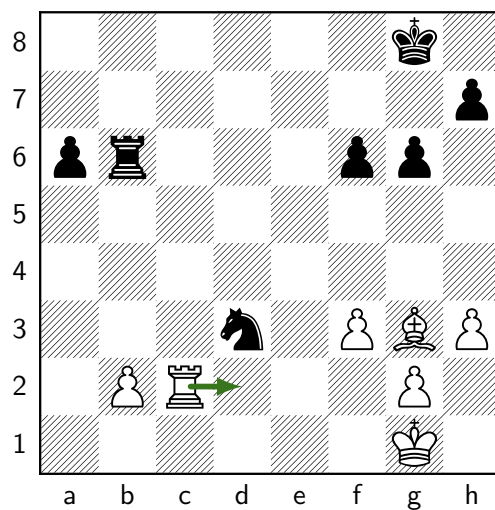


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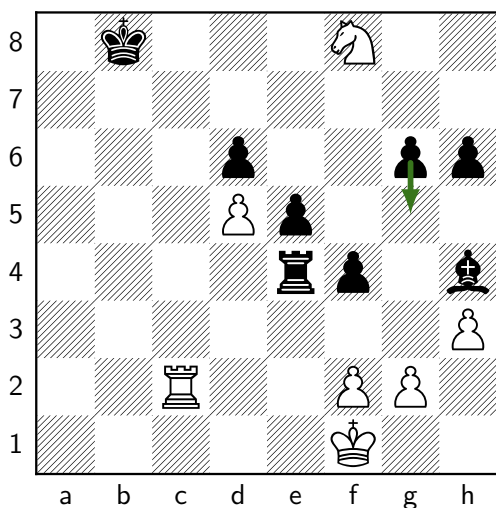
Figure 42: **Move 48, Black: Rf8**. (a) Our position; Melanie plays Rf8 (b8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#226` (ply 92) [structural distance $d=32$, QAP obj=53]: the master played Ka4; under μ our Kh2 plays the role of that game’s Kb3; cross-type roles: our nd2 as their rf2, our Bc6 as their Rd4; transferred move h7h6. (c) Retrieved from `ccr1_top_p0003#1675` (ply 83) [structural distance $d=27$, QAP obj=44]: the master played Re7; under μ our Bc3 plays the role of that game’s Rg2 (a **cross-type** correspondence); cross-type roles: our nd2 as their Bd8, our Re5 as their Nf6; transferred move c6a8. (d) Retrieved from `ccr1_top_p0003#745` (ply 91) [structural distance $d=32$, QAP obj=39]: the master played Rh3+; under μ our Bc3 plays the role of that game’s Rh2 (a **cross-type** correspondence); cross-type roles: our rh1 as their pb5, our Re5 as their Bc5; transferred move c6a4. Synthesis: no retrieved analog transferred to Rf8 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



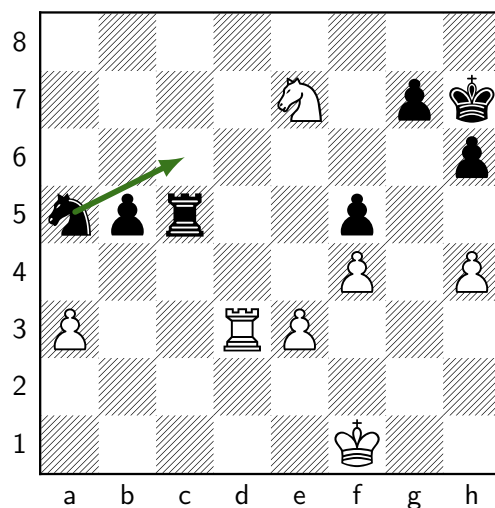
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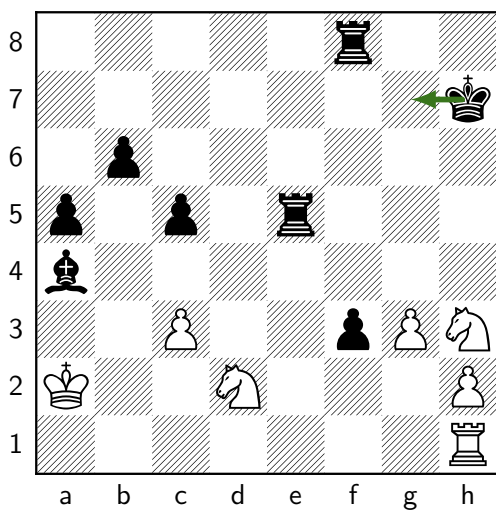


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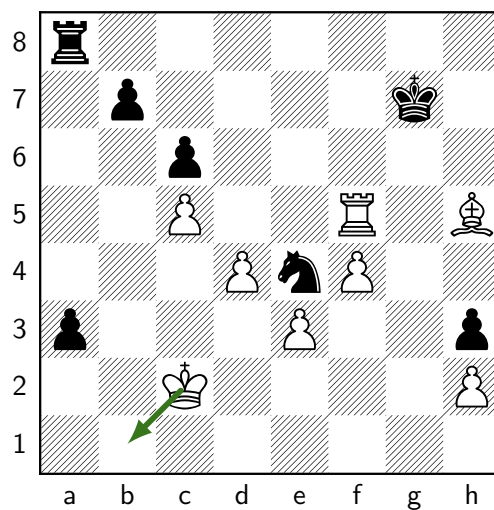


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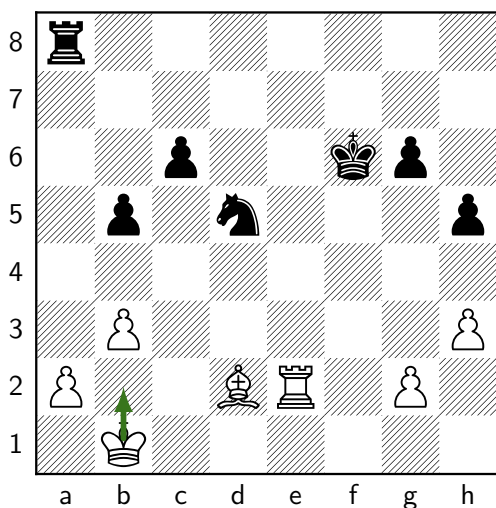
Figure 43: **Move 49, Black: Ba4**. (a) Our position; Melanie plays Ba4 (c6a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2368` (ply 84) [structural distance $d=26$, QAP obj=28]: the master played Rd2; under μ our Bc3 plays the role of that game’s Rc2 (a **cross-type** correspondence); cross-type roles: our Rf8 as their Bg3, our Re5 as their pa6; transferred move c6a4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2094` (ply 103) [structural distance $d=26$, QAP obj=31]: the master played g5; under μ our Pf6 plays the role of that game’s Pg3; cross-type roles: our Rf8 as their Ph6, our Pb6 as their Re4; transferred move f3f2. (d) Retrieved from `ccr1_top_p0000#2003` (ply 99) [structural distance $d=28$, QAP obj=24]: the master played Nc6; under μ our Bc3 plays the role of that game’s Na4 (a **cross-type** correspondence); cross-type roles: our Re5 as their Pb5, our nh3 as their pa3; transferred move c6e4. Synthesis: 2 retrieved analogs transferred to Ba4 (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



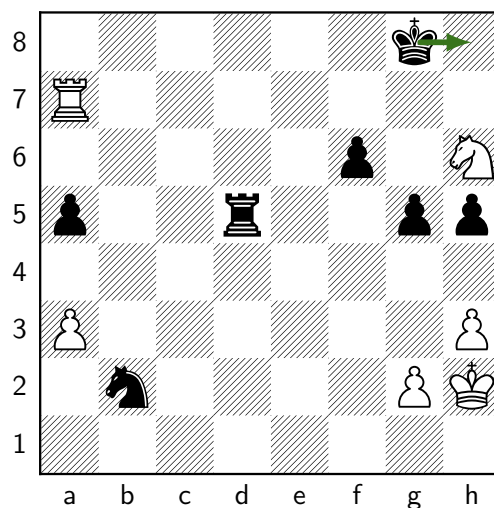
(a)



(b)

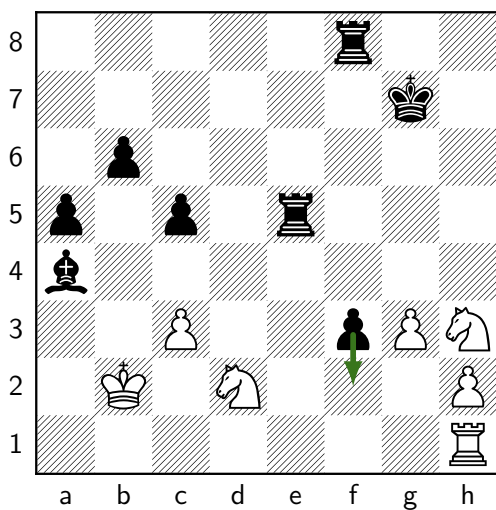


(c)

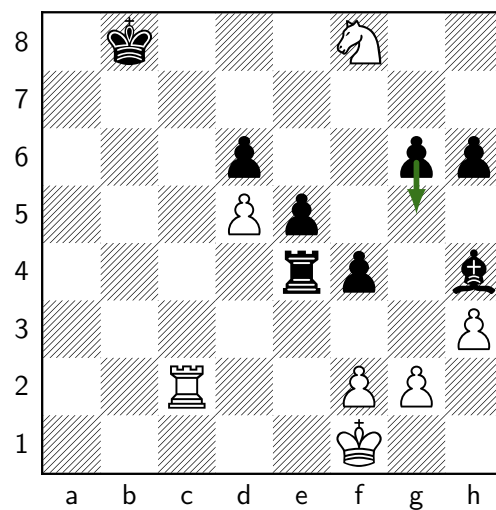


(d)

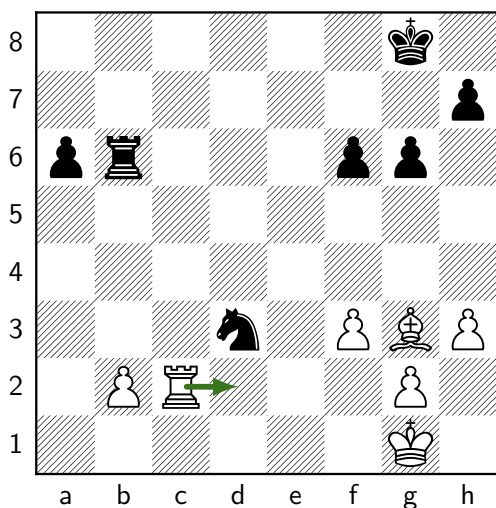
Figure 44: **Move 50, Black: Kg7 (h7g7)**. (a) Our position; Melanie plays Kg7 (h7g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2378` (ply 90) [structural distance $d=19$, QAP obj=22]: the master played Kb1; under μ our Kh2 plays the role of that game’s Kc2; cross-type roles: our rh1 as their pa3, our Ba4 as their Rf5; transferred move h7g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3300` (ply 90) [structural distance $d=29$, QAP obj=9]: the master played Kb2; under μ our Kh2 plays the role of that game’s Kb1; cross-type roles: our Re5 as their Bd2, our Ba4 as their Re2; transferred move h7g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#9` (ply 109) [structural distance $d=33$, QAP obj=4]: the master played Kh8; under μ our Kh2 plays the role of that game’s Kg1; cross-type roles: our Ba4 as their Ph5, our Rf8 as their Nb2; transferred move h7g7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kg7 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



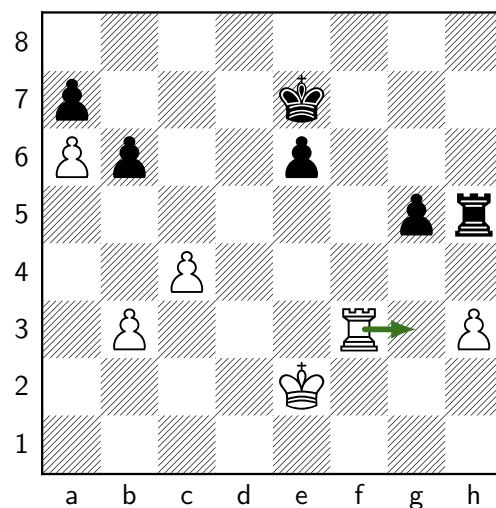
(a)



(b)

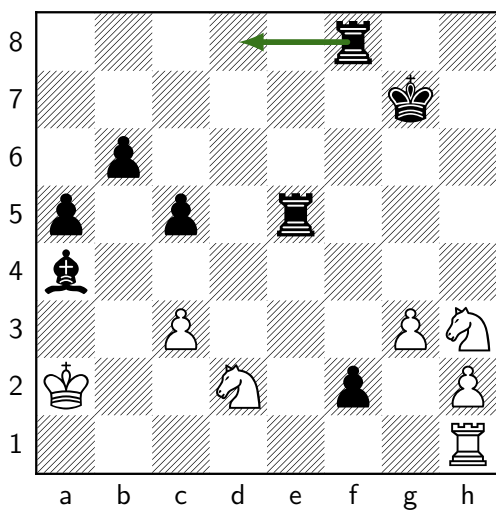


(c)

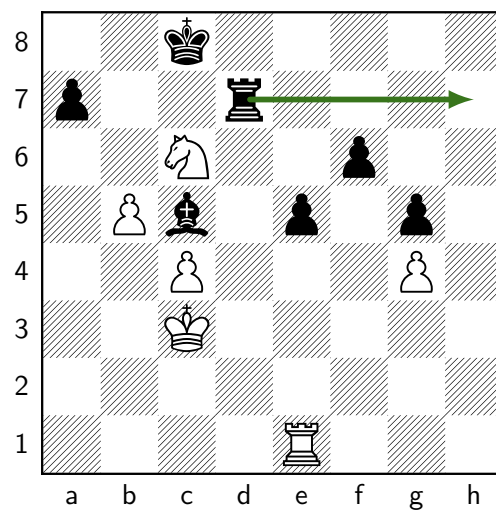


(d)

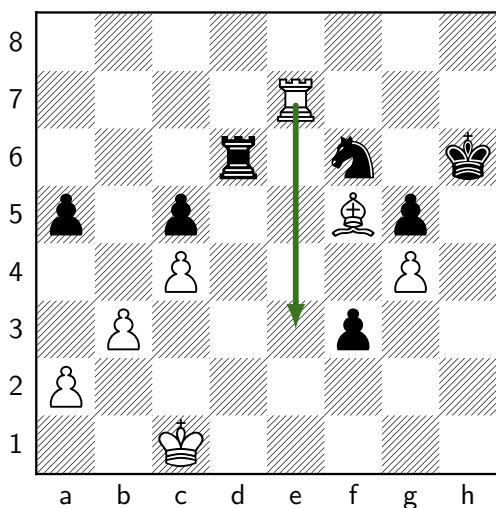
Figure 45: **Move 51, Black: f2**. (a) Our position; Melanie plays f2 (f3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2094` (ply 103) [structural distance $d=22$, QAP obj=30]: the master played g5; under μ our *Pf6* plays the role of that game’s *Pg3*; cross-type roles: our *Ba4* as their *Pe5*, our *pg3* as their *Re4*; transferred move f3f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2368` (ply 84) [structural distance $d=24$, QAP obj=25]: the master played Rd2; under μ our *Rf1* plays the role of that game’s *Rc2*; cross-type roles: our *nd2* as their *rb6*, our *Re5* as their *pf6*; transferred move f8a8. (d) Retrieved from `ccr1_top_p0001#1225` (ply 106) [structural distance $d=31$, QAP obj=22]: the master played Rg3; under μ our *Rf1* plays the role of that game’s *Rf3*; cross-type roles: our *nh3* as their *pg5*, our *nd2* as their *rh5*; transferred move f8a8. Synthesis: 1 retrieved analog transferred to f2 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



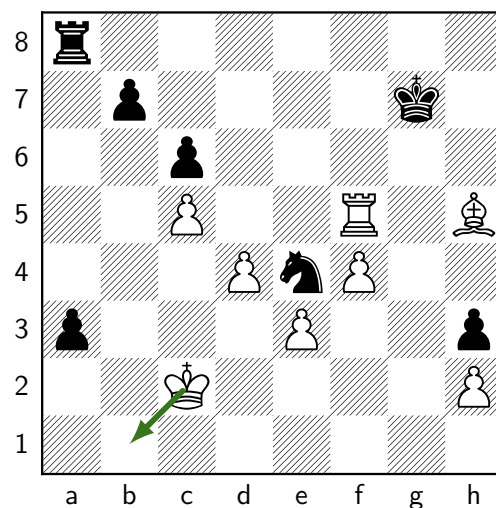
(a)



(b)



(c)



(d)

Figure 46: **Move 52, Black: Rd8.** (a) Our position; Melanie plays Rd8 (f8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#745 (ply 89) [structural distance $d=33$, QAP obj=40]: the master played Rh7; under μ our Rf1 plays the role of that game’s Rd2; cross-type roles: our rh1 as their pb5, our ph2 as their Bc5; transferred move f8d8 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1837 (ply 110) [structural distance $d=34$, QAP obj=21]: the master played Re3; under μ our Re4 plays the role of that game’s Re7; cross-type roles: our nd2 as their pa5, our Ba4 as their pc5; transferred move e5e6. (d) Retrieved from ccr1_top_p0000#2378 (ply 90) [structural distance $d=18$, QAP obj=20]: the master played Kb1; under μ our Kg2 plays the role of that game’s Kc2; cross-type roles: our Re5 as their Pd4, our Pa5 as their Bh5; transferred move g7h7. Synthesis: 1 retrieved analog transferred to Rd8 (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

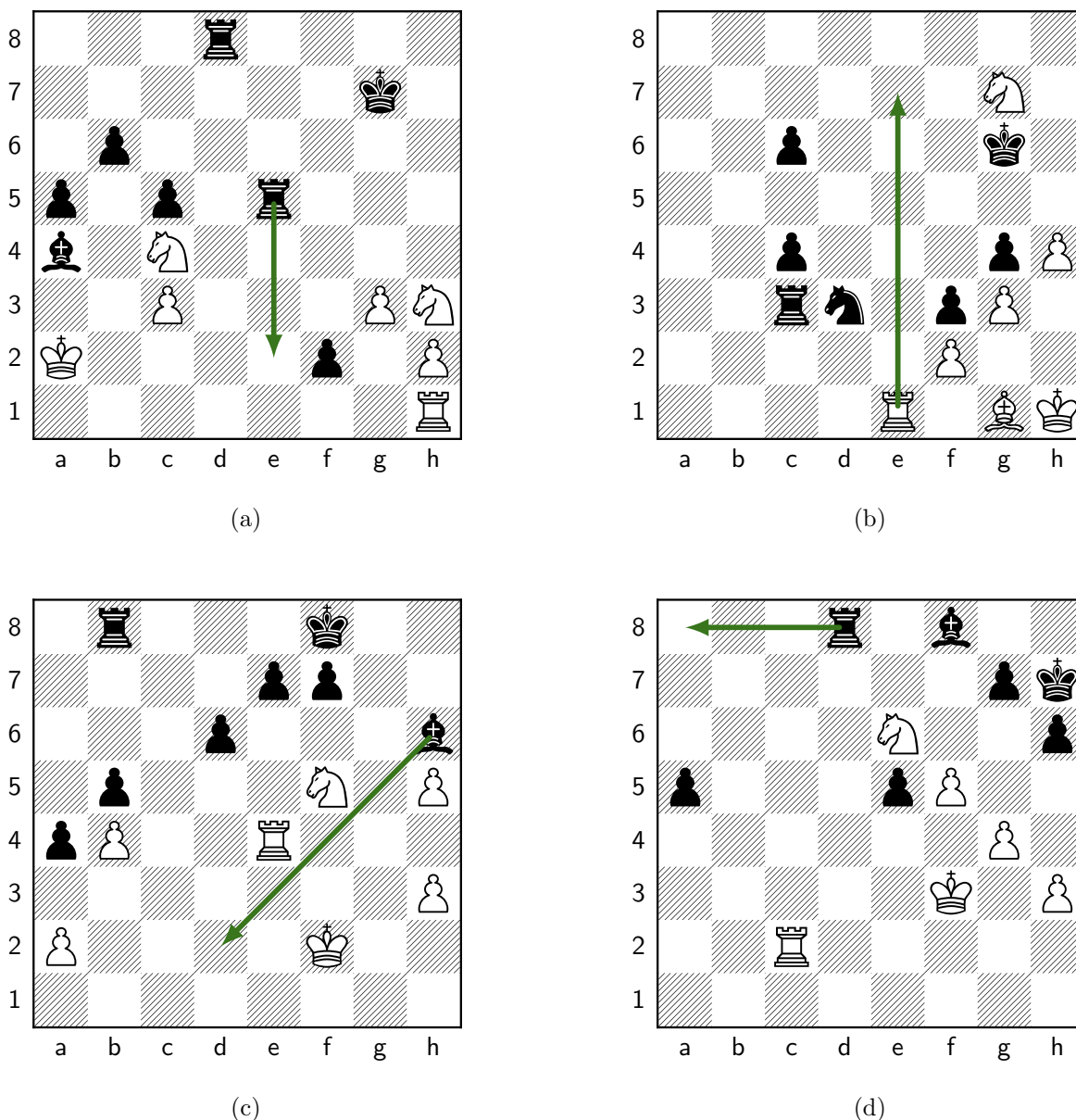
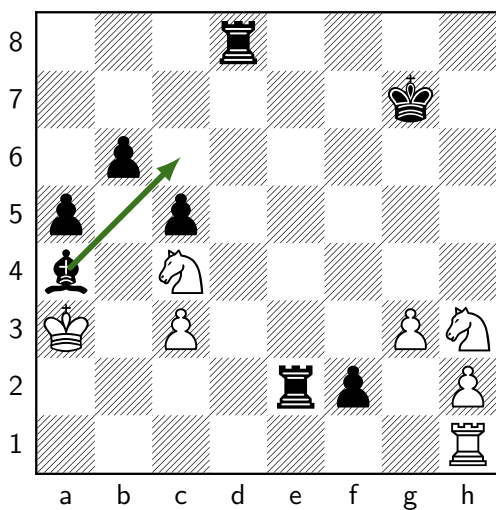
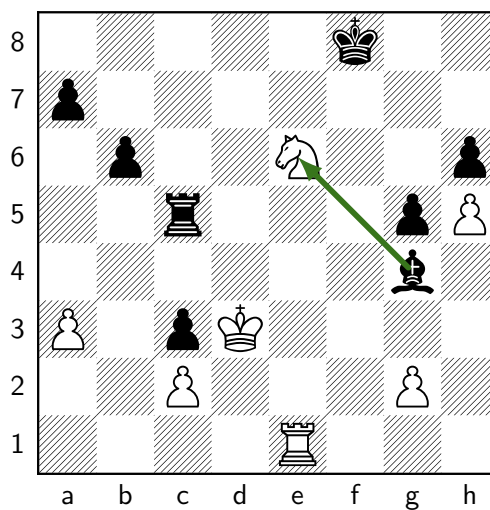


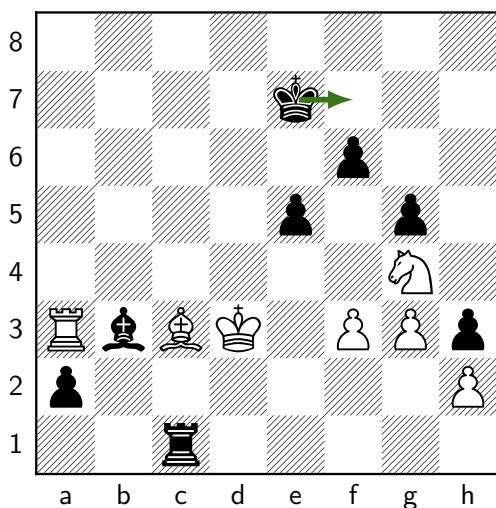
Figure 47: **Move 53, Black: Re2+.** (a) Our position; Melanie plays Re2+ (e5e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1500` (ply 116) [structural distance $d=45$, QAP obj=22]: the master played Re7; under μ our Re4 plays the role of that game’s Re1; cross-type roles: our Pc5 as their Bg1, our pc3 as their rc3; transferred move e5e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2689` (ply 99) [structural distance $d=48$, QAP obj=13]: the master played Bd2; under μ our Re4 plays the role of that game’s Bh3 (a **cross-type** correspondence); cross-type roles: our Ba4 as their ph3, our nh3 as their pa2; transferred move e5e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3609` (ply 93) [structural distance $d=49$, QAP obj=33]: the master played Ra8; under μ our Re4 plays the role of that game’s Rd1; cross-type roles: our Re5 as their Bf8, our Rd8 as their Pa5; transferred move e5e7. Synthesis: 2 retrieved analogs transferred to Re2+ (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



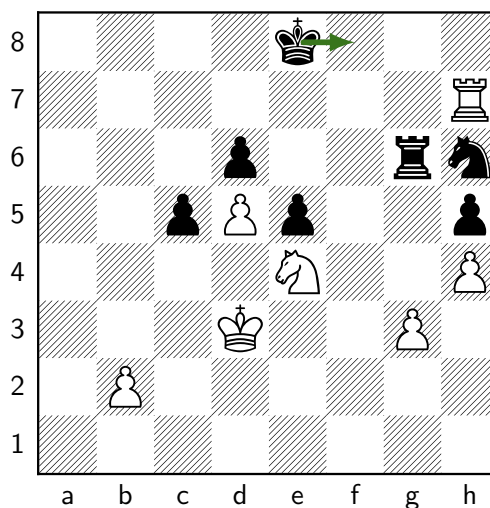
(a)



(b)

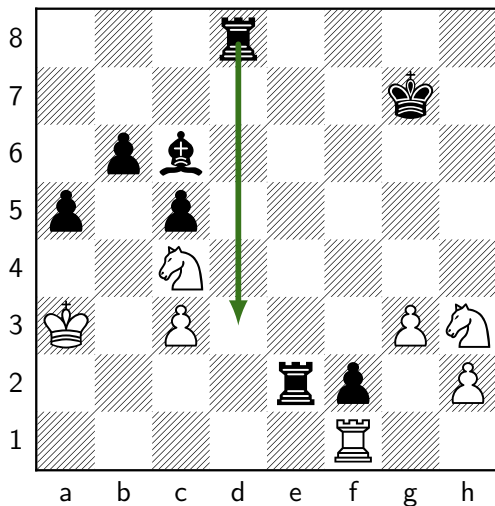


(c)

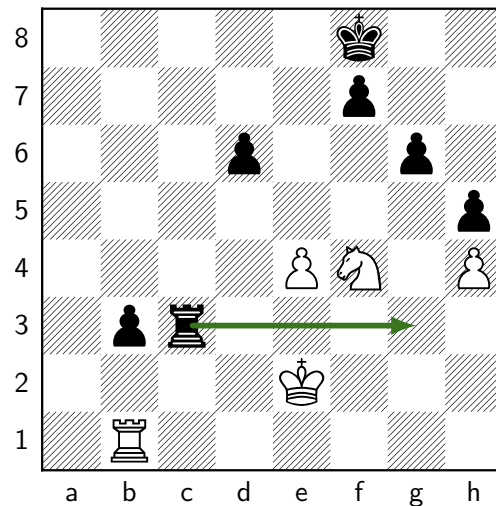


(d)

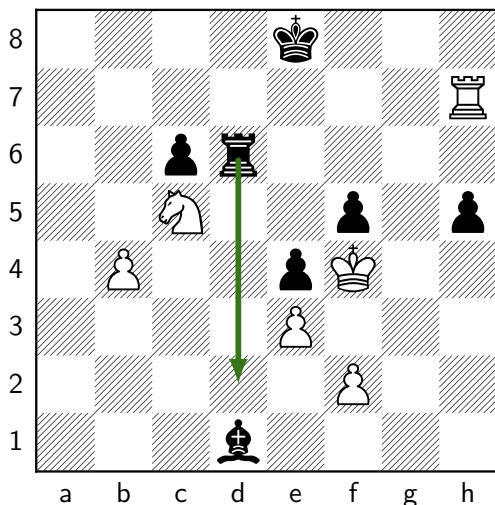
Figure 48: **Move 54, Black: Bc6**. (a) Our position; Melanie plays Bc6 (a4c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1553` (ply 103) [structural distance $d=42$, QAP obj=3]: the master played Bxe6; under μ our Ba5 plays the role of that game’s Bg5; cross-type roles: our Pa5 as their Rc5, our Re2 as their Pg5; transferred move a4c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3339` (ply 109) [structural distance $d=48$, QAP obj=24]: the master played Kf7; under μ our Kg2 plays the role of that game’s Ke2; cross-type roles: our ka3 as their ra3, our pg3 as their bc3; transferred move g7g8. (d) Retrieved from `ccr1_top_p0002#2290` (ply 107) [structural distance $d=41$, QAP obj=22]: the master played Kf8; under μ our Kg2 plays the role of that game’s Ke1; cross-type roles: our ka3 as their rh7, our Ba4 as their Nh6; transferred move g7h7. Synthesis: 1 retrieved analog transferred to Bc6 (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



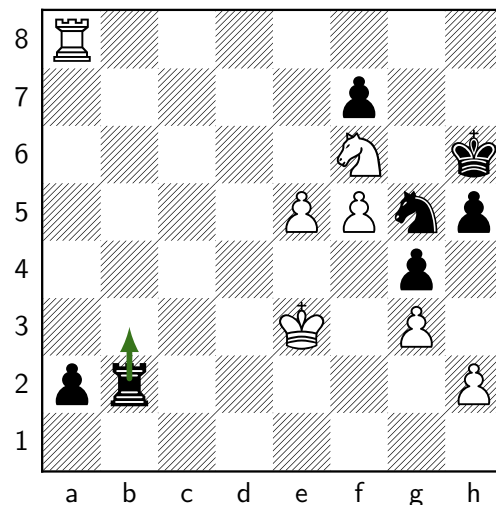
(a)



(b)



(c)



(d)

Figure 49: **Move 55, Black: Rd3.** (a) Our position; Melanie plays Rd3 (d8d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#660` (ply 109) [structural distance $d=41$, QAP obj=51]: the master played Rg3; under μ our *Re7* plays the role of that game’s *Rc6*; transferred move e2c2. (c) Retrieved from `ccr1_top_p0002#3560` (ply 101) [structural distance $d=44$, QAP obj=38]: the master played Rd2; under μ our *Rd1* plays the role of that game’s *Rd3*; cross-type roles: our *Re2* as their *Bd1*; transferred move d8d1. (d) Retrieved from `ccr1_top_p0002#265` (ply 99) [structural distance $d=50$, QAP obj=38]: the master played Rb3+; under μ our *Re7* plays the role of that game’s *Rb7*; cross-type roles: our *Bc6* as their *pe5*, our *pg3* as their *Ng5*; transferred move e2c2. Synthesis: no retrieved analog transferred to Rd3 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

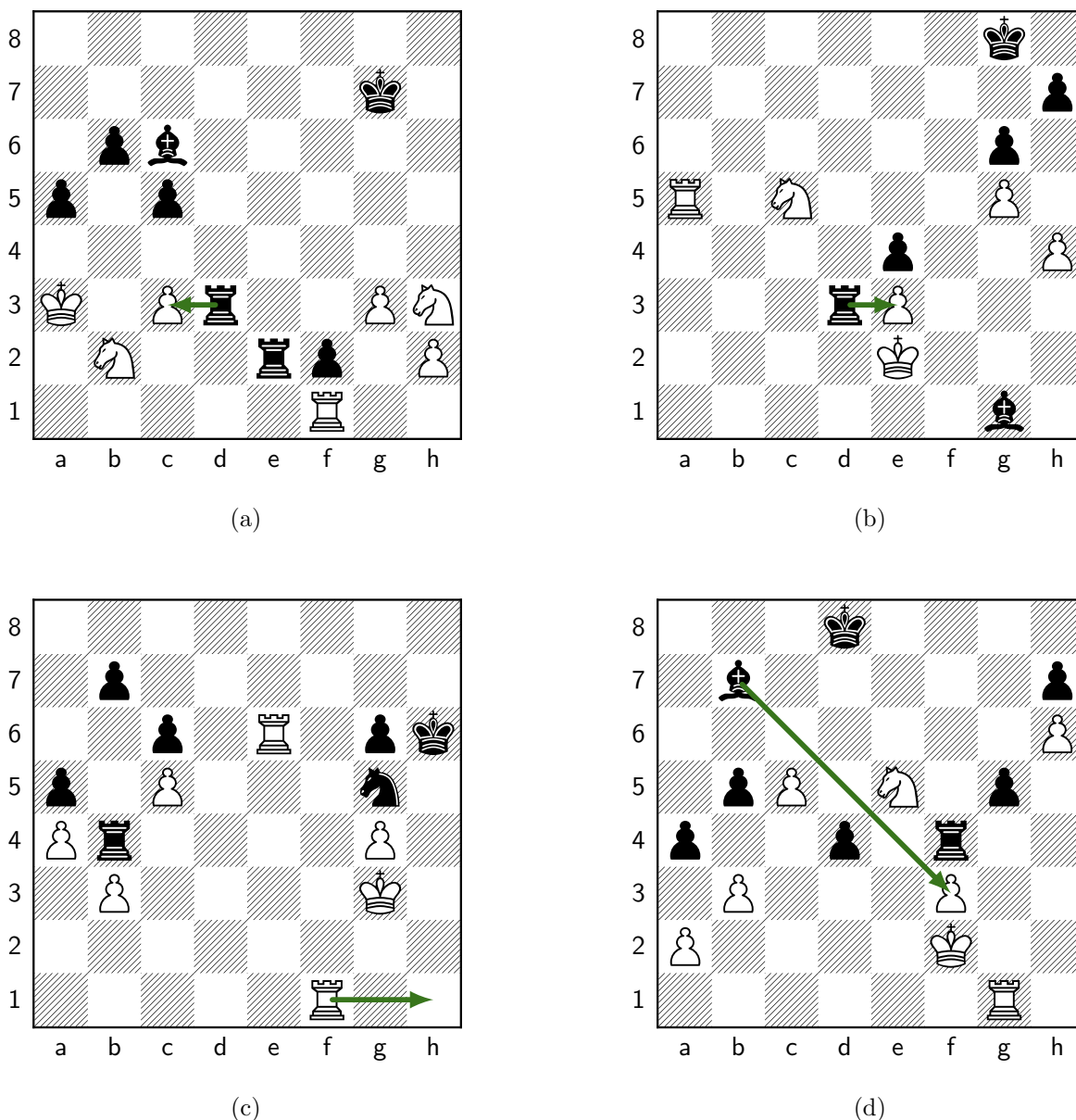
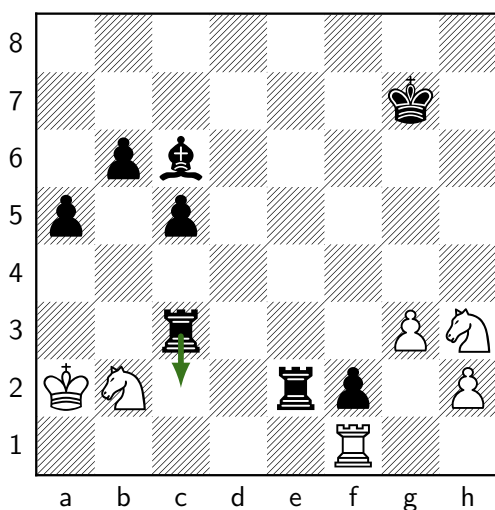
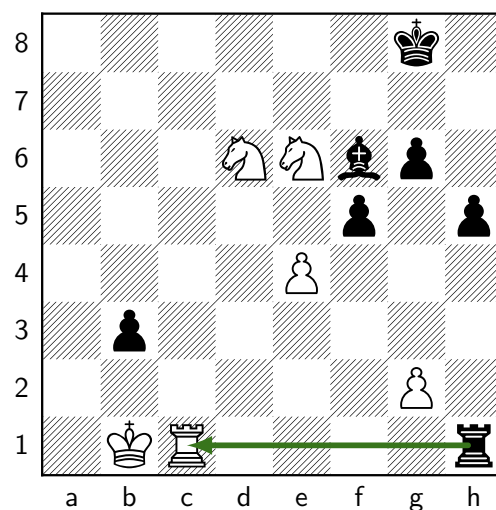


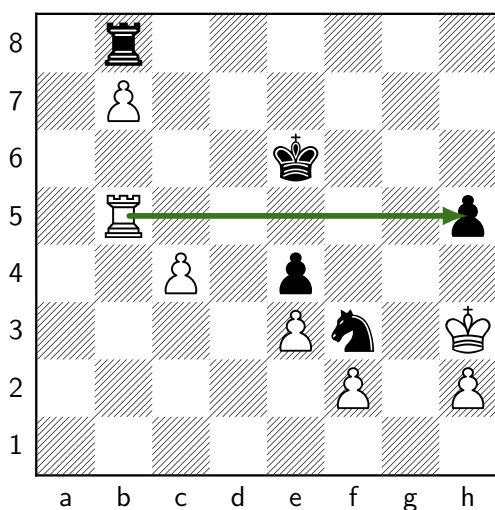
Figure 50: **Move 56, Black: $Rxc3+$.** (a) Our position; Melanie plays $Rxc3+$ ($d3c3$). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2725` (ply 95) [structural distance $d=48$, QAP obj=17]: the master played $Rxe3+$; under μ our $Rd6$ plays the role of that game's $Rd6$; cross-type roles: our $rf1$ as their $Ph7$, our $nh3$ as their $Pg6$; transferred move $d3c3$ (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2827` (ply 100) [structural distance $d=44$, QAP obj=66]: the master played $Rh1+$; under μ our $Bc3$ plays the role of that game's $Rf1$ (a **cross-type** correspondence); cross-type roles: our $nh3$ as their $pa5$; transferred move $c6b7$. (d) Retrieved from `ccr1_top_p0002#2505` (ply 103) [structural distance $d=50$, QAP obj=37]: the master played $Bxf3$; under μ our $Re7$ plays the role of that game's $Bb2$ (a **cross-type** correspondence); cross-type roles: our $rf1$ as their $Pa4$, our $Bc6$ as their $pa2$; transferred move $e2b2$. Synthesis: 1 retrieved analog transferred to $Rxc3+$ (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



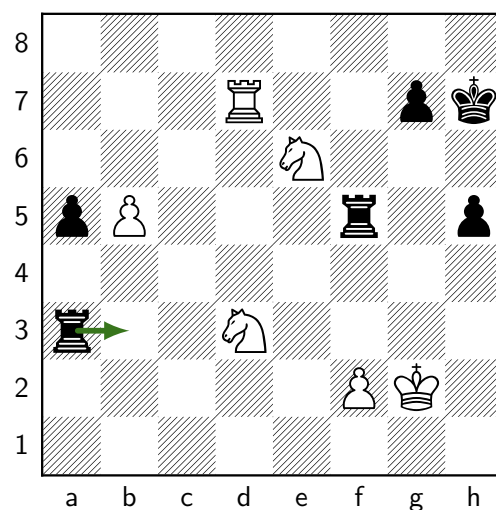
(a)



(b)



(c)



(d)

Figure 51: **Move 57, Black: Rcc2.** (a) Our position; Melanie plays Rcc2 (c3c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#232` (ply 115) [structural distance $d=36$, QAP obj=69]: the master played Rxc1+; under μ our *Re7* plays the role of that game’s *Rh8*; cross-type roles: our *ph2* as their *ne6*, our *Rc3* as their *Bf6*; transferred move e2b2. (c) Retrieved from `ccr1_top_p0001#1415` (ply 114) [structural distance $d=38$, QAP obj=47]: the master played Rxh5; under μ our *Rc6* plays the role of that game’s *Rb5*; cross-type roles: our *rf1* as their *ph5*, our *nb2* as their *Pb7*; transferred move c3c1. (d) Retrieved from `ccr1_top_p0002#919` (ply 97) [structural distance $d=41$, QAP obj=45]: the master played Rb3; under μ our *Re7* plays the role of that game’s *Ra6*; transferred move e2d2. Synthesis: no retrieved analog transferred to Rcc2 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

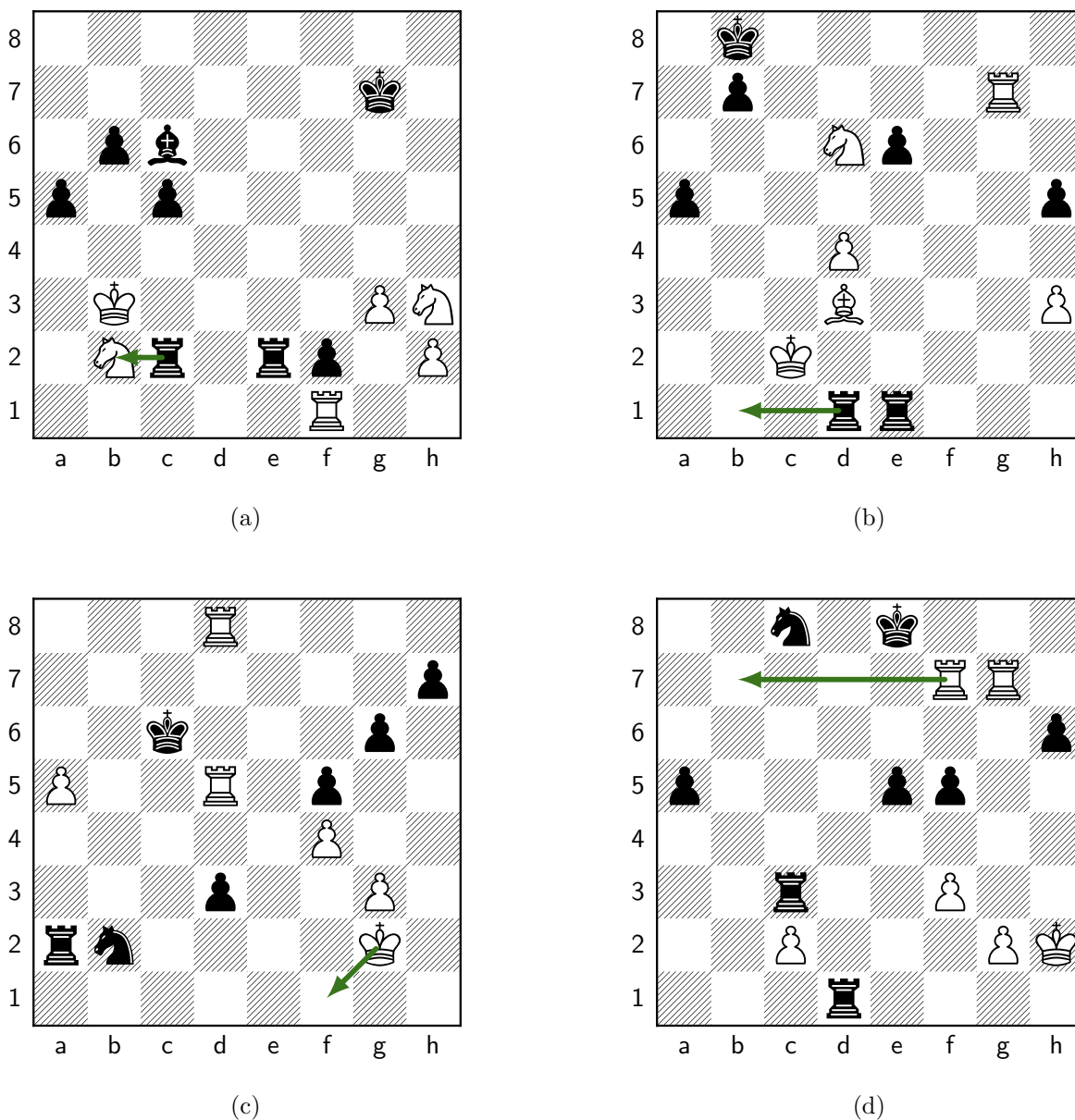
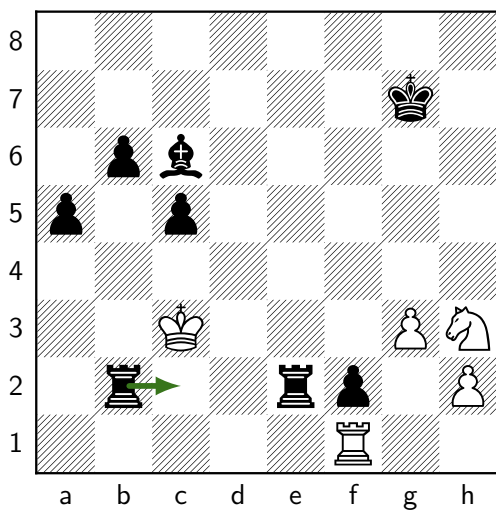
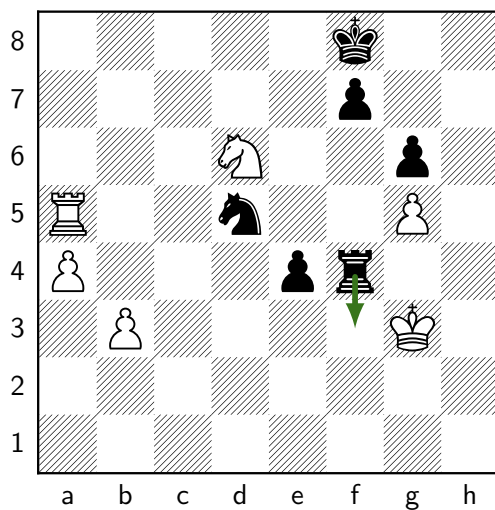


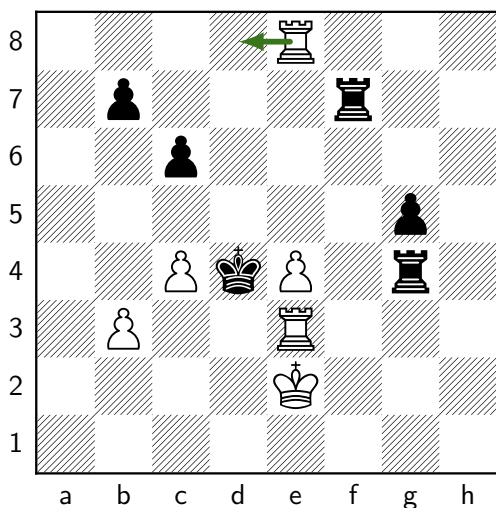
Figure 52: **Move 58, Black: Rxb2+.** (a) Our position; Melanie plays Rxb2+ (c2b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1082` (ply 111) [structural distance $d=41$, QAP obj=45]: the master played Rb1; under μ our Rc7 plays the role of that game’s Rd8; cross-type roles: our Bc6 as their nd6, our nb2 as their bd3; transferred move c2b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2657` (ply 124) [structural distance $d=54$, QAP obj=26]: the master played Kf1; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our ph2 as their ra2, our nb2 as their pf5; transferred move g7g8. (d) Retrieved from `ccr1_top_p0003#517` (ply 110) [structural distance $d=39$, QAP obj=25]: the master played Rb7; under μ our Rc7 plays the role of that game’s Rf7; cross-type roles: our Pb6 as their rd1, our nb2 as their pf5; transferred move c2d2. Synthesis: 1 retrieved analog transferred to Rxb2+ (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



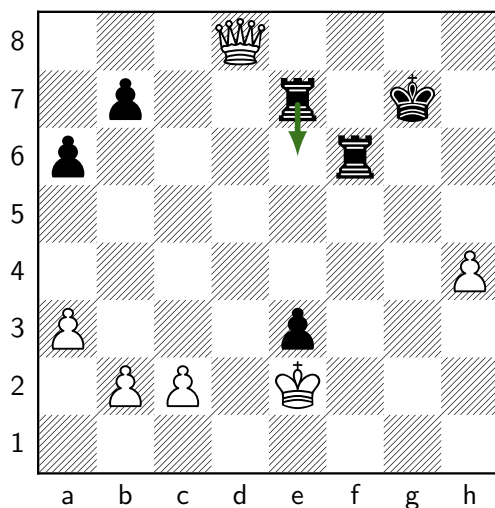
(a)



(b)



(c)



(d)

Figure 53: **Move 59, Black: Rbc2+.** (a) Our position; Melanie plays Rbc2+ (b2c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1013` (ply 121) [structural distance $d=39$, QAP obj=22]: the master played Rf3+; under μ our Rb7 plays the role of that game’s Rf5; cross-type roles: our rf1 as their nd6, our nh3 as their ra5; transferred move b2b3. (c) Retrieved from `ccr1_top_p0000#3474` (ply 114) [structural distance $d=45$, QAP obj=21]: the master played Rd8+; under μ our Re7 plays the role of that game’s Re8; cross-type roles: our pg3 as their rg4, our nh3 as their pc6; transferred move e2e3. (d) Retrieved from `ccr1_top_p0003#1283` (ply 133) [structural distance $d=37$, QAP obj=16]: the master played Ree6; under μ our Rb7 plays the role of that game’s Re2; cross-type roles: our kc3 as their qd8, our Bc6 as their ph4; transferred move b2a2. Synthesis: no retrieved analog transferred to Rbc2+ – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

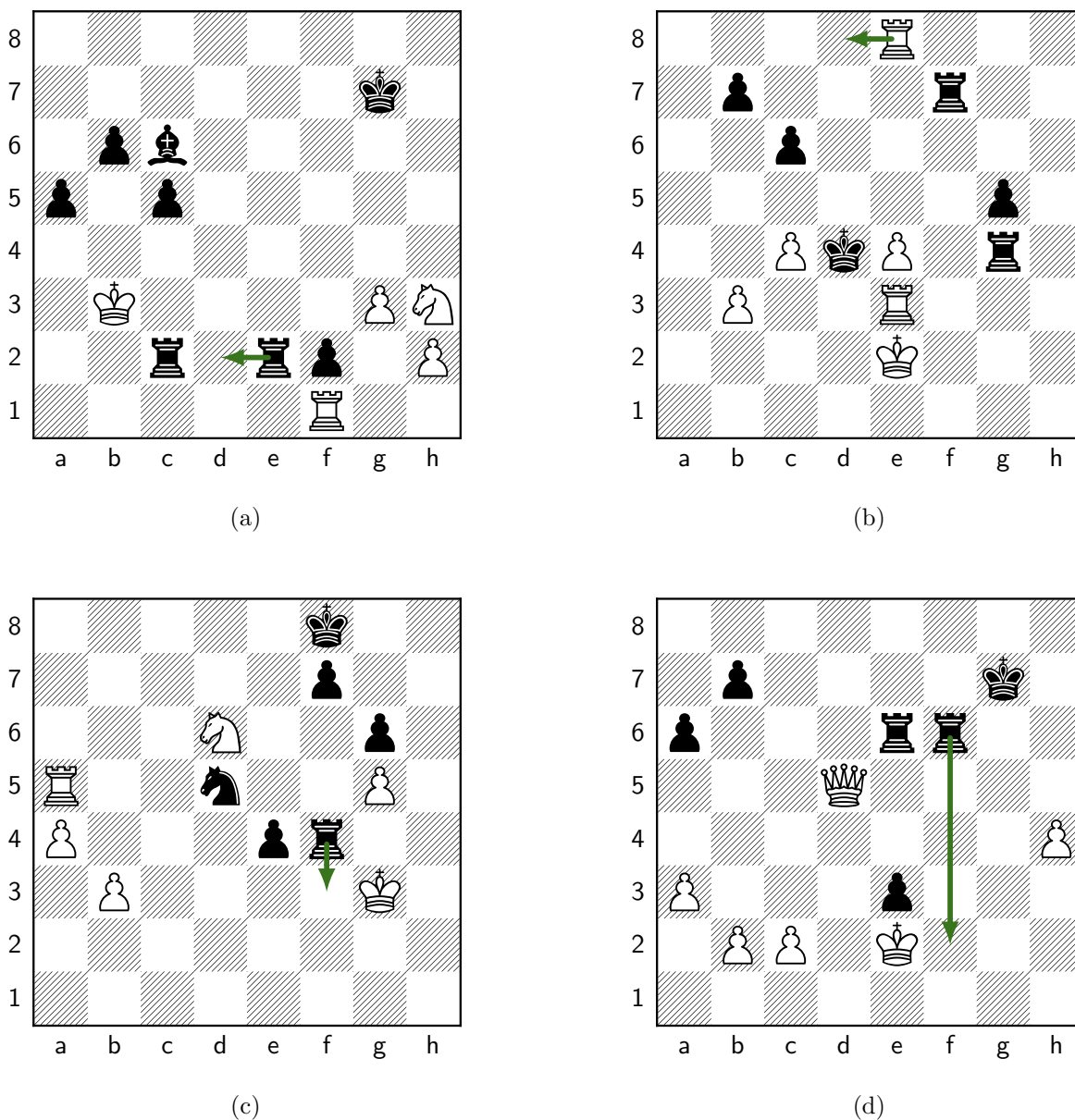
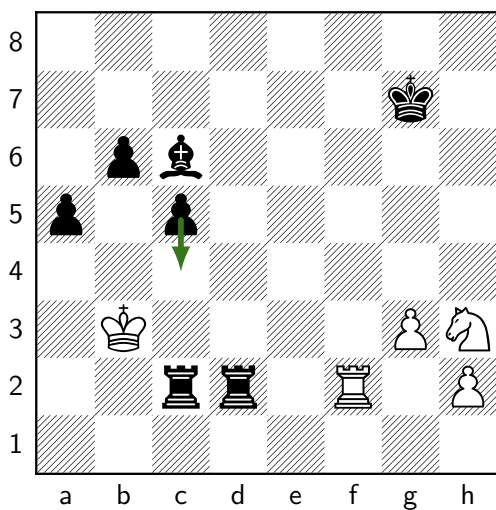
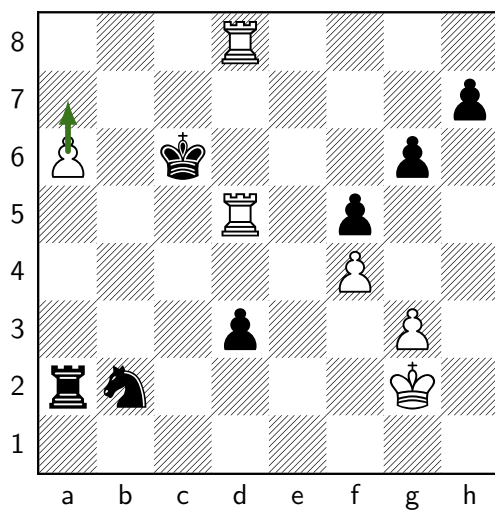


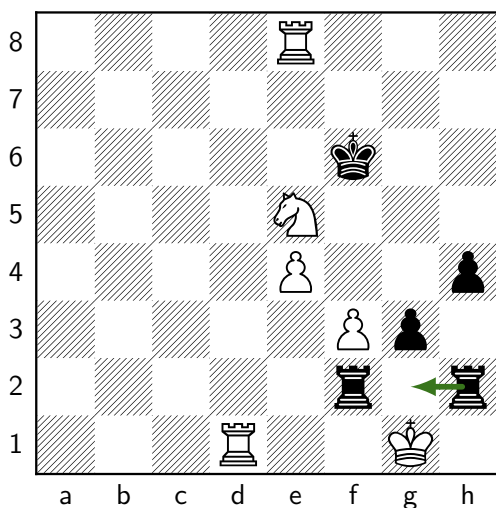
Figure 54: **Move 60, Black: Red2.** (a) Our position; Melanie plays Red2 (e2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3474` (ply 114) [structural distance $d=46$, QAP obj=21]: the master played Rd8+; under μ our Re7 plays the role of that game’s Re8; cross-type roles: our pg3 as their rg4, our nh3 as their pc6; transferred move e2d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1013` (ply 121) [structural distance $d=41$, QAP obj=23]: the master played Rf3+; under μ our Rc7 plays the role of that game’s Rf5; cross-type roles: our rf1 as their nd6, our nh3 as their ra5; transferred move c2c3. (d) Retrieved from `ccr1_top_p0003#1283` (ply 135) [structural distance $d=36$, QAP obj=19]: the master played Rf2+; under μ our Re7 plays the role of that game’s Rf3; cross-type roles: our kb3 as their qd5, our Bc6 as their pc2; transferred move e2e1. Synthesis: 2 retrieved analogs transferred to Red2 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



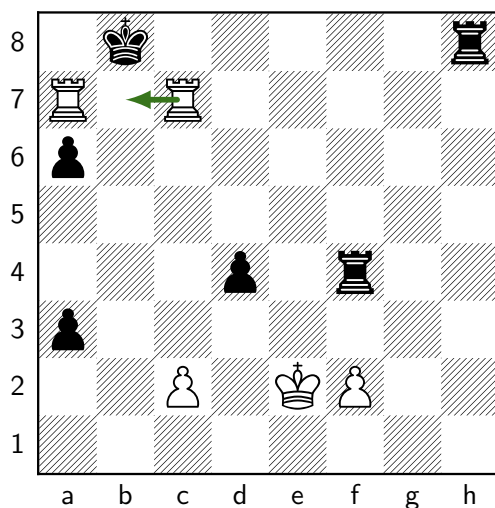
(a)



(b)

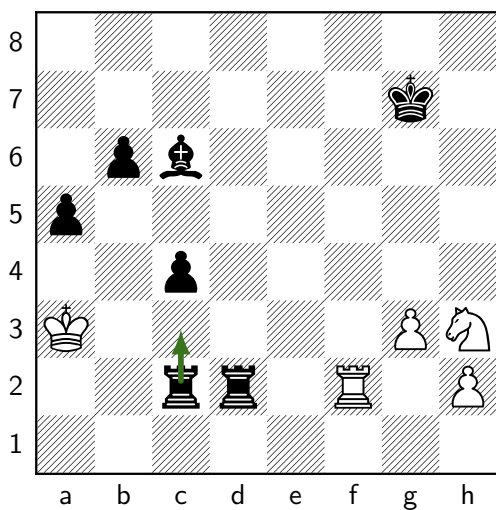


(c)

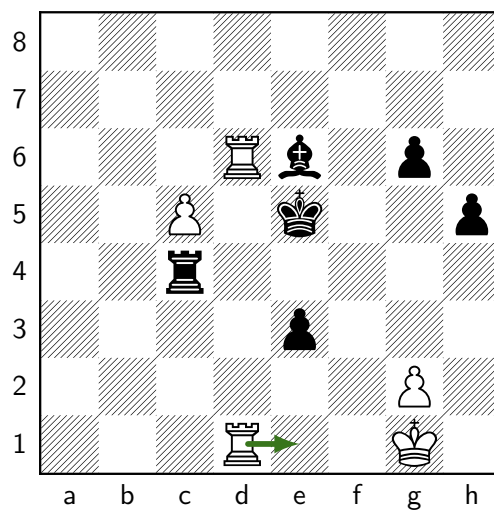


(d)

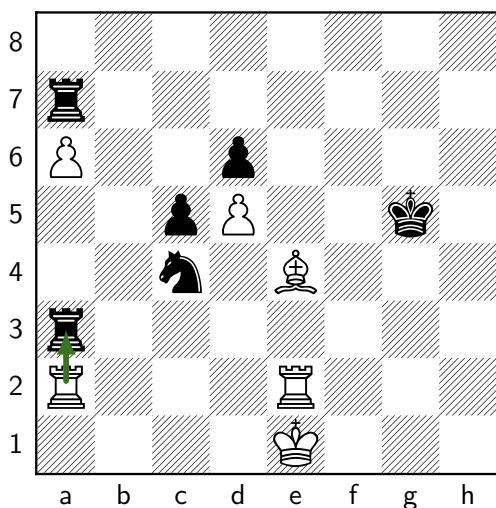
Figure 55: **Move 61, Black: c4+.** (a) Our position; Melanie plays c4+ (c5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2657 (ply 134) [structural distance $d=36$, QAP obj=25]: the master played a7; under μ our Pc_4 plays the role of that game’s Pa_6 ; cross-type roles: our nh_3 as their ra_2 , our rf_2 as their nb_2 ; transferred move c5c4 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#2642 (ply 111) [structural distance $d=30$, QAP obj=34]: the master played Rhg2+; under μ our Rc_7 plays the role of that game’s Rh_7 ; cross-type roles: our Pb_6 as their re_8 , our Rd_2 as their Ph_4 ; transferred move c2b2. (d) Retrieved from ccr1_top_p0000#3366 (ply 114) [structural distance $d=37$, QAP obj=34]: the master played Rcb7+; under μ our Rc_7 plays the role of that game’s Rc_7 ; cross-type roles: our Rd_2 as their Pc_2 , our nh_3 as their pd_4 ; transferred move c2b2. Synthesis: 1 retrieved analog transferred to c4+ (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



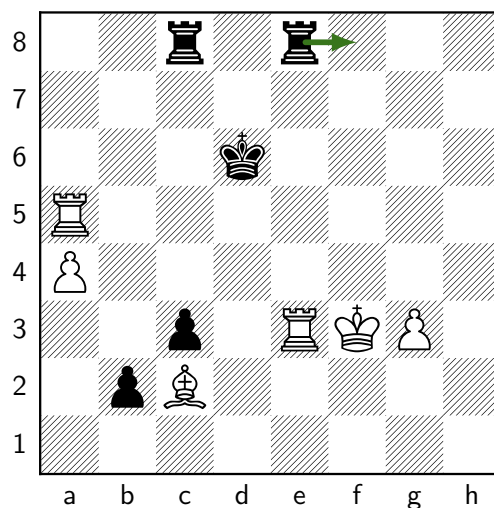
(a)



(b)



(c)



(d)

Figure 56: **Move 62, Black: Rc3#.** (a) Our position; Melanie plays Rc3# (c2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3502 (ply 116) [structural distance $d=36$, QAP obj=11]: the master played Re1; under μ our Rc7 plays the role of that game’s Rd1; cross-type roles: our nh3 as their Pg2, our Bc6 as their rc4; transferred move c2c3 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#2661 (ply 124) [structural distance $d=38$, QAP obj=25]: the master played Rxa3; under μ our Rd7 plays the role of that game’s Ra2; cross-type roles: our pg3 as their Be4, our Bc6 as their pc5; transferred move d2d7. (d) Retrieved from ccr1_top_p0001#2254 (ply 117) [structural distance $d=35$, QAP obj=18]: the master played Rf8+; under μ our Rd7 plays the role of that game’s Re1; cross-type roles: our pg3 as their ra5, our Bc6 as their pg3; transferred move d2d7. Synthesis: 1 retrieved analog transferred to Rc3# (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.